UNITED NATIONS DEVELOPMENT PROGRAMME

PROJECT DOCUMENT
The United Republic of Tanzania

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Project Title: SE4ALL Implementation in Tanzania Programme (SITAP)
Project Number: 00098492 (Award ID: 00094384)
Implementing Partner: Ministry of Energy and Minerals (MEM)
Start Date: 01 July 2017 **End Date:** 30 June 2022 **PAC Meeting date:** 12 April 2017

Brief Description

Despite the abundance of clean renewable energy (RE) potential resources, the energy sector in Tanzania is dominated by the use of biomass fuels, primarily wood, which constitutes around 85% of the primary energy mix[1]. The negative impacts of unsustainable biomass fuel use are being felt on health and the environment, through air pollution and land degradation. It also has detrimental effects on the socio-economic well-being of the households, particularly women and children who are the primary collectors and users of traditional biomass.

Tanzania's per capita electricity consumption is at very low level. It is about 137 kWh per person per year. The access to electricity stands at around 41% (Power System Master Plan (PSMP), 2016). However, about one third of the electricity sector relies on hydropower generation, which is becoming increasingly vulnerable to the effects of climate change, leading to significant strains on power generation in meeting the ever-increasing demand. To partly meet this increasing demand, diesel generators are being relied on for commercial and residential power generation. This leads to increased carbon emissions and higher costs for the end user, which ultimately hampers the national development.

The low level of electricity access and the lack of reliable energy supplies are some of the major challenges faced by the Tanzanian energy sector. They are among the major factors holding back Tanzania's achievement of its overall development goals outlined in the National Five Year Development Plan and the Sustainable Development Goals (SDGs). The situation now warrants action and a transition to more sustainable forms of energy that will ensure a decent minimum standard of living. Therefore, the demand for affordable, reliable, sustainable and efficient energy services for all developmental sectors is increasing in Tanzania.

To address some of the challenges in the energy sector, the Government of Tanzania (GoT) has committed to the global Sustainable Energy for All (SE4ALL) initiative in 2012, which was officially launched in 2016. The launch was to kick-start the activities in Tanzania's Action Agenda (AA) and Investment Prospectus (IP) that were developed based on the Rapid Assessment and Gap Analysis (RAGA) on the focus sectors i.e. grid expansion, sustainable charcoal production and usage, renewable energy based mini-grids and home systems, energy efficiency and clean cooking solutions.

In line with the SE4ALL initiative and the national priorities, this five year (s) full-fledged programme is designed to, *inter alia,* support the GoT attain its desired development goals, among other things, through enhancement of institutional capacity in RE; increased share of RE power generation; facilitating investment in RE generation; making use of the country's RE resources and reduced GHG emissions with the aim of supporting the socio-economic development of the country.

Contributing Outcome (UNDAP II 2016-2021):
Improved environment, natural resources, climate change governance, energy access and disaster risk management

Indicative Output(s):
Select LGAs and communities have their capacities strengthening in climate change governance and sustainable energy access

Total resources required ($):		**10,000,000**
Total resources allocated ($):		
	UNDP TRAC:	1,000,000
	Donors:	9,000,000
	Government:	
	Private sector:	
	In-Kind:	
Unfunded ($):		9,000,000

[1] http://www.iea.org/stats/WebGraphs/TANZANIA4.pdf

Agreed by (signatures)[2]:

Ministry of Energy and Minerals	UNDP
Print Name: *Dr Juliana Pallangyo (Deputy PS – Energy)*	Print Name: *Mr. Yahya Ba (Country Director a.i)*
Date: 23/06/2017	Date: 22-06-2017

[2] Note: Adjust signatures as needed

ABBREVIATIONS

ABPP	:	Africa Biogas Partnership Programme
AfDB	:	African Development Bank
APR	:	Annual Project Review
ATC	:	Arusha Technical College
BEST	:	Biomass Energy Strategy
BLEN	:	Biogas, LPG, Ethanol &Natural gas
BTOR	:	Back to Office Report
CAMARTEC	:	Centre for Agricultural Mechanization and Rural Technology
CDM	:	Clean Development Mechanism
COSTECH	:	Commission on Science and Technology
CPD	:	Country Programme Document
DfID	:	Department for International Development
EAC	:	East African Community
EEAP	:	Energy Efficiency Action Plan
EPZA	:	Export Processing Zones Authority
EU	:	European Union
EWURA	:	Energy and Water Utilities Regulatory Authority
FAO	:	Food and Agricultural Organization
FYDP	:	Five Year Development Plan
GACC	:	Global Alliance for Clean Cook stoves
GDP	:	Gross Domestic Product
GHG	:	Green House Gas
GIS	:	Geographic Information System
GLOLA	:	Global Off-grid Lighting Association
GoT	:	Government of Tanzania
ICS	:	Improved Cook Stove
IP	:	Investment Prospectus
IPP	:	Independent Power Producer
KSCL	:	Kilombero Sugar Company Limited
KSL	:	Kagera Sugar Limited
LDC	:	Least Developed Country
LGA	:	Local Government Agency
LNG	:	Liquefied Natural Gas
LPG	:	Liquefied Petroleum Gas
LRA	:	Legal, Regulatory and Administrative
LRTC	:	Lighting Rural Tanzania Competition
M&E	:	Monitoring and Evaluation
MDG	:	Millennium Development Goal
MEM	:	Ministry of Energy and Minerals
MEPS	:	Minimum Energy Performance Standards
MoFP	:	Ministry of Finance and Planning
MoIT	:	Ministry of Industry and Trade
MSEL	:	Mtibwa Sugar Estates Limited
MSME	:	Micro, Small and Medium Enterprises
MTOE	:	Million Tons of Oil Equivalent
NEMC	:	The National Environment Management Council
NGO	:	Non-Governmental Organization
NORAD	:	Norwegian Agency for Development Co-Operation
NPC	:	National Project Co-ordinator
NSGRP	:	National Strategy for Growth and Reduction of Poverty
O&M	:	Operation and Maintenance

PA	:	Project Assistant
PHC	:	Primary Health Care
PIR	:	Project Implementation Report
PM	:	Project Manager
PO - RALG	:	President's Office - Regional Administration and Local Government
PPP	:	Public Private Partnership
PPR	:	Project Progress Reports
PSC	:	Project Steering Committee
PSMP	:	Power System Master Plan
PSRGSP	:	Power Sector Reform and Governance Support Programme
PV	:	Photo Voltaic
RAGA	:	Rapid Assessment and Gap Analysis
REEE	:	Renewable Energy and Energy Efficiency
REIF	:	Renewable Energy Investment Facility
RERE	:	Renewable Energy for Rural Electrification
RRF	:	Results and Resources Framework
SACCO	:	Savings and Credit Cooperative Organization
SBAA	:	Standard Basic Assistance Agreement
SDG	:	Sustainable Development Goal
SE4ALL	:	Sustainable Energy for All
SEFA	:	Sustainable Energy Fund for Africa
SES	:	Social Environmental Standard
SEZ	:	Special Economic Zone
SME	:	Small and Medium Enterprises
SPPA	:	Small Power Purchase Agreement
SREP	:	Scaling-Up Renewable Energy Program
SSC/TrC	:	South-South and Triangular Cooperation
SSC	:	Sub-Saharan Countries
TANESCO	:	Tanzania Electricity Supply Company
TAREA	:	Tanzania Renewable Energy Association
TDBP	:	Tanzania Domestic Biogas Program
TBS	:	Tanzania Bureau of Standards
TEDAP	:	Tanzania Energy Development and Access Project
TFS	:	Tanzania Forest Services
TIB	:	Tanzania Investment Bank
TIC	:	Tanzania Investment Centre
ToR	:	Terms of Reference
TPC	:	Tanganyika Planting Company
TPDC	:	Tanzania Petroleum Development Corporation
TRA	:	Tanzania Revenue Authority
TREMIS	:	Tanzania Renewable Energy Information Management System
UDSM	:	University of Dar es Salaam
UNDAF	:	The United Nations Development Assistance Framework
UNDP	:	The United Nations Development Programme
UNIDO	:	The United Nations Industrial Development Organization
UNSMS	:	The United Nations Security Management System
USAID	:	The United States Agency for International Development
USD	:	United States Dollars
VETA	:	Vocational Educational and Training Authority
VPO-DoE	:	Vice-President’s Office – Department of Environment
WTE	:	Waste to Energy
WWF	:	World Wildlife Fund

I. DEVELOPMENT CHALLENGE

As per Tanzania's National Five Year Development Plan (2016/17 to 2020/21), energy production and distribution facilities are the main (the) efforts to improve socio-economic development in the country. The Plan encourages diversification of power generation sources to stand against the increasing unpredictability of weather and the growing negative impact of climate change. The Tanzanian economy is facing big challenges as majority of the population still lacks access to sustainable energy supplies. Currently, 41% of the population is connected to electricity, with electricity consumption per capita standing at137 kWh per person per year (Power System Master Plan (PSMP), 2016). This ranks very low compared to the world average per capita consumption of 3,104 kWh per annum and even the Sub-Saharan African's average per capita consumption of 488 kWh per annum[1]. The demand for affordable, reliable, sustainable and efficient energy services for all developmental sectors is increasing rapidly in Tanzania.Peak power demand is projected to increase rapidly from about 1,270MW at present to about 3,600 MW by 2025 and 5,870 MW by 2030[2]. Scarcity of energy is one of the underlying factors that will hinder the successful implementation of the National Five Year Development Plan and the achievement of the Sustainable Development Goals (SDGs).

In past, Tanzania's electricity sector had relied significantly on hydropower generation. The hydropower generation is vulnerable to the changing weather patterns and the expected power generation was less than the installed capacities during droughts. Blackouts and power rationing caused by such low electricity generation from hydropower plants had forced Tanzania Electricity Supply Company (TANESCO) to shift to the oil and gas fired generators and increasingly turn to the thermal projects to increase the future capacity. As of January 2017, total installed capacity (grid & mini-grid) of natural gas power plants is 640.84 MW (44%), large hydro is 561 MW (38%), other fossil fuel fired (heavy fuel oil and industrial diesel oil) is 225.25 MW (16%), biomass is 13 MW (1%) and small hydro is 12 MW (1%)[3]. This situation calls for diversifying the energy mix and tapping into renewable energy (RE) sources less vulnerable to climate change.

Lack of access to modern energy services increases rural poverty due to persistent limited production opportunities and social facilities. Available data reveals that the poor spend about 35% of their household income on energy while the better-off spend only 14%[4] for the same. And, even those with Liquefied Petroleum Gas (LPG)stoves still use biomass to reduce the high costs in meeting energy needs[5].Tanzania imports a large amount of fossil fuels for its cooking (LPG) and lighting (kerosene) needs, which burdens the country's economy and rural communities. This is mainly because the prices of imported fossil fuels are volatile and have been hindering the efforts to increase the rural energy access. Poor families are particularly more vulnerable to the volatile energy costs because they have few assets to fall back on and limited ability to afford to energy related costs and effects. Exposure to such expenses has been a trap to millions of people living in poverty and it undermines their efforts to overcome it[6].

The low use of electricity and LPG at the household level correlates with an over dependence on (of) wood fuel, which has detrimental effects on the socio-economic wellbeing of households, particularly women and children who are the primary collectors and users (women) of traditional biomass. Use of biomass energy has significant negative gender consequences; it is responsible for about 18,900 deaths of mainly women and children due to the pollution that it causes. The collection of firewood exhausts women and leavesthem with little time to engage in other important endeavours of education and other productive activities. Indeed, there is a direct relationship between the absence of adequate energy services and many poverty indicators such as infant mortality, illiteracy, life expectancy and total fertility rate. Inadequate access to energy also exacerbates rapid urbanisation in Tanzania, by driving people to seek better living conditions.

[1] http://data.worldbank.org/indicator/EG.USE.ELEC.KH.PC

[2] Power System Master Plan 2016, TANESCO, Tanzania

[3] Data provided by TANESCO, January 2017 (Refer Annex 6)

[4] https://www.worldfuturecouncil.org/file/2016/04/WFC_2016_100RE_and_poverty_reduction_in_Tanzania.pdf

[5] Refer Table 2 in Annex 7 for comparison of cooking fuel prices

[6] https://www.worldfuturecouncil.org/file/2016/04/WFC_2016_100RE_and_poverty_reduction_in_Tanzania.pdf

Given the importance of energy to Tanzania's economic growth and ambitious development goals, the country must continue to expand its power-generation capacity to meet the growing demands. But it must be done in a sustainable manner that diversifies the energy portfolio. The government recognises that a strategy that utilises indigenous sources of renewable energy has the potential to reduce the risk of supply disruptions, avoid price shocks, reduce pressure on the local and global environment and improve the socioeconomic development outcomes. PSMP 2016 suggests that the optimum future electricity generation expansion ratio as: 40% natural gas, 35% coal, 20% hydro and 5% other renewables. In the existing conditions (Jan 2017), the renewables constitute only around 2% of the overall grid power generation[7].

The root causes, problems and effects associated with the development of electricity access and modern cooking are represented in the figure 1.

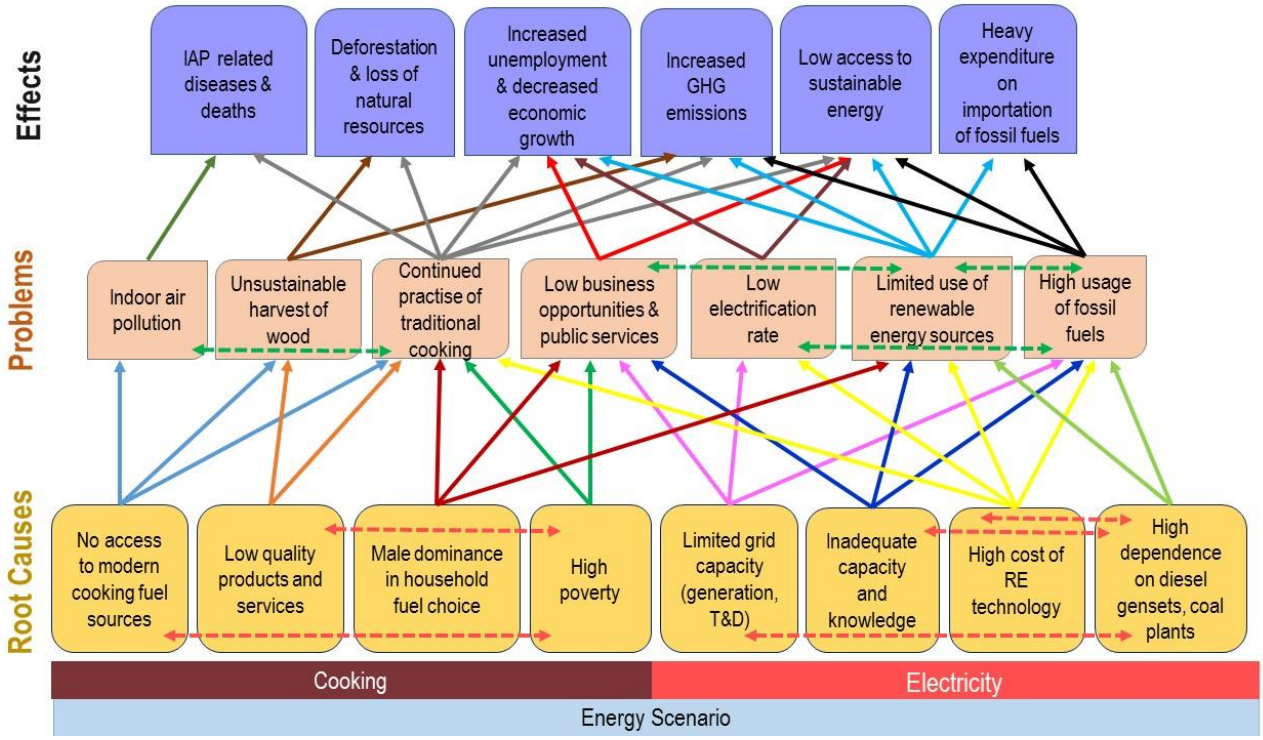


Figure 1: Problem Tree Analysis

Tanzania has a unique opportunity to propel its future socio-economic development by following a low-carbon pathway. Effective and sustained access to energy plays a significant role in improving people's living conditions, and contributes to economic and human development. Energy provides services to meet many basic human needs, particularly heat, mechanical power (e.g., water pumps and transport) and light. Business, industry, commerce and public services such as modern healthcare, education and communication are also highly dependent on access to energy services.

[7] Data provided by TANESCO, January 2017 (Refer Annex 6)

II. STRATEGY

The core aim of the SDG 7 is "Affordable and Clean Energy." UNDP Tanzania is committed to support the Government of Tanzania (GoT) in achieving the SDG of which this project's strategy will be rooted. As a trusted partner, UNDP has already supported the Government in integrating the environment and energy issues into national policy, as well as in building the local and regional capacity simultaneously. Tanzania's SE4ALL, under the leadership of the Ministry of Energy and Minerals (MEM) with high cooperation from UNDP and other development partners, seeks to integrate the multi-tier efforts of the country towards providing universal access to energy, increasing the energy efficiency (EE) and RE aspects.

Currently, there are several government policies, private sector initiatives, projects and programs that are being carried out in Tanzania energy sector. Significant development changes have been made possible through these initiatives in areas such as mini-grids for rural electrification, promotion of improved cook stoves (ICS) and biogas stoves, market development of solar home systems (SHSs), etc. However, each project is executed separately without or with limited co-ordination with each other. This often leads to the duplicated efforts, market development only in few concentrated regions, ineffective use of available funds/resources and deviation from the achievement of national targets.

Through a collaborative multi-stakeholder approach, this SE4ALL programme strategy is aimed to enhance coordination among stakeholders in order to oversee, collaborate and complement each other among multiple government policies, initiatives, projects and programs that are currently being carried out in the Tanzania's energy sector namely; (i) Second National Five Year Development Plan (2016/17 – 2020/21); (ii) Electricity Supply Industry Reform Strategy and Road map (2014-2025); (iii) Power Sector Reform and Governance Support Programme – Phases II & III (PSRGSP-II & III) (2015-2017); (iv) National Electrification Program Prospectus, Tanzania (2013-2022) and; (v) Power System Master Plan (2016-2040).

This programme will also align with important multi-lateral projects including; (i) Scaling-Up Renewable Energy Program (SREP), Tanzania (2013-2020); (ii) Africa Enterprise Challenge Fund (AECF); (iii) Power Africa; and (iv) Global Alliance for Clean Cook stoves (GACC). To oversee the programme a Secretariat will be formed. The simple schematic of SE4ALL implementation is given in figure 2.

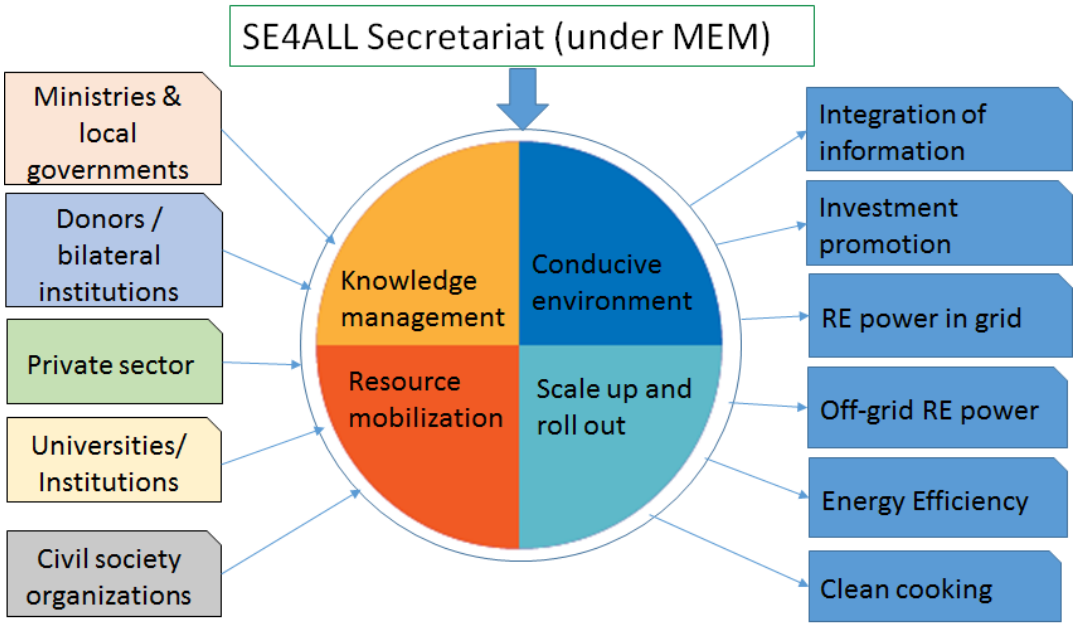


Figure 2: Schematic of SE4ALL implementation

To achieve the objectives of the SE4ALL initiative, it is important that all the stakeholders play a leadership role including but not limited to:

- **Government**– to provide supportive environment to encourage investments in focus sectors through its ministries, departments, local government authorities and institutions.
- **Private sector**– to provide business and technical solutions and drive investments.
- **Civil society organisations (CSOs)** and **Non-Government organizations (NGOs)** -to engage actively with people at the grass root level to increase awareness and promote RE & EE aspects.
- **Banks, financial institutions, micro-credit institutions -** to provide finance to scale up and roll out projects.
- **Multi-lateral funding agencies** - to bring in international expertise, finance and technology for knowledge transfer.
- **Universities/RE institutions**- to impart capacity and skill and bring innovative solutions in focus sectors through research and development (R&D).

The overall approach of the SE4ALL programme is to:

- Estimate the targets under different aspects for achieving SDG7.
- Collaborate with the stakeholders for accelerating the on-going project and plans.
- Identify the gaps in achieving the SDGs.
- Propose strategies for further scale up and roll out and bridge the gaps.

The first step is to consider the available latest information/data as the baseline and the SDG 7 as the key target achievement. The identified gap in between these two will show the opportunity available to take initiatives accordingly for filling up the gap to reach the target.

The baseline values for the SE4ALL goals are given in Table 1.

Table 1: Baseline values of SDG 7 goals of Tanzania

No.	Goals	Baseline values (%)	Source
1	Population with electricity access (at household level)[8]	24.00	Tanzanian energy sector under the universal principles of the Energy Charter, page 4 (Ref. year:2015)
2	Population with access to modern cooking solutions[9]	16.00	Tanzania SE4ALL AA, Table 1, page 16 (Ref year: 2012) and ICS Taskforce: Country action plan for clean cook stoves and fuels (Ref. year: 2014)
3	Rate of improvement in the energy intensity	-2.60	Tanzania SE4ALL AA, Table 6, page 24(Ref. year: 2012)
4	Renewable energy share in the overall electricity generation	2.00	TANESCO data, Annex 6(Ref. year:2017)
5	Renewable energy share in thermal (cooking and heating)	4.00	Tanzania SE4ALL AA, page 2, (box text) (Ref. year:2012)

The comparison of baseline values and target values (2030) for each goal of Tanzania SE4ALL is given in Table 2.

[8]Electricity access considering proximity to grid access is 41% and considering household connections is 24%. For SE4ALL target, household level connections are considered indicator for electricity access goal

[9]Includes charcoal ICS and clean cook stoves [Biogas, LPG, Ethanol and Natural Gas (BLEN)]

Table 2: Baseline and SDG 7 targets

No.	Goals	Baseline values (%)	Target (2030) (%)
1	Population with electricity access	24	75
2	Population with access to modern cooking solutions	16	75
3	Rate of improvement in energy intensity	-2.6	-2.6
4	RE share in the overall electricity generation	2	5
5	RE share in thermal (cooking and heating)	4	10

The details of all the on-going projects under different sectors (grid expansion, RE based mini-grids, EE and clean cooking) were collected from various sources, stakeholder meetings and validation workshops.

Data was collected on the various on-going and proposed projects under RE power plant, electricity access, modern cooking solutions, sustainable charcoal production, etc. Based on the data collected, the contribution from various on-going/proposed projects towards the SE4ALL targets and the identified gaps to be filled up are estimated. The gaps identified in achieving the targets are furnished in table 3.

Table 3: Baseline, contributions from the ongoing projects and gaps

No.	Parameter	Unit	Target 2030		Baseline[10]	Contribution from the on-going projects[11]	Identified gaps
			%	Value[12]			
1	Households with electricity access	Nos.	75	10,575,000	2,647,463	3,991,039	3,936,498
2	Households with modern cooking	Nos.	75	10,575,000	1,751,555	1,770,530	7,052,915
3	Average annual energy savings needed	MTOE	-2.6	7.18	0.38	0	6.80
4	RE share in overall electricity generation	MW	5	591	25	255	311
5	Annual sustainable wood supply	tons	10	11,359,143	2,009,664	0	9,349,479

To fill up these gaps and accelerate the energy access in focus sectors, the following strategies are proposed under this five-year programme.

[10]Based on baseline percentages identified in table 2

[11]Based on the data available so far from various on-going activities by GoT, donors and private sectors

[12]Estimated based on the target percentages

a) Strengthening capacity and knowledge management

The strategy will involve identification and integration of all information resources on RE and EE made available by various on-going initiatives of GoT, donors and the private sectors. The focus is to achieve maximum information exchange and co-ordination in all activities of RE and EE sector across the stakeholders. Quality training and knowledge transfer will be ensured through series of effective training approaches designed and monitored through existing education system which will incorporate RE and EE issues. The framework for promoting all the EE and energy conservation activities in the country will be developed.

b) Attracting investments in sustainable energy sector

The strategy is to come up with supportive environment for business development, develop market network, manufacture/make available quality products and attract investments from financial institutions and donor agencies. Quality standards and testing procedures will be developed to ensure quality of the products. The strategy will also involve evolving conducive environment for business by increasing the fund/finance availability, exploring the opportunity using different promotion avenues such as Tanzania Investment Centre (TIC), Special Economic Zones (SEZ), etc.

c) Accelerating sustainable energy initiatives

The strategy will involve further scaling up and rolling out of the existing projects and businesses in the energy sector. Private sectors will be encouraged to provide innovative business model to implement the projects in different sectors. Grid expansion and energy efficiency under public administration will also be improved. Partnerships between public, private & international agencies will be promoted wherever possible.

The theory of change based on above strategy is provided in figure 3.

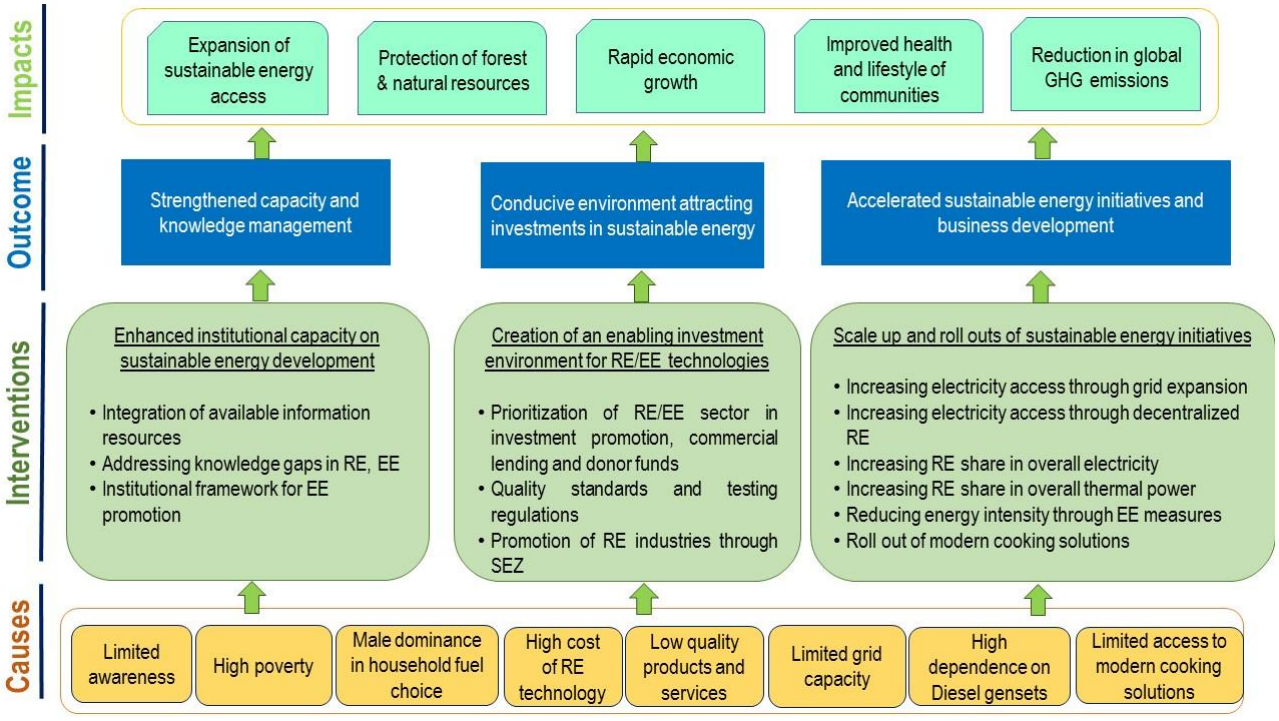


Figure 3: Theory of Change

III. RESULTS AND PARTNERSHIPS

Expected Results

The proposed interventions through the five year SE4ALL programme are detailed in this section.

Output 01: Enhanced institutional capacity on sustainable energy development

Activity 1.1: Integration of available information resources on renewable energy

Key existing initiatives:

MEM is developing a Tanzania Renewable Energy Information Management System (TREMIS), which is an exclusive website for renewable energy in the official website of MEM. UNDP is supporting MEM on this aspect. The website is expected to have details exclusively for all the subsectors namely solar, hydro, biomass, wind, etc. The plan is to have different levels of access to the portal for different stakeholders such as MEM, Government officials, donors, investors, private sectors, NGOs, educational institutions, common people, etc., depending upon the type and level of involvement. Currently, the web portal design is going on and all the key stakeholders are part of the development process.

Tanzania Investment Centre (TIC) has already established a one-stop-shop focused on business investments. TIC deals with all the enterprises whose minimum capital investment is not less than USD 500,000, if foreign owned or USD 100,000, if locally owned[13]. TIC also deals with RE business investments promotion. Linking information from the TREMIS will greatly facilitate promotion of RE investments and opportunities through TIC one-stop-shop.

An interactive mini-grids information portal (http://www.minigrids.go.tz) is available to provide comprehensive information for the investors interested in developing RE mini-grids in Tanzania. It was developed under Scaling-Up Renewable Energy Program (SREP) supported by the World Bank. This portal currently provides the countrywide geographic information system (GIS) mapping of potential sites for small hydro mini-grids and solar mini-grids. This portal will be handed over to REA, soon.

Similarly, ICS Taskforce, formed by the stakeholders working in cook stove sector, is working on establishing market information for ICSs in different regions of the country. Information on application formats, approval procedures and document checklists are available through National Environment Management Council (NEMC), Tanzania Bureau of Standards (TBS), Tanzania Revenue Authority (TRA), Electricity and Water Utilities Regulatory Authority (EWURA), etc.ICS market intelligent study has been conducted by Stichting Nederlandse Vrijwilligers (SNV) and Tanzania Renewable Energy Association (TAREA) so far in Arusha, Manyara & Mtwara regions.

Important institutions such as University of Dar es Salaam (UDSM), Centre for Agricultural Mechanisation and Rural Technology (CAMARTEC), Vocational Educational and Training Authority (VETA) and other Universities are offering courses and training in various RE sectors. Relevant donor activities, other related activities such as RE potential assessments, RE based initiatives, businesses, research, etc. are also being conducted by various stakeholders in the country.

[13] http://www.tic.co.tz/Team?l=en

Identified gaps:

- Though various initiatives and works are being done related to RE and EE within the country, there is no integration of information resources/services, so that the information is easily accessible and available to relevant stakeholders as per their requirement from one location.
- The web portal that is being developed through TREMIS requires to be updated and maintained regularly.
- TIC is unable to get the latest information from various relevant institutions to offer updated service accordingly. This is essential to attract investments in the country.
- Mapping of potential biomass and wind mini-grid sites need to be updated in the www.minigrids.go.tz portal.
- Conduct of baseline study on ICS/clean cook stove market demand assessment at the district and regional levels with special focus on women preferences. ICS market intelligent study has been conducted so far in Arusha, Manyara and Mtwara regions only.

Scope of interventions:

The TREMIS web portal under MEM can be integrated/linked with the web portals of TIC (one-stop-shop for investment), mini-grids information hub (www.minigrids.go.tz), ICS Task force and other relevant institutions. It should also include information such as NEMC procedures, TRA information, resource assessments and any such works/initiatives/reports done by the donors, NGOs, private sector, etc. Integration can be done in such a way that it covers all possible aspects of RE information & serves as an 'A-Z information portal for RE'. Any person (donors, private sector, NGOs, common people, research students, and entrepreneurs) should have access to the latest developments in the sector, as per their own requirement.

To achieve this objective, technical, financial and management support will be needed by MEM from all the relevant stakeholders.

Through this website, the gap in various activities in the country would be known and the activities can be planned accordingly to fill the gap. The website will reduce the time & financial resources that would be spent on the same activity done earlier. Trainings, workshops, conferences and trade fairs on renewable and clean energy technologies can be organised in co-operation with TIC. This will help market sensitisation and promotion of the SE4ALL objectives, proposed initiatives and opportunities available through strong partnership. In the long term, creation of a one-stop-shop exclusively for energy sector can also be considered. The initiatives taken for the next five years can provide a solid base for establishment of such one-stop-shop.

Approach for gender mainstreaming

To link the data and information on the available capacity to mainstream gender and women empowerment in energy projects/programmes for capacity strengthening.

Approach for engaging Local Government Agencies (LGAs)

- Awareness to be created for LGAs on RE technologies and the importance of sharing RE information.
- Mechanism for mutual sharing of information from LGAs to MEM on the on-going and upcoming projects in the respective regions along with their impacts to local communities.
- LGAs should give regular feedback on the latest information happening at the grass root level for regular update of the website.

Activity 1.2: Addressing knowledge gaps in RE, EE& modern cooking

Key existing initiatives:

Training and capacity building in RE/EE sector has already been imparted through various initiatives by the GoT, donors, NGOs and private sector. VETA and other high learning institutions are offering

technical courses and trainings on RE technologies. TAREA through its members' network is working on imparting skill and developing capacity in various levels like entrepreneurship, technical skill, etc.

Arusha Technical College (ATC) is offering training courses for skilled labourers/technicians in hydro power. It is working on developing turbine testing centre for hydro power technology. It also has a Pico size (less than 100 kW) Pelton turbine manufacturing facility. University of Dar es Salaam (UDSM) has a training centre for hydro turbines and has an axial flow turbine manufacturing unit for sizes up to 300 kW.

ICS Task Force has proposed to identify and support dedicated ICS and fuel training facilities in collaboration with GACC.



Figure4: Improved charcoal cook stoves

Mobisol, a solar based private company, has established an academy/training centre with the support of DfID in solar system service and maintenance.

CAMARTEC is offering certified courses and training on biogas digester installation, operation, maintenance, entrepreneurship, etc.

Identified gaps:

- No consolidated data/information is available on institutions, courses, curriculum, training facilities for RE/EE technologies such as ICS, LEDs, CFLs, BLEN[14], solar PV, solar thermal and small hydro power turbines, biomass and biogas technologies.
- No centralised approved minimum quality standard has been set for the various training courses, facilities, trainers' qualification, trainees' qualification, etc. The present status is that different institutions provide different training courses and different training facilities exist in the RE/EE and cooking sector. This leads to lack of quality work force in the RE sector.
- Limited training courses/facilities for EE sector.
- Need for capacity building for public, private sector, financial institutions, NGOs, etc., as per their requirements/involvement in RE/EE & modern clean cooking sector.

Scope for interventions:

The priority intervention will be to assess the existing facilities/resources available at the Universities and institutions in line with SE4ALL focus sectors. Assessment will also include capacity building needed for women and men in government ministries, LGAs, private sectors, financial institutions, entrepreneurs, artisans, NGOs, technical institutes, etc. For example, The NEMC has already

[14]Biogas, LPG, ethanol, natural gas

prepared a checklist for solar mini-grids and the checklist is in final stages to be published and it has expressed a need for technical capacity support for preparing such checklists for hydro and biomass power projects. The assessment will take into consideration, the capacity needed for various activities to be implemented under SE4ALL such as standards and labelling programme, quality testing labs for ICS / LEDs / CFLs / BLEN / SHS / Solar PV / Solar lantern / Solar thermal / small hydro power turbines / other RE sustainable charcoal production, etc.

A nodal training institute can be identified that will cover training and capacity building in each of the focus sectors of grid power generation/distribution, renewable energy such as solar PV, wind, small hydro, biomass, EE and mini-grid development. The nodal training institute will develop minimum standards for training and certification programs along with the detailed curriculum for training & certification courses based on the assessment in these focus sectors. Guidebooks, manuals, training materials, etc. will be prepared based on the curriculum and shared through online sources for access to anyone. Guidelines can be prepared for minimum infrastructure facilities required for a training centre. These nodal training institutes will then provide training to the trainers and other institutions. These trainers and the institutions can further set up training centres to train various stakeholders.

Partner channels[15] can be provided with franchise to set up training centres at cities, towns and villages. Technical & financial support may be provided in setting up these training centres and conducting training courses. This ensures that the training facilities reach to the target people to create maximum impact. The partner channels can be given a nominal grant for each trainee trained. This will also bring down the cost of training making it more affordable to the trainees. The grants can be reduced and brought to zero over a period, by which the partner channels will be self-sufficient to run their own quality training institutes. Establishment of a single centre of excellence for all energy technologies can be a long-term goal based on above experiences.

Quality assurance:

An accreditation team can be set up under Ministry of Education to assure the quality of the training and facilities provided by these training centres and partner channels from time to time. Training, capacity building and necessary assistance will be provided to the Ministry of Education in this aspect under this initiative. Similarly, the curriculum for training may be constantly reviewed and revised at least once in two years.

A simple depiction of activities for quality training centres is provided in figure 5.

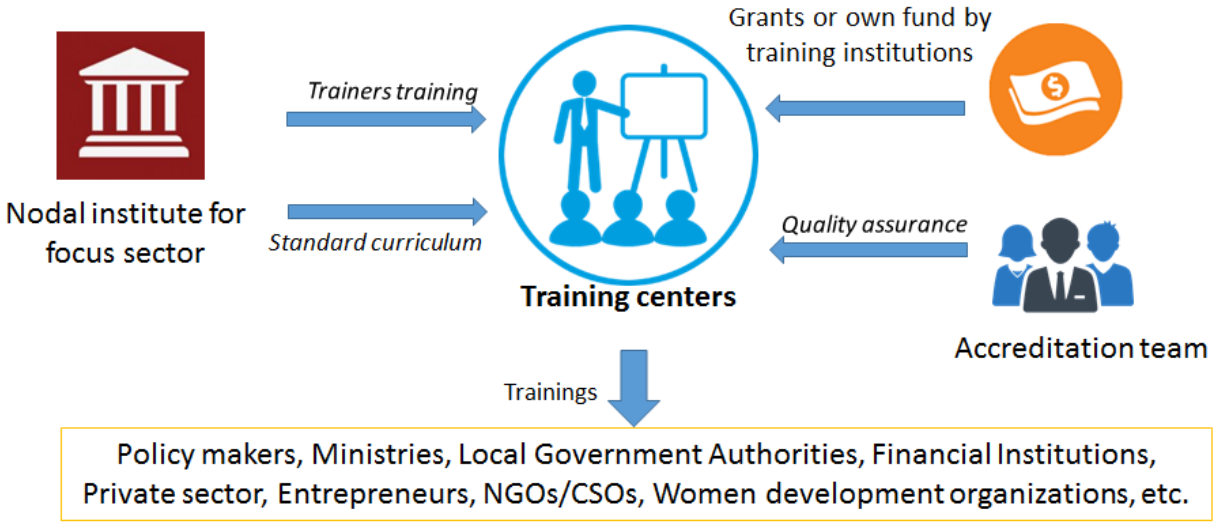


Figure 5: Schematic of quality assurance of the training centres

[15] Partner channels are authorized private sector partners who will set up training centers and provide trainings within the curriculum and minimum standard set by the nodal institute

Funds for these activities can be obtained from various stakeholders such as government ministries, Universities, institutions, partner channels, other training centres and donors. For cost effectiveness, the activities may seek co-ordination with other on-going training programmes to share the available resources.

Approach for gender mainstreaming

- Special fund/scheme can be created for female students & it can be linked with the universities and vocational training centres
- Engaging women consultants, experts, trainers, etc. in capacity assessment, guidebooks, manuals, training material preparations& in trainings, etc.
- Conducting dedicated training programs for women entrepreneurs& technicians in RE, EE & cooking sector.
- Special training schemes could be created, through already established institutes such as VETA, for school dropout girls/women on acquiring technical skills in RE, EE & cooking sector.
- At least 40% of the participants in all the training activities shall be women

Approach for engaging LGAs

- Capacity building for LGAs regarding their participation and contribution for effective implementation of RE/EE projects in their regions
- Engaging LGAs to identify local participants for the training programs
- Capacity building for the local government at district level to identify, develop and manage the available RE potentials (such as community biogas for cooking, agro-waste to power, sustainable wood harvesting, etc.)

Output 1.3: Institutional framework for energy efficiency promotion

Existing initiatives:

The National Energy Policy (NEP,2015) thrusts the promotion of energy conservation and EE practices. Benchmarking energy use practises in industries (such as mining, small and medium agro-based), transport, households and commercial building are recommended in the policy. Through various initiatives, Tanzania has already reduced the transmission and distribution losses in the TANESCO grid from 19.8% in 2010 to 16.5% in 2016. The PSMP 2016 targets to bring down the losses to 11.4% by 2025.

Currently, MEM is working on EE in Tanzania and implementing the EE targets in collaboration with TANESCO. The Ministry is in the preparatory stages of setting up responsible agencies/team and institutions to implement and oversee EE programs as indicated in the Ministry's strategic plans[16].

Several initiatives and projects are currently being undertaken in the EE sector through partnership between MEM/TANESCO and international organisations such as JICA[17], GIZ[18], USAID[19], Embassy of Sweden[20] and DANIDA/CTI[21]. Bank of Africa, Tanzania has proposed a credit line for implementing RE and EE projects in the country[22].

[16] http://rise.worldbank.org/data/exploreeconomies/tanzania/2014/energy-efficiency

[17] Capacity development of efficiency distribution and transmission systems - 2009/15

[18] Energy sector support programme – 2013/16

[19] Green buildings study 2014, Load management analysis for large energy consumers - 2015/16

[20] National Energy Efficiency Programme - 2014/15

[21] National energy audit programme 2014/16

[22] Roadmap for enhancing energy efficiency: Tanzania Country Report, MEM

Identified gaps:

- Institutional framework is needed for carrying out an integrated approach in promoting/enforcing nationwide EE activities across the sectors
- Need for capacity building among government ministries, Tanzania Bureau of Standards (TBS), private sectors, financial institutions, entrepreneurs, technical institutes, skilled labours, NGOs, etc. on EE and energy conservation, energy labelling etc. in their own context.
- Inadequate availability of EE and/or energy audit professionals
- Need for benchmarking baseline energy consumption in industries (such as mining, small and medium agro-based industries, etc.), transport, households and commercial buildings.

Scope for interventions:

Based on the above-mentioned baseline activities, it is proposed to establish a dedicated unit/team for EE under SE4ALL Secretariat within the existing legal and regulatory structure. The dedicated unit shall be the nodal agency for promoting all the EE related projects in the country. Design of operation, roles, responsibilities, budget and organisation structure will be prepared. In a longer run, this team may be developed into independent unit for EE. The thematic representation of the activities of the dedicated EE unit/team is given in figure 6.

Role of MEM, Dedicated Energy Efficiency Unit/team and TBS in energy efficiency:

Role of MEM:

- Prepare guideline for
  - Role of dedicated team for EE to implement EE projects in Tanzania
  - Identifying designated consumers[23] and their responsibilities
- Support planning, execution, monitoring of activities carried out by the dedicated team for EE.

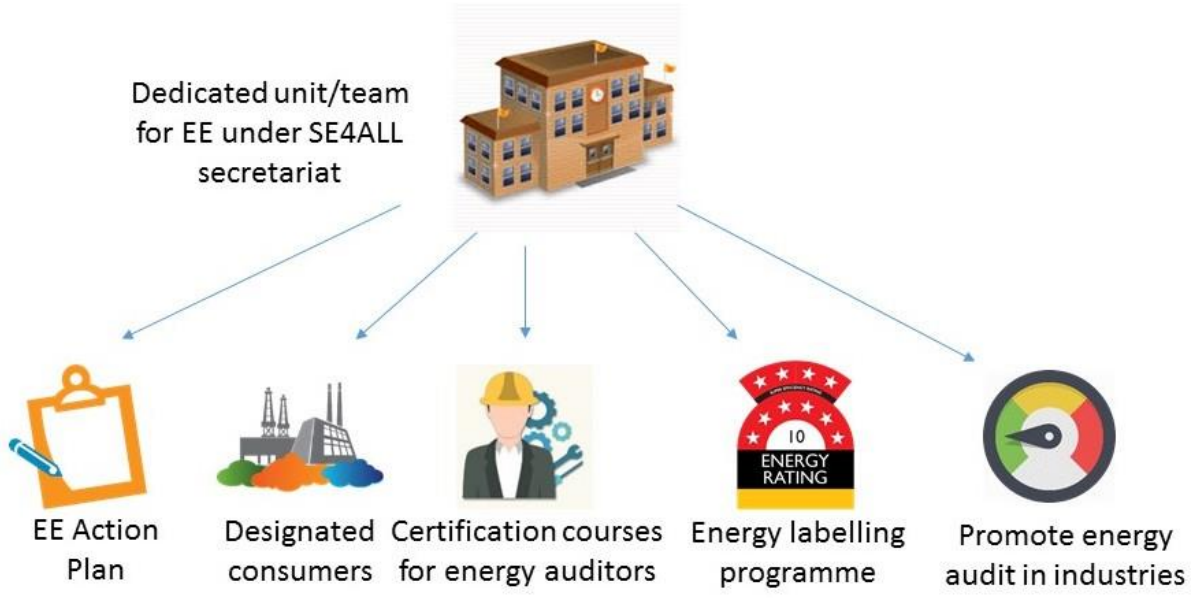


Figure 6: Proposed activities under EE unit

Role of dedicated unit/team for EE

- Prepare Energy Efficiency Action Plan (EEAP).
- Collaborate with the Ministry of Education, universities and other learning institutions to prepare syllabus for training courses and examination structure for certification of the energy auditors.
- Develop benchmark for annual energy consumption in the energy intensive sectors.
- Identify the designated consumers in the energy intensive sectors.
- Conduct workshops on EE for designated consumers.

[23]High energy consuming industries identified to report and reduce their energy intensity

- Conduct awareness campaigns on EE and energy conservation measures.
- Design and implement “Energy Performance Labelling Programme”.
- Approve manufacturers request for energy star labelling based on standards and testing procedures of TBS. Additional external testing facilities can be used for this purpose & the results can be finally sent to TBS for approval & energy star labelling.
- Monitor quality of star labelled products in the market and take actions on any default.

Role of TBS

- Prepare Minimum Energy Performance Standards (MEPS) for various products for “Energy Performance Labelling Programme”.
- Prepare standards/procedures for testing the MEPS of various products which the manufactures can use to test and estimate the energy star rating for approval from the dedicated unit/team for EE.
- Test the products from the manufacturers &provide star rating for energy consumption.
- Strengthen private testing facilities for conducting energy star rating tests for different products based on the manufacturers’ request.

Analysis of sector level energy consumption will be done for identification of benchmark energy consumption for different sectors. A central database will be created for this purpose. Based on these benchmarks, the high energy consuming sectors will be declared as energy intensive sectors. Designated consumers (i.e., industries consuming more energy than the benchmark value) will be identified from the energy intensive sectors. The identified designated consumers will be mandated to report their energy consumption through energy audits. Designated consumers will also be encouraged to take actions for reducing their energy intensity through energy saving measures.

A monitoring and evaluation mechanism will be developed under the EE unit to closely monitor and assess the energy conservation measures taken by various stakeholders and evaluate the EE improvement achieved from time to time. Thus, the EE unit will play a key role in achieving SDG 7 target of EE improvement in the country.

Energy auditor exam and certifications

A detailed curriculum for the certification courses of energy auditors and EE will be developed through the Ministry for Education and relevant universities/institutions. Guidebooks, manuals, training materials, etc. will be prepared based on this curriculum. Guidelines will be developed for applying to energy auditor courses, conducting examinations and awarding certificates. After completing these courses, an individual can become a certified energy auditor, eligible to conduct energy audit for the designated consumers. Similarly, certification courses will be designed for entrepreneurs, skilled and semi-skilled technicians, and engineers in various EE and energy saving measures. A website will be developed for the proposed dedicated unit/team for EE and all the resources will be made available online for easy access to anyone, from anywhere.

Approach for gender mainstreaming:

- Engaging of women consultants, experts for energy use data collection from the industries, energy consumption benchmarking, capacity assessment, guidebooks, manuals, training material preparation, etc.
- Encouraging participation of women in energy auditor exams
- Engaging certified women energy auditors in conducting energy auditing and conservation measures in the designated consumer industries

Approach for engaging LGAs:

Co-ordinating with LGAs in data collection, planning and implementation of potential EE activities in their regions (for example: in public buildings, street lightings, water supply systems, etc.)

Output 02: Creation of an enabling investment environment for RE/EE technologies

Activity 2.1. Prioritization of RE/EE sector in investment promotion, commercial lending and donor funds

Key existing initiatives:

Under Tanzania Energy Development and Access Project (TEDAP), a long term (15 years) fund was provided to the financial institutions that in turn lend to the eligible rural or RE projects. The program was administered by Tanzania Investment Bank(TIB) on behalf of Ministry of Finance and Planning (MoFP) under the direction of REA. Grants were provided to the private sector investments in RE related projects, which included USD 500 for each new mini-grid household connection.

REA is now developing the Renewable Energy Investment Facility (REIF)[24] with assistance from the African Development Bank (AfDB) hosted by Sustainable Energy Fund for Africa (SEFA). REIF will provide affordable finance to the private sector clean energy projects giving energy access to the rural communities in Tanzania.

Identified gaps:

- No special consideration is being given for the RE projects in the banks for providing credit facilities.
- Lack of long-term funding for RE projects is a major constraint.
- Inability of the project developers to match the high equity requirement (30% to 40% of the total investment) of the local banks.
- More documentation and longer process time for credit facilities.

Scope for interventions:

- Workshops and discussion forums between financial institutions, donors and RE/EE sector companies to come up with a trade-off for solving the above-mentioned gaps.
- Establish green funds for banks to re-finance RE/EE sector projects with long term tenure.
- Developing bank guarantees schemes for RE/EE projects

Under this five-year programme, UNDP will assist in procurement services of MEM, TANESCO, EWURA, REA, etc. UNDP can play a leading role in procuring services such as acquisition, purchase or lease of goods, works and/or services for RE power plants, grid distribution, mini-grids, vehicles, etc. These services will be carried out in strict compliance with UNDP global procurement mechanisms and in accordance with financial, technical and administrative international competitive bidding guidelines. It will be a win-win situation as UNDP can procure at a much lower rate, quickly and the saving from procurement can be used for other energy access activities. During this process, capacity and expertise of the personnel from ministries and departments will be developed in such procurement services for different project requirements. Around 10 million USD is proposed for this activity under this five-year programme.

Approach for gender mainstreaming

Priority financial services shall be provided for the women entrepreneurs to set up business in RE/EE sector such as low interest rates, longer tenure and low project developer equity requirements, etc.

Approach for engaging LGAs:

- Updating the LGAs about the existing credit facilities for subsidies available for RE/EE interventions.
- LGAs to give feedback on the project opportunities available in local areas.

Activity 2.2. Quality standards and testing regulations

[24] https://www.afdb.org/en/news-and-events/article/sefa-to-support-tanzania-in-establishing-a-renewable-energy-investment-facility-to-finance-rural-energy-access-projects-16613/

Key existing initiatives:

Tanzania already has standards for charcoal based cook stoves. However, due to poor enforcement, low quality products are made available in the market. Initiatives are now taken for developing standards on improved cook stoves and clean cook stoves in line with international ISO standards. TBS has also prepared draft recommendations for (i) RE and hybrid systems for rural electrification and (ii) Off-grid solar PV lighting kits which are kept for public consultation at present. Lighting Africa approved products are given import tax exemptions. This, to some extent, has reduced the low-quality products (which are not given any tax exemption) coming into the market.

Identified gaps:

- Inadequate RE products testing facilities
- Poor enforcement of product standards through market surveillance
- Inadequate consumer awareness in purchase of quality products
- Need for quality standards on:
  - Improved cook stoves
  - Biogas cook stoves
  - LPG stoves
  - Natural gas stoves
  - Ethanol stoves
  - Pressure cookers
  - Fuel grade ethanol
  - LED/CFL/Lightings
  - Small hydro turbines
  - Wind turbines
  - Solar PV panel and major components of Solar Home System (SHS)
- Need to study the existing standards in East African Communities (EAC) or Sub-Saharan Countries (SSC) or to study ISO standards& any other approved specification like Lighting Africa approved products and to follow a suitable approach in standardising a product. This can avoid unnecessary time and money spent on developing a new standard for Tanzania.

Scope for interventions:

Lack of approved quality products in the market is a serious constraint in promoting RE and EE products to the consumers. For promoting healthy market environment, it must be ensured that only quality assured products are brought into the market. There are several projects going on in Africa and the standards are being developed on regional basis, i.e., for EAC, SSC, etc. This programme will promote adopting such already available standards for Tanzanian market also. Only the products that are tested and approved through these standards can be made to avail benefits of tax/duty exemptions, funds and other government support under this programme.

Experts/consultants and institutions can be engaged to review the existing standards and to assess the need for the new standards. A tentative list of products for which quality standards need to be established are given below:

- Improved cook stoves
- Biogas cook stoves
- LPG stoves
- Natural gas stoves
- Ethanol stoves
- Pressure cookers
- Fuel grade ethanol
- LED/CFL Lightings
- Small hydro turbines
- Wind turbines
- LNG (Liquefied Natural Gas) fuel

The standards promoted through other regional programs (such as Lighting Africa, Global Off-grid Lighting Association (GLOLA)[25]) will be taken into consideration in drafting the new standards. To effectively enforce the quality testing of products, the presence of testing centres is necessary. The guidelines for establishing such testing labs and government approval process for the same will be prepared. The existing product testing centres at TBS, universities and other RE institutions will be strengthened. New testing centres can also be established as per requirement in these institutions. The institutions will be provided with the necessary technical assistance and support for setting up and operating these testing centres.

At present, there is no provision for getting TBS standard mark for the imported goods. Until such time comes and the standards amended, the private sectors will be encouraged to publish the quality approvals from the testing centres outside the country or from the country of origin of the products in their website/newspapers/advertisements/banners, etc. By this way, the consumers can acquire knowledge on the choices on the quality products.

Nation-wide awareness campaigns will be conducted to educate the people on the importance of buying the quality approved products. Advertisements in the newspapers, radio, TV channels, etc. will be done as (a) part of (the) awareness campaigns. Similarly, awareness will be raised among the product manufacturers to get the TBS mark certification.

These activities will also complement the promotion of energy efficient products under EE goals of SE4ALL initiative.

Approach for gender mainstreaming

- Engaging women consultants/experts in adaptation/development of standards, guidelines for testing of products& setting up of testing centres, etc.
- Engaging women in field testing of energy products.
- Engaging women development based NGOs and self-help groups in awareness programs on purchasing quality products

Approach for engaging LGAs

LGAs would be encouraged to participate and support the awareness programs conducted in rural regions in purchasing quality products.

Output 2.3. Promotion of RE industries/businesses

Key existing initiatives:

Apart from various other means discussed above, it is expected that due to SE4ALL programme initiative, several RE industries and services will flourish in Tanzania. To encourage this to happen, Export Processing Zones Authority (EPZA) could also be an ideal place for RE investment promotions. EPZA is now working on developing SEZ in the country.

Identified gaps:

- Limited industries in the local manufacturing/assembling, spare parts and O&M in RE/EE sector.
- Limited funds and resources for the development of infrastructure services for SEZ.
- Limited awareness and promotions in establishing RE related industries in the country.

[25] https://www.gogla.org

Scope for interventions:

Resource mobilisation and other incentives are needed for the establishment of industries and Small and Medium-sized Enterprises (SMEs) that would come up due to the increasing RE business in production, assembling, O&M, after sales services, etc.

Scale up of RE/EE sector in Tanzania needs a strong promotion of local assembling of components, manufacturing of products/spare parts, after sale services, operation and maintenance (O&M) related companies dealing with RE/EE components (for example: solar panels, batteries, SHS peripherals, energy efficient TVs, lighting products, other components used with the SHS; wind turbine related components, biomass power plant based components, biogas installation based components, spares of any such systems, relevant after sales services and O&M firms, etc.).

Establishment of RE industries in SEZs or other earmarked industrial areas will benefit from the infrastructure, continuous power supply and access road services provided within the premises. Such industries and business could also benefit from tax/duty exemptions for the sale of goods within the country.

Currently, Tanzania has exempted Value Added Tax (VAT) on the supply of solar panels, modules, solar charge controllers, solar inverters, solar lights, vacuum tube solar collectors, solar batteries and natural gas stoves. Import duty on stoves for natural gas fuel or for both natural gas and other fuels (together with their parts) is 10% instead of 25% (which is the current duty levied on all imports)[26].

These opportunities will be best utilized under SE4ALL initiative. The energy based companies such as cook stove manufacturers, LED manufacturers, solar PV modules, batteries, SHS peripherals and components, energy efficient TVs, lighting products, wind energy based components, biomass energy based components and services, biogas based components and services, etc., can develop plants under this facility. The overall objective will be to promote local manufacturing in order to bring down the market price of RE/EE products to end users and make them more affordable.

Approach for gender mainstreaming

Priority support will be given to women entrepreneurs to set up RE/EE industries through favoured access to loans, grants and technical support.

Approach for engaging LGAs

- LGAs can support & provide feedback in the land identification, allocation and infrastructure development while assisting in planning and setting up SEZ within their localities.
- LGAs can help in educating the local entrepreneurs about benefits of setting up RE firms in their regions.

Output 03: Scale up and roll outs of sustainable energy initiatives

Through various programs and development schemes, number of initiatives are already going on in Tanzania's energy sector. However, the on-going and planned projects alone may not be sufficient to achieve the SDGs7 goals by 2030. This SE4ALL programme will focus on further scaling up of these activities to close the gaps.

The existing baseline conditions in electricity access and the SDG7 targets to be achieved for the same by 2030are presented in table 4.

[26] As per information provided by Planning commission

Table 4. Electricity access - baseline, targets and gaps

No.	Parameter	Value
	Baseline condition	
1	Percentage of households with electricity connection	24
2	Respective number of households connected	2.64 million
	SDG 7 targets (2030)	
1	Percentage of population with electricity access	75
2	Respective number of households to be connected	10.57 million
3	Additional number of households to be connected to the grid	7.92million
4	Expected number of households to be connected by the on-going/proposed projects	3.99 million
5	Identified gap in the number of households	3.93 million

For this five-year programme, it is assumed that 50% of the target requirement of 2030 must be achieved by 2022. For this, it must be ensured that 5.28million households are provided with electricity access by 2022.

The following approach was suggested in the Action Agenda (page 27) for achieving the electricity access:

- Connection to the main grid (85%): Through densification and turnkey projects, electricity connection will be supplied to those settlements /villages/ households that are within 10 km of the grid.
- Rural Electrification/Off-grid (15%): Access to electricity is provided through rural electrification (mini-grids associated with small hydro, biomass power plants or solar PV system) to the settlements/villages/households located beyond 10 km from the main grid.

As per this strategy, the goal of electricity access can be met by (i) increasing electricity access through grid expansion and(ii) increasing electricity access through decentralised RE.

The existing baseline conditions of RE share in the overall power mix and the SDG7 targets to be achieved by 2030for the same are presented in table 5.

Table 5. RE share in overall installed power

No.	Parameter	Unit	Value
	Baseline condition		
1	Overall installed capacity of the power generation	MW	1,452
2	Installed RE power generation capacity	MW	25.11
3	Existing percentage of RE in the total capacity	%	2
	SDG 7 targets (2030)		
1	Target RE percentage in the overall power mix[27]	%	5

[27]Action Agenda goal states that 50% of overall power must be RE. However, PSMP 2016 has not considered such targets.

No.	Parameter	Unit	Value
2	Expected overall power demand by 2030	MW	11,829
3	Expected RE power generation capacity	MW	591
4	Additional RE capacity needed	MW	566
5	Expected RE capacity by on-going/proposed projects	MW	255
6	Identified gap	MW	311

For this five-year programme, it is assumed that 50% of the target requirement of 2030 must be achieved by 2022. For this, it must be ensured that around 300 MW of additional RE based power plants must be installed within 2022.

It is expected that the activities related to RE based power generation and electricity access to households/communities will contribute to address the SDG7 goals of improved electricity access and increased RE share in the power mix.

Activity 3.1. Increasing electricity access through grid expansion

Key existing initiatives:

REA Turnkey Phase III Project is envisaged to electrify a total of 2,996 villages which are located within 10 km from the existing MV lines and connect a total of 1,086,759 customers by the end of 2022. REA Turnkey Phase IV Project is envisaged to electrify 1,046 villages for connecting a total of 280,887 customers by the end of 2022. Under Urban Electrification Project, it is proposed to improve the electricity access in 11 regions with large number of customers.

The Rural Electrification Densification Programme is aimed at connecting around 900 villages in the regions of Mbeya, Iringa, Tanga, Pwani, Arusha and Mara. Preparation of the Rural Electrification Master Plan is in progress by REA.

To promote RE based grid power, REA has already implemented Tanzania Renewable Energy Program, a CDM PoA, to promote RE based small power plants less than 10 MW. World Bank's SREP programme is envisaged to install around 147 MW cumulatively including geothermal, through RE means by 2025. Other RE projects such as Shinyanga/Simiyu solar project (150 MW), Dodoma solar project (50 MW), Singida wind project (200 MW) and Njombe wind project (11 MW) are also under consideration by 2020.

Identified gaps:

- Inadequate finance/grants for the programs.
- Additional capacity building needed for TANESCO and REA for improved project implementation and monitoring.
- Inadequate Land allocation for transmission lines.
- Inappropriate synchronization of the grid with the RE supplying to the grid.
- No compensation for the losses caused to the RE based IPPs, when they are ready to export and could not be taken by TANESCO.
- Delayed payment from TANESCO to RE based IPPs.

It states that only 5% RE in the overall power by 2030. The same is considered in this Prodoc.

Scope for interventions:

Around 85% of the target electricity accessamong10.57 million household connections (refer table 4) by 2030, i.e., 8.98 million households, is proposed to be connected by grid expansion and densification strategies. These include mainly the settlements/villages/ households that are within 10 km from the existing grid infrastructure (Reference: AA, page 21).

Support will be provided to implement the existing TANESCO grid expansion plans to the new regions. In regions where grid connection is available nearby, densification will be encouraged, i.e., efforts will be taken to provide electricity connection to as many households as possible. Capacity of TANESCO and REA will be improved for effective management and monitoring to achieve timely execution of the projects. Assistance will be provided for land allocation for transmission/distribution lines and RE power plants.

Funds for grid expansion plans will be provided through international assistance. Loans for these grid expansion plans will also be arranged through available international credits. It is to be noted here that the UNDP procurement services proposed under activity 2.1 shall be effectively used by MEM, TANESCO, REA, etc., to purchase goods and services to achieve targets of grid connected electricity access.

The grid expansion achieved under this activity will contribute in achieving the electricity access targets as discussed in table 4. New RE based grid power plants installed under this grid expansion initiative will also contribute in addressing the gaps mentioned in table 5.

Approach for gender mainstreaming

Engaging women consultants/experts/staff in designing, data collection& MRV of grid expansion plans.

Approach for engaging LGAs

Engaging LGAs in land identification/allocation for grid transmission lines.

Activity 3.2. Increasing electricity access through decentralised RE

Key existing initiatives:

Renewable Energy for Rural Electrification (RERE) Project implemented under SREP envisages establishing 47 MW RE off-grid plants that provide access to around 500,000 people by 2025. The Sustainable Solar Market Packages (SSMP) phase 2 program is being implemented to encourage the installation of SHSs in remote villages, schools, health centres, etc. JUMEME – Rural Power Supply Limited is working on the initial phase of constructing 28 mini-grids to electrify more than 100,000 people and 2,340 shops and small businesses. Currently, 100 Wp SHS costs approximately around USD 400. Under SSMP phase 2 program, REA provides a subsidy of around 5 USD/Wp.

The Rural Electrification Expansion program of REA also includes components for off-grid electrification. Currently, Tanzania Renewable Energy Program, a CDM PoA, is under implementation to promote RE based small power plants. Preparation of the Rural Electrification Master Plan is in progress by REA.

Phoenix Power, a private company is currently working on solar battery recycling and trying to explore the best possibilities on this aspect in Tanzania. For example, batteries can be taken back to the manufacturers at the end of their lifetime for safe recycling and disposal. This will prevent piling up of batteries in waste disposal sites. In Bangladesh, where the world's highest number of SHSs has been successfully installed, similar recycling and disposal procedure is being practiced.

Identified gaps:

- No detailed assessment of potential sites with RE resources (municipal abattoir waste water treatment, agro-processing industries like coffee pulp washing, coffee curing, cashew processing industries, kitchen wastes in residential institutions, etc.), load demand, time for reaching grid to the region, etc.
- No successful business models for mini-grids.
- Inadequate standalone electricity access to schools, health centres, irrigation water pumps, domestic water supply pumps, rural business units, etc.
- Inadequate utilization of biomass for community power generation.
- Inadequate availability of skilled technicians for after sale servicing in each locality.
- Inadequate funds/grants for the projects.
- Inadequate availability of technical expertise in relevant RE technology.
- Inadequate land allocation for the project sites.
- Inadequate utilization of municipal solid wastes, waste water disposal from municipal abattoirs, wastes from kitchens, etc.
- High cost of products and low affordability by majority of the people



Figure 7: Solar street lighting in Dar es Salaam



Figure8: Solar battery with GPS tracking

Scope for interventions:

Around 15% of target electricity access of 10.57 million household connections (refer table 4) by 2030, i.e., 1.6 million households, is proposed to be connected by decentralised RE technologies. These include mainly the settlements/villages/households located beyond 10 km from the main grid and not expected to be connected to grid until 2030 (Reference: AA, page 21). The priority activities under this initiative include:

- Conduct research and survey to collect latest statistics on power access in street lighting, irrigation water pumping, domestic water supply pumping, schools, primary health care units, etc. As per data available from Ministry of Water and Irrigation, more than 75% of the irrigation pumps rely on diesel fuel for its operation.
- Development of after sale service centres (for example: solar kiosks) or product servicing of all/most common brands under one roof in each region/town with special focus on women entrepreneurs. Through this, job opportunities will be created for local technicians. Customers will get easy access to the product services. It would be win-win for the customers, local technicians and private sector in effective utilization of time & money on this aspect.
- Innovative business models such as "pay-as-you-go", attractive micro-finance, GPS (Global Positioning System) based performance monitoring options etc. will be promoted. GPS based performance monitoring if possible to be implemented for all systems can help in centralized monitoring under SE4ALL activities. But currently, the private firms using GPS in SHS (fig 8), for their own monitoring are reluctant to make it as centralized one as they consider it as their own advantage over their competitors.
- Nation-wide awareness campaigns will be conducted to educate the consumers on the benefits of off-grid power solutions

The approaches to achieve the decentralised energy access may involve:

- Solar/wind/small hydro based community mini-grids supplying to communication towers, small businesses and households
- Solar/wind based power for schools, health care units, hospitals, etc.
- Solar based street lighting (fig 7), irrigation water pumping and domestic water supply pumping
- SHSs
- Waste to power in community kitchens, municipal abattoirs, agro-processing units like coffee pulp washing, coffee curing units, cashew processing industries, etc.

Co-operatives such as SACCOs and women self-help groups will be considered as potential target customers for the SHSs and Pico PV units including solar lanterns.

Solar kiosks for product servicing will be developed in each region/town(s). Entrepreneurship training will be provided to start these service kiosks. The local technicians/personnel can be trained to handle and service different brands. Hence, the consumers can have their solar products serviced in the remote places/locations irrespective of the brands. Educating and developing women entrepreneurs and women self-help groups, both at the service provider level and at the technician level will be focused under this activity. The Global Lighting Association (GLA) Standards will be referred wherever necessary for the products in SHS package. Collaboration with micro-finance institutions will be made for facilitating credits to the end users for overcoming the high initial cost of SHSs/Pico PV. Technical assistance (for resource assessments, feasibility study and development of business plans through international agencies), grants from donors and loans through financial partners will be arranged under SE4ALL initiative.

Though the solar systems are producing 'green' electricity, they are deemed to produce significant quantity of e-waste such as solar panels, hazardous wastes such as batteries over a period. This is a matter of serious concern. Hence, safe disposal of panels, LED/CFL lamps, electronic inverters and batteries are very important. A recent survey carried out by TAREA estimated that around 128,860 million batteries have been imported and are being used in Tanzania[28]. Under this SE4ALL

[28]Quantification of imported solar batteries – TAREA 2016

initiative, mechanism for solar panel and battery recycling either through the equipment supplier or by third party services need to be developed.

Through this five-year programme, at least USD 5 million funding is proposed for micro-capital grants to pilot RE initiatives in rural communities. Private sector and entrepreneurs can utilize this opportunity to raise finance for their rural electrification projects. Detailed design on fund mobilization, selection of pilot projects, fund management will be done during the implementation phase.

Capacity additions under this initiative will contribute in achieving both the electricity access targets and increased RE share targets as discussed in table 4 and table 5.

Approach for gender mainstreaming

- Promoting opportunities for women as entrepreneurs, managers, manufacturers, service providers, etc., across the SHS value chains.
- Conducting awareness workshops for women on the benefits of using SHS and solar lanterns.
- Involve as many women as possible in marketing & servicing the solar products.
- Promoting opportunities for women entrepreneurs, managers, manufacturers and service providers across the biogas value chains.
- Village women can be encouraged to become entrepreneurs for the proposed 'solar kiosks'. The solar kiosks would serve the local solar market with spares and after sale services required for all the brands. The entrepreneur could have the contacts of locally available trained & qualified solar service persons when kiosk is approached for such services, the entrepreneur could check with the available service person and send him/her to the required complainant. This will boost the women participation as well as provide employment to the local people. This would be win-win for the private sectors as well as the local people, thereby reducing the cost and time of servicing required by either.

Approach for engaging LGAs

- Encouraging and educating LGAs to install solar, biogas power generation units in their premises, wherever possible.
- Encouraging LGAs to explore other RE power generation opportunities in their areas.

Activity 3.3. Increasing RE share in thermal requirements

Biomass-to-energy, mostly for heat production is responsible for 90% of the total primary energy consumption in Tanzania, that too mainly for the household consumption (firewood, charcoal, crop residues, etc.). The rest of the biomass demand (10%) is for commercial, institutional and industrial sectors[29]. It was identified that only 4% out of these biomass is sourced from sustainable sources.

The following table provides the baseline, target and gaps identified in the heating energy consumption.

Table 6. Heating energy - baseline, targets and gaps

No.	Parameter	Unit	Value
	Baseline condition		
1	Total primary energy supply for heating and cooking	MTOE	19.8
2	Current RE share (sustainable source)	%	4

[29]Action Agenda Page 15

No.	Parameter	Unit	Value
3	Current RE energy supply	MTOE	0.792
4	Corresponding wood supply	tons	2.00million
	SDG 7 targets (2030)		
1	RE share from sustainable sources	%	10
2	Expected sustainable wood supply	tons	9.35million
3	Estimated contribution from on-going/proposed projects	tons	0
4	Identified gap in achieving the targets	tons	9.35million

For this five-year programme, it is assumed that 50% of the target requirement of 2030 is to be achieved by 2022. For this, it must be ensured that 4.67 million tons of wood supply from sustainable sources must be achieved by 2022.

Key existing initiatives:

MEM, along with the support of EU, is developing Biomass Energy Strategy (BEST). The strategy will identify the means of ensuring a sustainable supply of biomass energy; increasing the efficiency with which the biomass energy is produced and utilised; promoting access to alternative energy sources wherever appropriate and affordable; and ensuring an enabling institutional environment for implementation. This strategy will be implemented soon.

Transforming Tanzania's Charcoal Sector phase II (2015 – 2019) is under implementation with Swiss fund through Tanzania Forest Conservation Group (TFCG), Tanzania Community Forest Conservation Network(MJUMITA) and Tanzania Traditional Energy Development Organization (TaTEDO).

Since 2012, TFCG, MJUMITA and TaTEDO have been supporting communities in Kilosa, Mvomero and Morogoro Districts to integrate sustainable charcoal production into the management of their forest reserves, based on the ecologically sound harvesting principles, production and legal transportation. The project aims to incentivize ecologically sustainable production by facilitating the re-structuring of the charcoal value chain by securing exemption from the Central Government royalties for charcoal produced sustainably from village forest reserves; and by allowing communities to set and retain their own fees with these funds then being made available for community development and sustainable natural resources management.

TaTEDO has been promoting improved cook stoves since its establishment in 1990. Solid Biofuel technologies (products) promoted are (i) efficient charcoal production kilns (ii) energy efficient firewood stoves (iii) efficient charcoal stoves and ovens and (iv) efficient tree planting and management practices. TaTEDO has developed three types of kilns. These are Improved Basic Earthmound Kiln (IBEK), Half Orange Charcoal Kiln (HOCK) and Retort Kiln. Energy efficient firewood stoves are of three types. These are improved brick made firewood stoves, low cost stove and Kunimbili stoves. Energy efficient charcoal stoves and ovens can be categorized into improved charcoal saving stoves (Straight, Bell-bottom and Sazawa), Double Plates (Stand and Box), Meat Roasting Ovens (Teksawa and Nyama Choma) and Baking Ovens (Households and SMEs).



Figure 9: Types of stoves promoted by TaTEDO

The major gaseous biomass technologies promoted by TaTEDO are biogas plants. Biogas typically refers to a gas produced by the biological breakdown of organic matter in the absence of oxygen. The biogas plants promoted by TaTEDO include Vacvina, Fixed Dom Plant and Art types.



Retort Kiln for carbonizing bio-wastes



Briquettes



Biogas Plant

Figure 10: Agro-waste conversion methods

TaTEDO has come up with the means of converting agro-wastes into useful resource for generating income, producing alternative cooking fuels and increasing agricultural productivity. The agro-wastes pilled around these plants are filled in the retort kiln semi-drums. The agro-wastes are carbonized by using retort kiln through process called pyrolysis to produce carbonized residue (bio-char). The carbonized residues are processed into briquettes which are enterprises for increasing rural employment, income generation,

TaTEDO and other organisations working in this sector also came together and formed the Clean Cook Stoves and Fuels Alliance of Tanzania (CCFAT) and the ICS Taskforce. Key members of this ICS Taskforce include SNV, TAREA, TaTEDO, COSTECH, REA, MEM and few universities. Key objective of ICS Task Force is to develop a commercially viable sub-sector through supply and demand side strengthening, research and development, resource mobilization, market intelligence and enhanced public policy dialogue.



Figure11: ICS manufacturing unit in Arusha

A national tree planting and management strategy is also proposed (2016-2021) for promoting sustainable wood plantation.

Identified gaps:

- No nationwide data available on the demand and utilization of firewood, charcoal, use of ICS, etc.
- Inadequate finance/grants for the programs.
- Inadequate stakeholder engagement at all levels (ministries, local governments, communities, NGOs, etc.).
- Inadequate awareness to end users on the benefits of using good quality and efficient improved charcoal stoves.
- Lack of sustainable charcoal supply chain networks.
- Inadequate effective land use planning for sustainable harvesting of wood for fuel (fire wood or charcoal).
- Unexplored waste to energy potential in municipal abattoirs, agro-industries, etc.
- Inadequate village land use planning in all villages.
- Limited awareness raising on the sustainable forest use, sustainable harvesting, importance of preserving local bio diversity, zero grazing, sustainable harvesting of fuel wood, etc.

Scope for interventions:

The priority interventions will be to increase the sustainable wood and charcoal production capacity through energy plantations, sustainable harvesting, biomass briquetting, etc. Effective land use planning in co-operation with local government authorities will be encouraged to identify and allocate lands for sustainable biomass cultivation. Assistance will be provided in this regard to cover more villages within a shorter period.

In co-ordination with Vice-President's Office – Department of Environment (VPO-DoE) and Tanzania Forest Services agency (TFS), the programme will implement energy plantations to produce renewable biomass. Sustainable harvesting will be promoted in these energy plantations. Detailed resource assessments will be carried out to identify such potential sites. Skills will be imparted on energy efficient charcoal production.

Charcoal production, supply and demand master plan will be developed for the main supply and demand centres. This master plan will include tree cover inventory, classification of harvesting zones, development of conditions and technical specifications for charcoal production and use.

Increasing the improved cook stove production and expanding the market network are equally important to manage the market demands. The SE4ALL initiative will seek synergy with activities of

GACC and ICS Task Force to improve ICS market. Nationwide awareness campaigns will be conducted to increase the consumer awareness in using the energy efficient ICSs. The targets achieved under this initiative will also contribute to achieving the gaps identified in table 8 under activity 3.5 “Roll out of modern cooking solutions”.

Potential of waste-to-energy (WTE) from coffee pulp washing, coffee curing, cashew processing industries, local municipality abattoirs, etc. is not yet assessed in Tanzania. Under this programme, resource assessment study will be conducted on the potential of WTE in the above-mentioned sectors and feasibility studies will be conducted for installation of these power plants at the identified potential sites. Local municipalities and private sector investments will be encouraged to develop these plants. Under this activity, community biogas power generation will also be promoted in local municipality abattoirs and agro-processing units. Technical assistance, grants and loans will be arranged under SE4ALL initiative.

Activity 3.4. Reducing energy intensity through EE measures

EE and conservation refers to measures aimed at reducing energy consumption without sacrificing productivity, level of service or increasing costs. The baseline energy intensity and expected EE improvements under SDG 7 goals are provided in the table 7.

Table 7. Annual energy savings –baseline, targets and gaps

No.	Parameter	Unit	Value
	Baseline condition		
1	Existing annual energy savings	MTOE	0.38
2	Energy intensity improvement	%	-2.6
	SDG 7 targets (2030)		
1	Energy intensity improvement	%	-2.6
2	Annual energy savings needed (considering future energy demand also)	MTOE	7.18
3	Contribution from the existing projects	MTOE	0
4	Identified gap	MTOE	6.80

As per Action Agenda[30], the energy intensity improvement was -2.6% in the baseline condition. The same was proposed for each year until 2030. So, every year around 2.6% of the overall energy consumption (MTOE) must be saved to be in line with SE4ALL goals. For the year 2022, the energy savings target will be around 2.45 MTOE.

The energy intensity of the country shall be reduced through various approaches. The important sectors of focus will be the transmission and distribution (T&D) losses in the grid, the energy consumption in industries, commercial/public buildings (both power and fossil fuel) and energy consumption in household level. Accordingly, this programme design includes several initiatives in these key sectors. A simple approach is presented in figure 12.

[30]Table 6, page 24

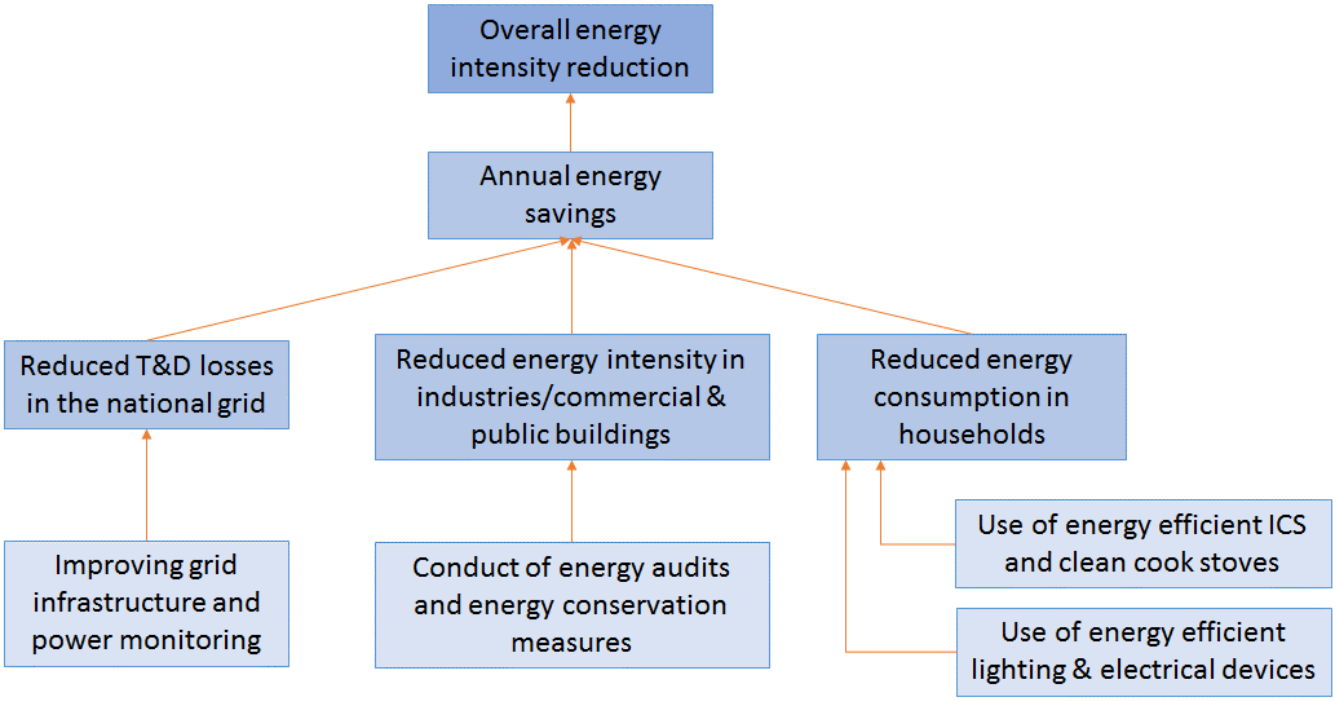


Figure 12: Approach for overall energy intensity reduction

Key existing initiatives:

TANESCO is working on the target of reducing the transmission and distribution losses in the national grid to 12% by 2020. World Bank-funded Lighting Rural Tanzania Project in partnership with the REA facilitated the development of sustainable business models to bring efficient, clean and affordable lighting services to the Tanzanian poor. MEM along with USAID is studying the "Load Management Analysis for Large Energy Consumers". Bank of Africa, Tanzania has proposed credit line for implementing RE and EE projects. With assistance from EU, MEM is preparing for the energy performance certification for large buildings in Tanzania.

Various demand side management activities including very few energy audits have been carried out so far in the country.

Identified gaps:

- Inadequate finance/grants for the programs.
- Inadequate stakeholder engagement at all levels (ministries, local governments, communities, NGOs, etc.).
- Need for Minimum Energy Performance Standards (MEPS).
- Awareness to the end users on the benefits of using energy efficient products.
- Technical assistance on energy efficient technologies.

Scope for interventions:

Technical assistance (for resource assessments, feasibility study and development of business plans through international agencies), grants from donors and loans through financial partners to TANESCO will be arranged under SE4ALL initiative.

Energy efficient lightings will be promoted and will be implemented in alignment with other activities such as SHSs, solar lanterns, grid connection to households under RE mini-grids and EE improvement in industries/commercial buildings. Quality of LEDs and CFLs will be ensured through the standards emphasised under SE4ALL.

Under this programme, MEPS will be prepared for different products such as electric water heaters, diesel generators, LPG/Natural gas cook stoves, TVs, fans, LEDs, motors, transformers, pumps,

etc. Draft MEPS prepared will be shared with the respective stakeholders for review and feedback. Based on these feedbacks, the MEPS shall be further revised and submitted for approval to GoT. Initially, labelling of performance rating can be voluntary and later it can be made mandatory once the market conditions are well established.

The MEPS will be prepared by TBS (or by any institute& finally approved by TBS) for different products. It will also prepare the testing procedure for reporting MEPS for each product. The manufacturers can use these standards and testing procedure to develop products with a specific energy rating. The manufacturers can then submit the test reports to the dedicated EE unit/team for approval towards using the energy star label in the product cover. The dedicated EE unit/team will check the application requests and testing reports and will grant approval. Once the products are in the market, the dedicated EE unit/team will do a sample testing of product performance. If there are any deviations, then the respective manufacturer will be penalized.

Nationwide awareness campaigns will be conducted to educate consumers to buy performance labelled products. Product manufacturers will be educated through workshops and forums to produce energy efficient products and get them certified for the performance rating. The performance rating standards for various products will be made available to public through MEM website. Clear guidelines will be provided for the manufacturers to apply and get their products certified. Certification process will be transparent. A minimum of 100 products will be certified with the performance labelling. These products will be available in the market for purchase and use by the consumers.

Industries and commercial buildings will be encouraged to use these performance labelled products. The utilization of this scheme and products under other activities of SE4ALL such as LED/CFL distribution programme, energy auditing and conservation in industries, promotion of energy efficient cook stoves, etc., will also be encouraged.

EE team will prepare clear guidelines for accreditation of the energy audit firms. Only these firms can conduct energy audit in industries of designated consumers. Minimum 10 energy audit firms will be accredited under this scheme. The designated consumers shall conduct energy audit of their plant premises to identify EE and energy saving measures. A minimum of 50 preliminary audits will be supported by SE4ALL initiative.

Through this SE4ALL initiative, subsidy will be provided to the designated consumers for implementing energy conservation measures identified through the energy audit. The designated consumers will also be requested to submit a post implementation energy audit report to identify the energy savings achieved.

Approach for gender mainstreaming

- Involving as many women as possible in awareness campaigns
- Involving women participants in training and certification programs
- Educating and involving as many women staff as possible in marketing the energy efficient products

Approach for engaging LGAs

- Involving LGAs in awareness campaigns
- Involving LGAs in identification of site for energy plantations and their continuous management

Activity 3.5. Roll out of modern cooking solutions

Access to efficient heating and cooking systems such as LPG and biogas, is currently limited to better-off families and communities that can afford these. Under these existing conditions, the charcoal based improved cook stoves take a major share in the overall cooking use in Tanzania. So, the cook stove technologies considered under this initiative are; improved charcoal stoves & BLEN. The following table summarizes the baseline and targeted cooking scenario in Tanzania.

Table 8. Modern cooking – baseline, targets and gaps

No.	Parameter	Value
	Baseline condition	
1	Percentage households with access to modern cooking[31]	16
2	Respective number of households	1.75 million
	SDG 7 targets (2030)	
1	Percentage households with access to modern cooking	75
2	Respective number of households	10.57 million
3	Additional number of modern cook stoves to be retailed	8.82 million
3	Estimated number of cook stoves provided through on-going/proposed projects	1.77 million
4	Identified gap in number of cook stoves to be retailed	7.05 million

For this five-year programme, it is desired that at least 50% of the target requirement of 2030 is to be achieved by 2022. To achieve this, it must be ensured that at least 5.28 million households are using clean cook stoves by 2022.

The promotion of charcoal based ICSs is already discussed under activity 3.4. “Increasing RE share in thermal requirements”. The ICSs market development and usage in households will also contribute for the gaps identified in the table 8. Therefore, the initiatives for promoting other technologies such as LPG stoves, ethanol stoves, natural gas cooking and biogas digesters are discussed in this section.

Key existing initiatives:

Tanzania Domestic Biogas Program (TDBP) through The Africa Biogas Partnership Programme (ABPP) which was, launched in 2009 aims at developing a sustainable domestic bio digester market and supporting sector in the respective countries. The programme is in its second phase that started in 2014 and will run until 31st March 2019. The ABPP is financed by Directoraat Generaal Internationale Samenwerking (DGIS), the Ministry of Foreign Affairs in the Netherlands and implemented by Humanistisch Instituut Voor Ontwikkelings Samenwerking (HIVOS) as fund manager and SNV as provider of advisory services. Since the program commencement in 2009, more than 15,000 biogas digesters have been constructed. As part of the Government contributions toward the programme, REA is supporting construction of 10,000 more digesters. Through the program and CAMARTEC initiatives, several local artisans have been trained on various biogas technologies, Furthermore, a number Biogas Construction Entrepreneurs (BCEs) have been formed and have taken biogas as a business opportunity. As initial biogas investment cost is one major impeding factors, CAMATEC is continuously conducting research including redesigning the existing plants to bring down the cost for one system (fig 13).

[31]Includes ICS, LPG, biogas, kerosene and electricity



Figure13: CAMARTEC new biogas plant design to reduce cost

In the mason incentive scheme by EnDev, incentives will be provided to the masons on construction of each biogas digesters, but in three term payments. The first payment will be given immediately after the construction of digester. Second payment will be made at end of three months after confirming with the end user on its successful operation. The final payment will be made after one year, again after confirming with the end user on its successful operation. This ensures that all the digesters constructed are of best quality and could be used without any issues by the end users. REA subsidy will be sought to overcome high initial cost of the digesters to the end users.



Figure14: Cooking using LPG

LPG sector is currently developed and dominated by private players only. LPG stoves are being promoted through various means such as subsidy, pay-as-you-go, etc. Fig 14 shows cooking using LPG in Tanzania.

UNIDO is working using GEF fund on the roll out of ethanol based cooking solutions.

Through TPDC, pipeline for natural gas (NG) connection is envisaged for around 30,000 households in Dar es Salaam. Preliminary feasibility study towards NG cooking in Lindi and Mtwara has been conducted and is in draft stage.

Identified gaps:

- Inadequate grant/fund for promoting BLEN & compensating for the high cost involved in using these technologies.
- Lack of assessment of exact number of modern cook stoves in use and market potential for different clean cooking options
- Inadequate micro-finance for the end user to overcome high cost of modern cooking solutions (towards stove & fuel purchase).
- No successful business model for the development of supply chain network for modern cook stoves and related fuel supply.
- No quality assessment for biogas digesters.
- No quality standards for cook stoves and cooking fuels (like LPG stoves, ethanol stoves, biogas stoves, NG stoves, etc. and quality standards for ethanol cooking fuel, NG for cooking, etc.).
- Inadequate awareness on health and environmental benefits of modern cooking solutions.
- Inadequate education to women on decision making for choosing appropriate cooking options.
- Lack of data on LPG penetration rate, which could help in the information to the reduction on wood fuel.
- Inadequate promotion on biogas cooking solutions in residences, schools, prisons, communities, etc.
- High cost of clean cooking products and low affordability by majority of the people

Scope for interventions:

Through this programme, biogas digesters for household cooking will be promoted. A quality assessment/assurance system like EnDev's mason incentive scheme will be developed and implemented to ensure that only quality digesters are installed under this programme.

In LPG sector, private sector companies will be encouraged to develop market network in remote village areas. This way, consumers can have easy access to fuel refilling and servicing. Innovative business models such as "pay-as-you-use" can be developed to overcome high initial cost of LPG systems. This model is being used in Kenya. The LPG system sale package may also include pressure cookers (with separators) for promotion purpose. A model pressure cooker with separators is shown in the fig 15. The pressure cooker facilitates efficient cooking by the end users. Separators will help in cooking multiple items in a single go. For e.g., rice, beans, vegetables, etc. can be cooked at a time. This will save lot of cooking time for women, especially when they should attend to school going children in the morning & they themselves must go for work. Assessment of LPGs penetration in various areas and particularly to the big cities and towns will be carried out to gauge whether there has been reduction in wood fuel usages.



Figure 15: Pressure cooker with separators

It is important that the pressure cookers comply with the approved standards. All LPG systems (stoves, pressure cookers) under the SE4ALL programme must comply with the standards proposed

by this programme. Nationwide awareness campaigns will be conducted to educate people on the benefits of LPG cooking. Close monitoring will be done to ensure compliance of quality and composition of gas supplied by various service providers.

TPDC has already conducted feasibility studies for the installation of NG pipeline from Lindi and Mtwara to Dar es Salaam, mainly for fuel supply to household cooking. Through this project around 30,000 households will be provided with NG based cooking. Similarly, TPDC has also conducted feasibility studies for installation of NG distribution network around Lindi and Mtwara region to supply gas to the surrounding households. Through this project, around 3,500 households will be provided with NG based cooking. Technical assistance, grants and loans will be arranged for these projects under SE4ALL initiative.

RE based clean cook stove activities may also benefit from micro-capital grants proposed in activity 3.2 under this five-year programme. A key objective of these program assistances will be to bring down the market price of modern cooking solutions to make it more affordable to masses.

Approach for gender mainstreaming

- Involving as many women as possible in awareness campaigns, trainings, etc.
- Encouraging women entrepreneurs in manufacturing clean cook stoves.
- Educating and involving as many women staff as possible in marketing of clean cook stoves.
- Involving women self-help groups, NGOs and SACCOs in the sale and retail of clean cook stoves and fuel.

Approach for engaging LGAs

- Enhancing existing capacities to LGAs on health & environmental benefits of clean cooking solutions.
- Involving LGAs in the awareness campaigns.

Gender Mainstreaming:

It is recognized that gender equality and the empowerment of women have a significant positive impact on the sustained economic growth and inclusive industrial development, which are the key drivers of poverty alleviation and social progress. In fact, due to diverging needs and rights regarding energy consumption and production, women and men are expected to be affected differently by the project (in terms of their rights, needs, roles, opportunities, etc.). The programme aims at the demonstration of good practices in gender mainstreaming aspects wherever possible and avoiding negative impacts on women or men due to their gender, ethnicity, social status or age. Consequently, the programme includes the gender dimension during the entire programme cycle. To mainstream gender into the project, an in-depth gender analysis will be done and various entry points for mainstreaming project outcomes, outputs as well as activities will be identified. Engaging women in various activities will be targeted.

The guiding principle of the programme will be to ensure that both women and men are provided with equal opportunities to access, participate and benefit from the project, without compromising the technical quality of the project results. In practical terms,

- Gender-sensitive recruitment will be practised at all possible levels especially, in the selection of project staff. Gender responsive TORs will be used to mainstream gender in the activities of consultants and experts. In cases where the project does not have direct influence, gender-sensitive recruitment will be encouraged. Furthermore, whenever possible, the existing staff will be trained in awareness on gender issues.

- All decision-making processes will consider gender dimensions. At the programme management level, meetings will invite observers to ensure that gender dimensions are represented. Also, at the project activity implementation level, effort will be made to consult with the stakeholders focusing on gender equality and women empowerment issues.

- Efforts will be made to promote participation of women in the training activities, both at the managerial and technical levels.

When data-collection or assessments are conducted as part of the project implementation, gender dimensions will be considered. This can include sex-disaggregated data collection, performing gender analysis as part of socio-economic impact studies, etc.

Resources Required to Achieve the Expected Results

a) International/local expert assistance (for the project activity design, resource assessment, feasibility study, business plan preparation, etc.)

Important barrier in achieving the energy access is the inadequate knowledge and skill in RE technologies, resource assessments, techno-economic feasibility studies, policy planning, project finance management, etc. The programme will engage international/local expert services wherever required to fill in these knowledge gaps. Capacity building and knowledge sharing will be used effectively to increase knowledge and skills of the relevant stakeholders for sustained programme implementation.

b) Funds for programme initiation/implementation through grants, loans and investments

To create an enabling market environment, the focus sectors will need fund flow to overcome the existing market barriers. Tanzania being an LDC does not have much resource either with the Government or with the private sectors. There is a need to create an investment strategy to access finance for the proposed interventions from the international donor agencies, financial institutions, climate change funds, regional co-operations, etc. The funds shall be utilised in various ways such as grants for capacity building activities/policy planning, loans for project developers and investment in infrastructure and business development.

c) Partnerships with international/regional/national agencies/institutes

Various initiatives in the energy sector are implemented in Tanzania with the support from several international donor agencies and UN organisations. The programme will enhance partnerships with these international donor agencies such as AfDB, EU, DfID, GIZ, NORAD, etc. Similarly, there are several regional programmes implemented in focus sectors such as Lighting Africa, Global Alliance for Clean Cook stoves (GACC), etc., where Tanzania is also a stakeholder. National institutions such as CAMARTEC, UDSM, TIB, etc., have also been a part of these activities. This proposed programme will seek partnership with all the relevant stakeholders and with other on-going global and regional programs.

d) Infrastructure facilities and testing equipment

Setting up of SE4ALL Secretariat, dedicated EE unit/team, RE portal with A – Z information etc. require provision of infrastructure facilities depending on their formation within the country's legal and administrative framework. The infrastructure could be anything like an office set up, furniture, software, staff allocations for regular functioning of these units, transport vehicles, website publishing, etc. Similarly, existing product quality testing centres at ATC, UDSM, CAMARTEC, COSTECH, TBS, etc. need to be strengthened. New testing facilities need to be created as per programme requirements for testing of solar PV panels, LED lightings, modern cook stoves, fuel, etc.

e) UNDP Inputs

UNDP, through its country office, will provide technical and advisory support for setting up the SE4ALL Secretariat. It will be part of the Project Steering Committee (PSC) and will have a key say in all the important decision makings. UNDP will also support the monitoring and evaluation (M&E) of the programme throughout its 5-year duration. UNDP is planning to mobilize at least USD 15 million from various sources for implementation of various activities of the programme, including

high impact activities like solar irrigation pumps, solar electricity for rural health centres, schools, etc.

Partnerships

Table 9 shows the list of tentative stakeholders and their roles and responsibilities under this programme. There may be additional stakeholder partnerships from GoT, ministries and private sectors during actual programme implementation as per requirements.

Table 9: Stakeholders and partnerships (all the stakeholders need to be a part of the Project Steering Committee to ensure their fullest co-operation and participation)

No.	Stakeholder	Partnership contribution	Risks and assumptions
1	Ministry of Energy and Minerals (MEM)	• Establish SE4ALL Secretariat • Facilitate government support in the implementation of project activities. • Policy guidance wherever needed. • Facilitate issuance of licenses, approvals, permits to project developers/private sectors from other ministries and institutions. • Operation and maintenance of TREMIS portal and development of RE portal with A – Z information	• Continued support throughout the programme duration irrespective of any change in the government set up. • Assistance from UNDP
2	Ministry of Finance and Planning (MoFP)	• Allocate funds/resources to engage experts • Approve funds for infrastructure development for SE4ALL Secretariat, functioning of A-Z renewable energy portal, one-stop-shop for Energy (in the long run), dedicated EE unit and increasing the capacity of the testing centres at TBS or any other institution • Resources to programme (in-kind, subsidy, cash)	• Continued support throughout the programme duration irrespective of any change in governments.
3	Ministry of Industry and Trade (MOIT)	• Process requests for licenses, approvals, permits for project developers/private sectors • Policy support, if needed • Support incorporation of energy investments in the established SEZ. • Support development of Ministry of Micro, Small and Medium Enterprises (MSMEs) in focus sectors • Resources to programme (in-kind, subsidies, cash).	• Continued support throughout the programme duration irrespective of any change in government

No.	Stakeholder	Partnership contribution	Risks and assumptions
4	Tanzania Petroleum Development Corporation (TPDC)	• Enhance construction and development of natural gas pipeline for household cooking, hotels and other institutions • Resources to programme (in-kind, subsidies, cash).	• Continued support throughout the programme duration irrespective of any change in government • Technical assistance from the international experts through SE4ALL initiative
5	Vice-President's Office – Department of Environment (VPO-DoE)	• Provide policy support in the implementation of sustainable charcoal production project. • Facilitate access to climate change funds • Facilitate implementation of the BEST • Resources to programme (in-kind, subsidies, cash).	• Continued support throughout the programme duration irrespective of any change in government
6	Tanzania Forest Service Agency (TFS)	• Support in the implementation of sustainable charcoal production project, forest management, etc. • Facilitate implementation of the BEST • Support initiatives aimed at reduction in the consumption of charcoal and firewood. • Policy guidance on biomass feedstock • Resources to programme (in-kind, subsidies, cash)	• Continued support throughout the programme duration irrespective of any change in governments
7	Tanzania Bureau of Standards (TBS)	• Develop standards for quality testing of various products promoted under SE4ALL • Streamline TBS quality marking under SE4ALL for imported products • Development of MEPS for products • Set up testing facilities for solar PV and LED lightings • Prepare guidelines for setting up and approval of testing labs in other institutions • Take initiatives on increasing their capacity in various aspects like technical knowledge, testing facilities, etc. as per the growing demand	• Technical assistance from international experts through SE4ALL initiative • Funds/grants available for setting up of the testing facilities
8	Tanzania Electricity Supply Company (TANESCO)	• Grid expansion plans, improving EE of distribution network, installation of grid connected RE	• Technical assistance from international experts through SE4ALL initiative

No.	Stakeholder	Partnership contribution	Risks and assumptions
		plants, establishment of mini-grids. • Facilitate provision of necessary energy data to the prospective developers • Provide technical guidance wherever required • Resources to programme (in-kind, subsidies, etc.)	• Funds/grants available for infrastructure development
9	Renewable Energy Agency (REA)	• O&M of mini-grids portal • Establishment of mini-grids in rural areas, promotion of biogas systems, SHS, ICS, clean cooking solutions, etc. • Support rural electrification programmes • Provide energy data for the prospective investors and developers • Resources to programme (in-kind or subsidies)	• Technical assistance from international experts through SE4ALL initiative • Funds/grants available for infrastructure development
10	President's Office - Regional Administration and Local Government (PO – RALG)	• Coordinate LGAs in implementing SE4ALL • Facilitate availability of local investment facilities such as land etc. • Support in identification, development, monitoring of RE based power generation • Support monitoring, reporting and verification of various programme initiatives • Resources to programme	• Capacity building and awareness on sustainable energy • Funds/grants available for co-coordinating resources
11	Universities and technical institutions	• Facilitate development of training curriculum for various stakeholders • Carry out innovative research related to RE and EE • Conduct research on various energy feedstock appropriate to the local environments • Set up quality testing facilities for various products • Conduct training (s) and courses on the focus sectors to various stakeholders • Contribute for one-stop-shop for energy in the long run	• Technical assistance from international experts through SE4ALL initiative • Funds/grants available for setting up of testing & training facilities

No.	Stakeholder	Partnership contribution	Risks and assumptions
12	Tanzania Investment Centre (TIC)	• Promote investments in RE business • Increase efficiency in processing investment licenses and timely linking of investors to the relevant sectors/institutions • Facilitate availability of RE information at TIC • Contribute to one-stop-shop for energy in the long run	Continuous support throughout the programme duration
13	Ministry of Water and Irrigation	• Support potential assessment of water resources in particularly in small rivers for mini-hydro development • Link with A-Z renewable energy portal by TREMIS • Contribute to one-stop-shop for energy in the long run	Continuous support throughout the programme duration
14	International donor agencies	• Facilitate establishment of SE4ALL Secretariat • Facilitate regional co-operation in implementing SE4ALL initiative • Provide technical assistance, grants/loans for programme activities • Provide awareness creation, capacity building and community engagement	Effective monitoring and reporting of programme progress and achievement of targets
15	Private sectors	• Contribute to the development of training curriculum by providing training needs relevant to their operations • Provide technical training avenues on focus sectors to various stakeholders • Contribute towards improvement of policy framework and regulations • Support RE resource assessment • Investment and market development for various programme activities	• Technical assistance from international experts through SE4ALL initiative • Funds/grants for development of training facilities • Funds/grants for market development of solar home systems
16	Financial institutions (Banks, SACCOs, micro-finance institutions, etc.)	• Loans to project developers • Manage funds/grants from the donors for focus sectors • Micro-credit to the consumers of SHS and modern cook stoves	Effective monitoring and reporting of programme progress and achievement of targets

No.	Stakeholder	Partnership contribution	Risks and assumptions
17	NGOs, CSO, Women development organisations	• Provide awareness creation to public on modern clean cooking solutions and other renewable energy technologies • Implement projects related to RE and EE • Educate on gender mainstreaming aspects	Access to funds/grants
18	Export Processing Zones Authority (EPZA)	• Identify areas for promoting RE resources • Support establishment of business space for focus sectors	Access to funds/grants
19	Ministry of Education	• Support establishment of accreditation team to assure the quality for training and facilities • Co-ordinate with dedicated EE team in developing training and certification courses for energy auditors • Facilitate establishment of training centres in Universities and institutions	Training and capacity building in setting up accreditation team

Risks and Assumptions

Table 10 shows the potential risks and corresponding mitigation measures. More details on the same are furnished in Risk analysis (annex 3).

Table 10: Risks and mitigation measures

No.	Risk	Mitigation measures
1	Change of government plans and priorities	GoT has already signed a letter for implementation of SE4ALL initiative. If there is change in the government plans and priorities, the Project Steering Committee (PSC)) will discuss with the government representatives and do the needful for continued support and co-operation from the government for this programme.
2	Lack of financial resources to address all the project components	Financial commitments will be obtained from various partners for the given activities before the start of the project implementation.
3	Government policy changes in relation to the programme	PSC to convene regularly and identify specific measures for overcoming any potential implications
4	Operational limitations causing delay in programme implementation	Utmost care and efforts will be taken to ensure timely actions and quality planning for all the project activities, including procurement of services and goods. Regular monitoring and corrective actions will be taken accordingly in all the activities.
5	Natural disasters of medium to large scale	All the feasibility studies for the power plants and construction will be made on appropriate precautions for any natural disaster like

No.	Risk	Mitigation measures
		flood or drought. If anything, unforeseen happens, immediate actions will be taken accordingly based on PSC's decision.

Stakeholder Engagement

The target stakeholder group under this programme includes government ministries, financial institutions, banks, micro-credit institutions, universities, RE institutions, private sectors, business entrepreneurs, NGOs, CSOs, women development organisations and households. The estimated number of beneficiaries under different target groups at the end of 2030 is provided in table 11.For year 2022, it is assumed that 50% of 2030 target. So, the beneficiaries will also be 50% of the values mentioned in table 11.

Table 11: Programme outputs, target group& beneficiaries by 2030

No.	Programme Outputs	Target activity	Beneficiaries
1	Enhanced institutional capacity on sustainable energy development	• Training government officials and other relevant stakeholders on different aspects (policy, management and technical) • Training at least 1,000 trainers • Carrying out training exchange visits • Setting up of minimum 50 training centres through partnership • Procuring office facilities such as office equipment, computers and vehicles that will enhance service delivery • Facilitating participation of Secretariat and relevant official in the SE4ALL and other related international energy forums	• At least 10 financial institutions • Minimum 20 NGOs and 5 women development organisations • Minimum of 100 staff from the government ministries • Minimum 5,000 private sector, entrepreneurs, SMEs, etc. • Minimum 30,000 skilled labours/artisans with 40% of them being women
2	Creation of an enabling investment environment for RE/EE technologies	• Prioritizing RE/EE sector in the investment promotion/funding • Developing quality standards for products • Promoting RE industries under SEZ and other avenues • Supporting development of RE information system which will facilitate data exchange and information • Developing high-level forums to periodically	• Minimum of 100 products from different product manufacturers quality assured and labelled and available for end users in the market • At least 15 testing centres benefitted through SE4ALL scheme to set up quality testing labs • At least 50 RE industries supported in setting up factory/firms within the SEZ

No.	Programme Outputs	Target activity	Beneficiaries
		discuss RE development in the country • Facilitating accessibility of RE data including research data and potential sites for RE investments	
3	Scale up and roll outs of sustainable energy initiatives	Increasing electricity access through grid expansion	• Around 9 million households provided with electricity access • Minimum of 100,000 job opportunities created
		Increasing electricity access through decentralised RE	• Around 1 million households provided with electricity access • Minimum of 50,000 job opportunities created
		Increasing RE share in thermal requirements (mainly with ICS)	• Around 7 million households benefitted with access to improved cook stoves • At least 200,000 jobs created
		Reducing energy intensity through EE measures	• Minimum of 50 sustainable charcoal production units established • Minimum of 30 ICS production units established • Around 100 energy intensive industries benefitted through energy conservation • At least 100,000 jobs created under grid loss reduction, efficient lightings, energy auditing, EE improvement, etc.
		Rolling out of modern cooking solutions (excluding ICS)	• Around 2 million households benefitted with access to modern cooking solutions • Around 50,000 jobs created in clean cook stove market

Awareness campaigns, workshops, discussion forums, etc. will be organised through SE4ALL Secretariat for different target groups. A – Z RE portal will be created and enhanced for accessing information on different projects. Facility to obtain feedback will also be made in the information system. LGAs will be highly involved in the awareness campaigns, training programs and demonstration projects to garner their full support and co-operation.

Surveys will be conducted as (a)part of M&E process through NGOs, CSOs, women development organisations as required under different projects to collect the end user preferences, challenges, impact of projects, improvement in health and lifestyle, creation of direct and in-direct jobs, etc.

Projects that are identified to have significant impact on the surrounding environment such as installation of RE based power plants will require environmental impact assessment (EIA), stakeholder consultation meetings, etc., to obtain the concerns of the would be affected people. Steps will be taken to mitigate the impacts identified, if any.

South-South and Triangular Cooperation (SSC/TrC)

Tanzania is one of the first countries in SSA to prepare SE4ALL programme design. Most(of) other countries are still either preparing or have completed the AA and Investment prospectus (IP). However, the Tanzania SE4ALL will seek co-operation in knowledge sharing and technology know-how from similar SE4ALL programmes in other African countries during the development and implementation stages. Since UNDP has been in the forefront in the promotion of SE4ALL programmes globally, it can assist in enhancing these south-south co-operations.

Knowledge

A central database will be created by SE4ALL Secretariat with all key information and easy accessibility. The database will have a quarterly update of resources/database on the latest development in each focus sectors. Newsletters will be prepared and circulated among the stakeholders on the latest happening in the energy sector. A website will be created and all the available resources will be updated online for access to anyone from anywhere in their own levels.

The progress made in programme targets, barriers overcome, market development achieved, issues faced, lessons learnt, etc., will be captured through monitoring and evaluation set up under SE4ALL and will be published through reports. These reports will also be made available at the A-Z RE portal developed by TREMIS.

The materials prepared for awareness campaigns through government, private sectors, institutions, etc., will help in documenting the knowledge created. The information about the programme will also be disseminated though websites, radio talks, TV shows, newspapers, magazines, etc.

Sustainability and Scaling Up

Sustainability of the projects under focus sectors will be ensured through the following approaches:

- ***Financial:*** The involvement of the private sector through investment in RE will be enhanced through an environment that is conducive to sustain the operations with appropriate rate of returns.
- ***Institutional:*** The SE4ALL project will develop capacity development programs that will be institutionalised into the existing training institutions to provide continued RE-based training as part of its curriculum.
- ***Technical:*** More skilled labour(s) and technicians will be trained through various training programs in solar, wind, small hydro, biomass, EE and clean cooking sector. This will ensure the availability of skilled work force at affordable cost for service, operation and maintenance of the power generation systems. The capacity development programme(s) with local training institutions will provide a continued avenue to train and maintain the required technical skills in RE. Ensuring the quality standards of the solar PV products will build confidence among the end users in solar PV products. This will boost the solar PV market both at the household level and community level power generation systems. Banks will be more open for financing the solar PV projects.
- ***Policy:*** The SE4ALL project will bring-about an improved approach to energy sector activities through the formulation of an overarching framework, including the necessary implementing rules and regulation.
- ***Environmental and Social:*** A sustainable charcoal production plan is to be implemented through co-operation from VPO-DoE, TFS and all other relevant stakeholders. This will ensure the availability of quality charcoal fuel for consumers who purchase ICS under this programme. The environment that is conducive to enhance the use of RE technologies will cushion any environmental impacts resulting from additional use of fossil fuel for power generation. In addition, the anticipated boost of IPPs will provide additional employment. An excellent synchronisation of TANESCO grid with the small RE based IPPs is envisaged. This will avoid losses to IPPs and ensure operation of the plants for the maximum power generation.

In general, the sustainability of the programme is ensured through following design approaches:

1. Continuous review of project progress, deviations in trends and course-corrections:

SE4ALL Secretariat will have a vigilant monitoring and evaluation system on the project implementation. Any deviations will be identified and challenges faced will be discussed with the relevant stakeholders. Based on these discussion and review, corrective actions will be taken to keep the project implementation on track.

2. Capacity building to the target stakeholders:
All the stakeholders involved in different focus sectors will be trained on their respective roles and responsibilities. Especially, the government ministries and policy makers will be trained in energy sector planning, prioritizing development projects, fund management, project management, latest technologies, public-private partnerships, etc. As most of the focus sectors come under the management of ministries, they will be better equipped to steer the projects towards successful implementation.

3. Creation of project ownership:
All the projects implemented will encourage the ownership creation at each level. Each stakeholder will have a share in the successful implementation of the project towards reaching the targets. This will also improve the overall co-operation among the stakeholders.

Scale up of the projects will be promoted through the following activities:

- Setting up many 'solar kiosks' will encourage people to buy more solar PV products. This will open the market segment for scaling up.
- Clarity on integration of mini-grid into the national grid & approach of national grid in that location will boost confidence among the investors in developing RE based mini-grids (this aspect is being worked on at present)
- Resource availability studies for wind, solar, hydro and biomass potential will help entrepreneurs to develop power plants at remote locations and agro-industries
- Testing centres and training centres can be operated sustainably by collecting a nominal fee for their services

In general, the scaling up of the programme is ensured through the following design approaches:

1. Creation of revolving funds wherever required:
Revolving funds will be created under the management of banks and financial institutions, wherever possible. The project developers can access these funds for business development activities. Once the developers pay them back, then the fund will be available to other developers for further scale up of the projects

2. Conducive environment created:
Attractive credit and finance mechanisms will create conducive environment for investment flow into the energy sector. Effective enforcement of quality standards and further supports under SEZ schemes will encourage more entrepreneurs to get into the RE/EE sectors.

3. Availability of market/resource data:
Through various resource assessment studies, the available potential will be clearly visible. This will encourage entrepreneurs and private sectors to start new business. Financial institutions will have confidence in financing those projects.

IV. PROJECT MANAGEMENT

Cost Efficiency and Effectiveness

The programme strategy is built on the previous and on-going projects and sustainable development initiatives. The cost efficiency and effectiveness of the programme design is ensured through the following considerations.

a) Maximum use of existing infrastructure:

Wherever possible, the programme is designed to utilize the existing office buildings of ministries, institutions and other relevant stakeholders to the maximum. The SE4ALL Secretariat will be housed within the office premises of MEM. Similarly, the dedicated EE team/unit can be established within the Secretariat. The physical location of office infrastructure, staff, mode of operation etc. can be designed to fit within the Legal, Regulatory and Administrative (LRA) framework of MEM.

b) Alignment within the programme outputs and with national priorities and action plans:

The strategy is designed to create synergies between the efforts taken under different focus sectors thus creating maximum impact from the funds and resources allocated. The integration of data resources will serve to fill up the gaps in all the focus sectors. The stakeholders from any of the focus sector can assess the integrated data links to get the latest happenings, market trends, etc. This eliminates the need for separate information centre for each of the focus sectors and avoids separate resource allocations.

The development of quality standard and testing for various heating, electrical and electronic goods will be best utilised under several demonstration projects such as promoting clean cooking solutions, increasing SHS use in households, reducing charcoal consumption through improved cook stoves, promotion of LEDs and increasing EE in industries and commercial buildings. Similarly, the efforts such as awareness campaigns, trainings and market enabling activities in one sector will complement and leverage the successful implementation of activities in other sectors.

Since the strategy is aligned with the national priorities and plans, the actions taken by the government through its budget/resource allocations will also support effective implementation of SE4ALL programme.

c) Strategy in-line with other regional and international programmes

The programme will seek co-operation with Regional and international programmes such as GACC, Africa SE4ALL Energy hub, Lighting Africa, etc. for maximising the transfer of lessons learnt, knowledge and technology transfer, etc. Hence, significant resource and budget allocation for separate research and development of technologies/skills will be avoided.

Thus, it is ensured that the programme has effectively considered cost efficiency and effectiveness in its strategy design.

Project Management

The projects under different focus sectors will be implemented by co-ordinating public or private sector partners. Renewable energy (solar, wind, small hydro and biomass) based power plants will be set up in identified potential sites for power supply to the national grid and dedicated mini-grids.

Clean cook stove solutions such as improved charcoal stoves, LPG stoves, natural gas stoves and efficient lightings such as LEDs/CFLs will be disseminated to rural and urban households. Biogas digesters will be constructed in rural households. Under the programme, electricity access will be increased to more rural and urban households through grid expansion, mini-grids, SHSs and

increased generations from other sources. Thus, the programme will be operationalized across the country in rural and urban regions.

Under the programme, new business start-ups in the focus sectors will be supported through MoIT, MEM and respective ministries through applicable procedures for approval of licenses and permits. SE4ALL Secretariat will help in achieving high co-ordination among the ministries in this regard. Business and market development will be the responsibility of the private sector partners. The partners will set up the dedicated branch offices, create dealer networks, source products and sell their products to the end users through marketing offices. They may also conduct training (s) to their staff and end users. However, the collaborations may happen in training and awareness campaigns conducted collectively for the staff of all the partners through identified institution. Technicians can be trained to set up service centres where consumers can get their products of any brand serviced. Thus, the partners need not invest in the dedicated service centres for their brands, instead could benefit from a shared service centres. This would be win-win for the private sector as well as the consumer in reducing their time and cost required for after sale service.

UNDP Project office will work closely with SE4LL Secretariat in overall coordination and management of the programme. They will provide technical and advisory services in the procurement of international expertise for programme needs and in creating partnerships with other international agencies and programmes.

A SE4ALL Secretariat will be established under MEM in collaboration with UNDP and AfDB. The Secretariat will take care of the overall management of all programme activities under this five-year plan. A detailed design of SE4ALL Secretariat operation, its roles and responsibilities, annual budget and the organisation structure will be done during the initial implementation phase of the project. The SE4ALL will consist of Project Steering Committee (PSC) whose members will be from various ministries, donors, private sectors, institutions, NGOs, etc. (see fig 16). The PSC members and administrative staff for the Secretariat will then be appointed by MEM in discussion with UNDP and AfDB. A National Programme Coordinator (NPC) will be appointed by the PSC to oversee operations of SE4ALL Secretariat. The PSC members and administrative staff will be trained on project management, co-ordinating with various stakeholders, communication, monitoring and reporting and programme data management.

Once formed, the Secretariat will prepare its annual work plan and budget for the first year. The annual work plan shall be approved by the PSC. All the required infrastructure facilities such as office rooms, furniture, software, office equipment including printer/copier, etc., will be provided to the team.

The Secretariat will then convene a programme kick-off meeting with the relevant stakeholders from each focus sectors. Continuous follow-up communication, meetings and co-ordination will be done for effective stakeholder engagement throughout the programme duration. The Secretariat will be the lead advocacy and managerial unit that would,

- Define high level programmes, develop strategies and assess funding needs and gaps.
- Develop the portfolio of investment and indicative funding needed for the inventory of programmes.
- Define matchmaking/identification of multi-stakeholders that will partner with each programme.
- Design the resource mobilisation strategy and implementation of the prospectus.
- Monitor SE4ALL related projects and studies, which should remain under the responsibility of the relevant government agencies and institutions.
- Take the responsibility for realising SE4ALL goals, implementing the interventions and applying the SE4ALL Monitoring, Evaluation and Reporting (MER) mechanism.
- Create and increase awareness.
- Act as focal point for exchanging information with the SE4ALL global initiative, especially with the SE4ALLGlobal Facility Team, the SE4ALL Africa Hub and the SE4ALL Thematic Hubs.
- Present draft modifications to the AA, to be reviewed by the SE4ALL PSC and GoT authorities, propose any other “mid-course” adjustments that may be needed to keep Tanzania on its path to achieve its SE4ALL goals.

Given the overarching nature of the SE4ALL initiative and GoT's commitment to the global initiative, a special protocol will be developed to facilitate and streamline the capacity of the Secretariat to reach across public and private sector stakeholders. The Secretariat will be led by the SE4ALL Project Manager and supported by a small team of experts with experience and expertise in each of the SE4ALL goals. The personnel of the Secretariat will be employed by MEM. It is not envisaged that SE4All would have financial autonomy; rather, it would initially draw its financial resources from a budgetary line from MEM's budget, and will incorporate additional funding from the programmes related to the Secretariat's activities and as appropriated by the mid-term planning process.

The Project Manager will support the implementation of AA and IP resources in general and advise on the best opportunities for delivering the SE4ALL outcomes and on how the required investments to be made and managed to ensure maximum returns. She/he will be responsible for monitoring and working with the project implementation partners identified. The Project Manager will facilitate project initiatives, coordinate agencies and institution activities such as, but not limited to, project implementation, feasibility studies, consumer and end-user research, work plan preparation and monitoring results. International and national experts can be engaged to provide technical advisory services to the SE4ALL Secretariat on:

- institutionalization process and providing on-demand support.
- leading, identifying, implementing and monitoring SE4ALL activities.
- supporting the Secretariat throughout the AA and IP's integration to the Mid Term Planning (MTP) process and through the early stages of the transformational phase.

The SE4ALL will have a dedicated team to keep track of various project implementation activities going on in each sector. The team will be in constant discussions with relevant ministries, banks, donor agencies, private sector partners, NGOs, etc. Audit of funds and investments made in the focus sectors will be conducted through the fund management partners such as banks and financial institutions. They will submit reports to SE4ALL Secretariat on the funds and investments made by different stakeholders from time to time.

V. RESULTS FRAMEWORK[32]

Intended Outcome as stated in the UNDAF/Country [or Global/Regional] Programme Results and Resource Framework:

Increased access to affordable and sustainable energy

Outcome indicators as stated in the Country Programme [or Global/Regional] Results and Resources Framework, including baseline and targets:

- percentage population with electricity access
- percentage population with access to modern cooking solutions
- RE share in the overall power mix
- RE share in cooking and heating energy
- Reduced energy consumptions through EE measures

Applicable Output(s) from the UNDP Strategic Plan:

1. Enhanced institutional capacity on sustainable energy development
2. Creation of an enabling investment environment for RE/EE technologies
3. Scale up and roll outs of sustainable energy initiatives

Project title and Atlas Project Number:SE4ALL Implementation in Tanzania – Project Number00098492 (Award ID: 00094384)

EXPECTED OUTPUTS	OUTPUT INDICATORS[33]	DATA SOURCE[34]	BASELINE		TARGETS (by frequency of data collection)						DATA COLLECTION METHODS & RISKS[35]
			Value	Year	Year 1	Year 2	Year 3	Year 4	Year 5	FINAL	

[32]UNDP publishes its project information (indicators, baselines, targets and results) to meet the International Aid Transparency Initiative (IATI) standards. Make sure that indicators are S.M.A.R.T. (Specific, Measurable, Attainable, Relevant and Time-bound), provide accurate baselines and targets underpinned by reliable evidence and data, and avoid acronyms so that the external audiences clearly understand the results of the project.

[33]It is recommended that projects use output indicators from the Strategic Plan IRRF, as relevant, in addition to project-specific results indicators. Indicators should be disaggregated by sex or for other targeted groups where relevant.

[34] Considered as data source for indicator

[35] Considered as verification/data collection for the indicator

Output 1: Enhanced institutional capacity on sustainable energy development	**1.1** SE4ALL Secretariat established	Secretariat operation reports	No Secretariat in place	2016	*To guide the implementation of the SE4ALL Programme*						Review of reports, interviews, observation
	1.2 Availability of key RE data in the country	TREMIS renewable energy portal	No platform for information sharing	2016	*To manage, update and keep latest data as per annual work plans*						Accessing TREMIS RE portal for RE based information, reports, documents and manuals prepared
	1.3 Number of certified/accredited training centres	List of accredited centres in TREMIS RE portal	No accredited training/certification centres	2016	5	5	5	5	5	25	Physical visit to training centres
	1.4 Number of persons trained in focus sectors	• Trainings conducted and attendance list • Progress report • Final evaluation report	Inadequate training	2016	100	3,000	5,000	5,000	4,900	18,000	Verification of trainings conducted & participants attended
	1.5 Dedicated unit/team for EE set up	• Annul work plan • Progress report/final evaluation report	No dedicated unit/team to promote EE	2016	*To plan, organise and implement EE activities as per annual work plans*						Physical site visit to team office and collection of reports prepared
Output 2: Creation of an enabling investment environment for RE/EE technologies	**2.1.** Increased RE investments in the country (million USD)	• Bank records, agreements • TIC annual investment reports • New registered and licensed energy companies from EWURA records	0	2016	50	100	100	100	100	450	Review of transaction records

	2.2. Number of products for which quality standards are developed and approved	• TBS website • Final evaluation report	-	2016	0	5	5	5	5	20	Review of quality standard documents prepared by TBS
	2.3. Number of testing centres created	• Testing centres registered with TBS • Progress report • Final evaluation report	-	2016	0	1	2	2	2	7	TBS website, TREMIS RE portal
	2.4. Number of designated consumers identified	Reports from energy efficiency team/unit	0	2016	0	5	5	5	5	20	Review of assessment reports for identification of designated consumers
	2.5. Number of RE based business firms established in various SEZs in the country	• Annual reports and records from the MOIT and SEZs • Progress report • Final evaluation report	-	2016	0	2	3	5	10	25	• Data from EPZA • Physical verification at SEZs
Output 3: Scale up and roll outs of sustainable energy initiatives	**3.1.** New household electricity connections (or % of population with access to electricity) and energy services (millions) – through grid, mini-grid, off-grid	• Records of TANESCO • Progress report • Final evaluation report	2.7	2016	1	1	1	1	1	5	Review of records of TANESCO and other implementing partners
	3.2. Capacity of RE based power plants (MW)	Records of TANESCO, REA and private sector	25	2012	0	50	50	100	100	300	Site visit to power plants

	3.3. Number of SHSs, solar lantern/Pico PV systems retailed	• Progress report • Final evaluation report • Sale records	10,800	2014	20,000	80,000	100,000	100,000	100,000	400,000	Physical site visit to households
	3.4. Number of Solar kiosks created	• Data available in TREMISRE portal • Progress report/evaluation report	-	2016	5	10	10	10	15	50	Physical site visit to service kiosks
	3.5. Total installed capacity of biomass based power plants (MW)	• Data available in TREMIS RE portal • Progress report/evaluation report	-	2016	0	0.5	0.5	1	1	3	• Physical site visit to power plants • Plant commissioning report
	3.6. Annual production of sustainable charcoal (tons)	Records of TFS and VPO-DOE	-	2016	0	50,000	100,000	200,000	400,000	400,000	Physical site visit to plantations and charcoal production units
	3.7. Number of ICS constructed and retailed in the country	Sale records of concerned firms/organization	28,500	2014	10,000	50,000	100,000	100,000	200,000	460,000	Physical site visit to households and survey reports
	3.8. Number of substations updated with proper monitoring	• Records of TANESCO • TREMIS energy portal • Progress report/final evaluation report	-	2016	2	3	5	5	5	20	Physical site visit to substations

	3.9. Number of LED/CFL lamps distributed	Private sector payment records	7 million	2014	10,000	90,000	200,000	200,000	500,000	1,000,000	Physical site visit to households and review of Private sector payment records
	3.10. Installed capacity (MW) of thermal biogas/gasifier plants established	• Plant commissioning reports • Progress report/final evaluation report	-	2016	0	0.5	0.5	1	1	3	Physical site visit to biogas plants
	3.11. Preparation of MEPS for number of products	Reports on MEPS of products	0	2016	1	4	5	5	5	20	Review of MEPS reports prepared
	3.12. Number of products tested and performance labelled	• Data with the dedicated EE unit/team • Website of the dedicated EE unit/team	0	2016	10	20	20	20	30	100	Review of products available in market and document available with dedicated EE unit/team
	3.13. Number of audit firms accredited	• Data with dedicated EE unit/team • Progress report/final evaluation report	0	2016	0	1	3	3	3	10	Physical site visit to audit firm and review of document available with dedicated EE unit/team
	3.14. Number of audits done in industries	Data with dedicated EE unit/team and industries	0	2016	0	5	10	15	20	50	Plant energy audit reports
	3.15. Number of EE measures implemented	Data with dedicated EE	0	2016	0	10	20	20	20	70	Plant energy audit report after implementation

		unit/team and industries									
	3.16. Number of biogas digesters installed	• Payment records and households' records • Progress report/final evaluation report	13,037	2015	5,000	5,000	5,000	5,000	5,000	25,000	Physical verification of biogas plant sites
	3.17. Number of LPG stoves retailed	• Sale records • Private sector feedback forms to MEM • Programme progress report/final evaluation report	90,000	2012	10,000	20,000	50,000	50,000	50,000	180,000	• Physical site visit to households • Review of private sector annual reports
	3.18. Number of ethanol stoves retailed	• Sale records • Private sector feedback forms to MEM • Programme progress report/final evaluation report	0	2016	0	10,000	30,000	50,000	100,000	190,000	• Physical site visit to households • Review of private sector annual reports
	3.19. Number of natural gas stoves retailed	Sale records	57	2014	1,000	2,000	2,000	3,000	5,000	13,000	• Physical site visit to households • Review of private sector payment records

VI. MONITORING AND EVALUATION

In accordance with UNDP's programming policies and procedures, the project will be monitored through the following monitoring and evaluation (M&E) plans: *[Note: monitoring and evaluation plans should be adapted to project context, as needed]*

SE4ALL Secretariat will be responsible for M&E, including organising project evaluations, approving annual implementation work plans and budget revisions, monitoring progress, identifying problems and suggesting remediating actions, facilitating timely delivery of project outputs and supporting the coordination and networking with other related initiatives and institutions in the country and in the region. The setting up of the SE4ALL Secretariat along with its operational frameworks will be initial activities in the implementation of this programme.

The programme will be monitored through the following M&E activities.

Programme start:

A programme inception workshop will be held within the first 2 months of the project start, with assigned roles in the project organization structure, UNDP country office and other stakeholders. The inception workshop is crucial in building the ownership for the project results and for the first-year annual work plan. The inception workshop should address number of key issues including:

- Assisting all the partners to make them fully understand and take ownership of the project.
- Detailing the roles, support services and complementary responsibilities of SE4ALL Secretariat, UNDP and the project team.
- Discussing the roles, functions, and responsibilities within the project's decision-making structures, including reporting and communication lines, and conflict resolution mechanisms.
- Finalizing the first annual work plan, based on the project results framework.
- Reviewing and agreeing on the indicators, targets and their means of verification, and recheck assumptions and risks.
- Providing a detailed overview of reporting and M&E requirements.
- Scheduling the agreed M&E work plan and budget.
- Discussing financial reporting procedures, obligations and arranging for the annual audit.
- Planning and scheduling the PSC meetings.
- Defining the roles and responsibilities of the entire project organisation structure.
- Conducting the first PSC meeting within first 12 months following the inception workshop.

An inception workshop report is a key reference document and must be prepared and shared with the participants to formalize various agreements and plans decided upon, during the meeting.

Training will be provided to all relevant stakeholders on data collection and reporting for effective capture of project conditions.

Quarterly:

- SE4ALL Secretariat should track progress made in each focus sectors
- The risk log shall be regularly updated, based on the initial risk analysis submitted.
- Project Progress Reports (PPR) will be prepared and submitted to the PSC based on the information collected.

Annually:

Annual Project Review/Project Implementation Reports (APR/PIR): This key report will be prepared to monitor the progress made since project start and for the reporting period). The APR/PIR includes, but is not limited to, reporting on the following:

- Progress made towards the project objective and project outcomes - each with indicators, baseline data and end-of-project targets (cumulative)
- Project outputs delivered per project outcome (annual).
- Lesson learnt/good practices.
- Risk and adaptive management

Periodic monitoring through site visits:

SE4ALL Secretariat and UNDP will conduct visits to the project sites, power plants and households based on the agreed schedule in the project's Inception Report/Annual Work Plan to assess first hand project progress. Other members of the PSC may also join these visits. A Field Visit Report will be prepared by SE4ALL Secretariat and will be circulated within one month after the visit to the project team and the PSC members.

Approach for gender mainstreaming:

Wherever possible, women consultants/experts will be engaged for training, monitoring data collection, survey, etc. Women participation will be encouraged throughout. The monitoring will focus on capturing gender benefits and their feedback on the project interventions.

Engagement of LGAs:

LGAs will be engaged in monitoring the implementation of various initiatives in their regions such as grid, mini-grid, off-grid and cooking solutions. Their views will be taken into consideration during survey, consultations, data collection and evaluation of the programme. They will play major role in monitoring and conveying the information to the SE4ALL Secretariat. All the activities mentioned in output 3 needs monitoring and reporting and should start from LGAs. The information should flow to the SE4ALL Secretariat in a pre-determined smooth way.

Mid-term evaluation:

The project will undergo an independent Mid-Term Review at the mid-point of project implementation. The Mid-Term Review will determine progress being made towards the achievement of outcomes and will identify the course of correction if needed. It will focus on the effectiveness, efficiency and timeliness of project implementation; it will highlight issues requiring decisions and actions; and will present initial lessons learnt about the project design, implementation and management. Findings of this review will be incorporated as recommendations for enhanced implementation during the final half of the project's term. The organization, terms of reference and timing of the mid-term review will be decided after consultation with the concerned parties. The Terms of Reference (ToR) for the Mid-term review will be prepared by the SE4ALL Secretariat under the guidance of UNDP.

End of project evaluation:

An independent Terminal Evaluation will take place three months prior to the final PSC meeting and will be conducted under the guidance of UNDP. The terminal evaluation will focus on the delivery of the project's results as initially planned (and as corrected after the mid-term review, if any such correction took place). The terminal evaluation will look at the impact and sustainability of results, including the contribution to capacity development and the achievement of global environmental benefits/goals. The ToR for this evaluation will be prepared by SE4ALL Secretariat under the guidance of UNDP. Evaluation will include a survey to assess the increased availability of quality products in the market and increased purchase of quality products by the end users.

During the last three months, the project team will prepare the Project Terminal Report. This comprehensive report will summarize the results achieved (objectives, outcomes, outputs), lessons learnt, problems met and areas where results may not have been achieved. It will also lay out recommendations for any further steps that may need to be taken to ensure sustainability and replicability of the project's results.

Overall, mid-term and final evaluation activities will contribute in:

- Improving the programme effectiveness and delivery towards Tanzania's SE4ALL goals by 2030 through using the knowledge and lessons learnt from its implementation expressed in the country.
- Aligning the strategic activities of the AA and ensure that it remains relevant in addressing the country level objectives whilst also aligning with the global initiative on SE4ALL.

Learning and knowledge sharing:

Results from the project will be disseminated within and beyond the project intervention zone through the existing information sharing networks and forums. The project will identify and participate, as relevant and appropriate, in scientific, policy-based and/or any other networks, which may be of benefit to the project implementation through lessons learnt. The project will identify, analyse and share the lessons learnt that might be beneficial in the design and implementation of similar future projects.

The summary of monitoring plan and budget is provided below:

Monitoring Plan

Monitoring activity	Purpose	Frequency	Expected action	Partners	Cost (USD)
Track project progress	To capture programme progress data against the result indicators in the Results and Resources Framework (RRF) and assess the progress made in achieving the agreed outputs.	Quarterly	Slower than expected progress will be addressed by the project management.	UNDP, MEM and AfDB	100,000
Monitor and manage risk	To identify and monitor the risk management actions using a risk log. This includes monitoring measures and plans that may have been required as per UNDP's Social and Environmental Standards. Audits will be conducted in accordance with UNDP's audit policy to manage the financial risk.	Quarterly	Risks are identified by the project management and actions are taken to manage the risk. The risk log is actively maintained to keep track of the identified risks and the actions taken.	UNDP, MEM and AfDB	50,000
Annual project quality assurance	The quality of the project will be assessed against UNDP's quality standards to identify the project strengths and weaknesses and to inform management in decision making to improve the project.	Annually	Areas of strength and weakness will be reviewed by the project management and will be used to support decision making to improve the project performance.	UNDP, MEM and AfDB	50,000

Monitoring activity	Purpose	Frequency	Expected action	Partners	Cost (USD)
Review and make course corrections	Internal review of data and evidence from all monitoring actions to inform decision making.	At least annually	Performance data, risks, lessons and quality will be discussed by the PSC and used to make course corrections.	UNDP, MEM and AfDB	200,000
Project report	A progress report will be presented to the PSC, consisting of progress data showing the results achieved against pre-defined annual targets at the output level, the annual project quality rating summary, an updated risk long with mitigation measures and any evaluation or review reports prepared over the period.	Annually, and at the end of the project (final report)	Identify progress towards programme targets and make course corrections, if needed.	UNDP, MEM and AfDB	250,000
Project review	The PSC will hold regular project reviews to assess the performance of the project and review the Multi-Year Work Plan to ensure realistic budgeting over the life of the project. In the project's final year, the PSC shall hold an end-of project review to capture lessons learnt and discuss the opportunities for scaling up and to socialize project results and lessons learnt with relevant audiences.	At least annually	Any quality concerns or slower than expected progress should be discussed by the PSC and management actions be agreed to address the identified issues.	UNDP, MEM and AfDB	100,000
Knowledge sharing	Knowledge, good practices and lessons learnt will be captured regularly, as well as actively sourced from other projects and partners and integrated into the project.	At least annually	Relevant lessons are captured by the project team and are used to inform the management decisions.	UNDP, MEM and AfDB	250,000
			Total		**1,000,000**

Evaluation Plan[36]

Evaluation title	Partners (if joint)	Related strategic plan output	UNDAF/CPD outcome	Planned completion date	Key evaluation stakeholders	Cost and source of funding (USD)
Mid-Term evaluation	UNDP, MEM and AfDB	Enhanced institutional capacity on sustainable energy development, Creation of an enabling investment environment for RE/EE technologies, Scale up and roll outs of sustainable energy initiatives	Increased access to affordable and sustainable energy	June 2019	Stakeholders in all focus sectors	200,000
End of project evaluation	UNDP, MEM and AfDB	Enhanced institutional capacity on sustainable energy development, creation of an enabling investment environment for RE/EE technologies, scale up and roll outs of sustainable energy initiatives	Increased access to affordable and sustainable energy	April 2022	Stakeholders in all the focus sectors	300,000
				Total		**500,000**

[36]Optional, if needed

VII. MULTI-YEAR WORK PLAN [37,38]

All anticipated programmatic and operational costs to support the project, including development effectiveness and implementation support arrangements, needs to be identified, estimated and fully costed in the project budget under the relevant output(s). This includes activities that directly support the project, such as communication, human resources, procurement, finance, audit, policy advisory, quality assurance, reporting, management, etc. All services which are directly related to the project need to be disclosed transparently in the project document.

EXPECTED OUTPUTS	PLANNED ACTIVITIES	Planned Budget by Year (USD)						RESPONSIBLE PARTY	PLANNED BUDGET		
		Y1	Y2	Y3	Y4	Y5	Total		Funding Source	Budget Description	Amount
Output 1: Enhanced institutional capacity on sustainable energy development	**1.1.** Setting up of the SE4ALL Secretariat (Finalization of the Communication & Awareness Strategy; Monitoring, Evaluation & Reporting Plan, Regionalization Implementation Plan; and SE4ALL Structural & Operational Framework	400,000	300,000	100,000	100,000	100,000	1,000,000	UNDP, MEM, DPs	*Funding to be mobilized from DPs to top up UNDP TRAC contribution*		

[37]Cost definitions and classifications for programme and development effectiveness costs to be charged to the project are defined in the Executive Board decision DP/2010/32

[38]Changes to a project budget affecting the scope (outputs), completion date, or total estimated project costs require a formal budget revision that must be signed by the project board. In other cases, the UNDP programme manager alone may sign the revision provided the other signatories have no objection. This procedure may be applied for example when the purpose of the revision is only to re-phase activities among years.

EXPECTED OUTPUTS	PLANNED ACTIVITIES	Planned Budget by Year (USD)						RESPONSIBLE PARTY	PLANNED BUDGET		
		Y1	Y2	Y3	Y4	Y5	Total		Funding Source	Budget Description	Amount
	1.2. Integration of data resources	50,000	50,000	50,000	50,000	50,000	250,000	SE4ALL Secretariat/UNDP	*Funding to be mobilized from DPs*		
	1.3. Quality training / certification courses for RE, EE, modern cooking	600,000	700,000	900,000	900,000	1,900,000	5,000,000	UNDP/ SE4ALL Secretariat /Private sector/DPs	*Funding to be mobilized from DPs*		
	1.4. Institutional framework for EE promotion	500,000	500,000	500,000	500,000	500,000	2,500,000	UNDP/ SE4ALL Secretariat /Private sector/DPs	*Funding to be mobilized from DPs*		
	MONITORING	50,000	50,000	50,000	50,000	50,000	250,000	UNDP/SE4ALL Secretariat	*Funding to be mobilized from DPs*		
		Sub-Total for Output 1:9,000,000									
Output 2: Creation of an enabling investment environment for RE/EE technologies	**2.1.** Prioritization of RE/EE sector in investment promotion/funding (including resources for procurement services and increased procurement capacities within GoT agencies)	2,200,000	2,300,000	2,500,000	2,500,000	2,500,000	12,000,000	UNDP/SE4ALL Secretariat /Private sector/DPs	*Funding to be mobilized from DPs*		

EXPECTED OUTPUTS	PLANNED ACTIVITIES	Planned Budget by Year (USD)						RESPONSIBLE PARTY	PLANNED BUDGET		
		Y1	Y2	Y3	Y4	Y5	Total		Funding Source	Budget Description	Amount
	2.2. Quality standards and testing regulations	200,000	300,000	500,000	500,000	500,000	2,000,000	UNDP/DPs/SE4ALL Secretariat	*Funding to be mobilized from DPs*		
	2.3. Promotion of RE industries through SEZ	1,000,000	2,0000,000	2,000,000	2,000,000	3,000,000	10,000,000	SE4ALL Secretariat	*Funding to be mobilized from DPs*		
	MONITORING	50,000	50,000	50,000	50,000	50,000	250,000	UNDP/SE4ALL Secretariat	*Funding to be mobilized from DPs*		
		Sub-Total for Output 2: 24,250,000									
Output 3: Scale up and roll outs of sustainable	**3.1.** Increasing electricity access through grid expansion	682,483,122	682,483,122	682,483,122	682,483,122	682,483,122	3,412,415,610[39]	TANESCO/Private sector	*Funding to be mobilized from DPs*		

[39]Considering grid connection cost as 960 USD per household as per SE4ALL Action Agenda page 22 for around 3.4 million households

EXPECTED OUTPUTS	PLANNED ACTIVITIES	Planned Budget by Year (USD)						RESPONSIBLE PARTY	PLANNED BUDGET		
		Y1	Y2	Y3	Y4	Y5	Total		Funding Source	Budget Description	Amount
energy initiatives	**3.2.** Increasing electricity access through decentralised RE (including provision for at least USD5 million micro-capital grants to pilot renewable energy initiatives in rural communities)	268,998,000	268,998,000	268,998,000	268,998,000	268,998,000	1,344,990,000[40]	UNDP/SE4ALL Secretariat /Private sector/DPs/PO-RALG/REA	*Funding to be mobilized from DPs*		
	3.3. Increasing RE share in thermal requirements (including ICS promotion)	16,560,000	16,560,000	16,560,000	16,560,000	16,560,000	82,800,000	TANESCO/REA, private sector	*Funding to be mobilized from DPs*		
	3.4. Reducing energy intensity through EE measures	12,000,000	12,000,000	12,000,000	12,000,000	12,000,000	60,000,000	SE4ALL Secretariat/TBS/SDPs/Private sector	*Funding to be mobilized from DPs*		
	3.5. Roll out of modern cooking solutions	12,500,000	12,500,000	12,500,000	12,500,000	12,500,000	62,500,000	UNDP/DPs/Private sector	*Funding to be mobilized from DPs*		

[40]Considering off-grid connection cost as 2,140 USD per household as per SE4ALL Action Agenda page 22 for around 0.6 million households

EXPECTED OUTPUTS	PLANNED ACTIVITIES	Planned Budget by Year (USD)						RESPONSIBLE PARTY	PLANNED BUDGET		
		Y1	Y2	Y3	Y4	Y5	Total		Funding Source	Budget Description	Amount
	MONITORING	50,000	50,000	50,000	50,000	50,000	250,000	UNDP/SE4ALL Secretariat	*Funding to be mobilized from DPs*		
		Sub-Total for Output 3: 4962,955,610									
Evaluation *(as relevant)*	EVALUATION			200,000		300,000	500,000	UNDP/SE4ALL Secretariat	*Funding to be mobilized from DPs*		
General Management Support		50,000	50,000	50,000	50,000	50,000	250,000	UNDP/SE4ALL Secretariat	*Funding to be mobilized from DPs*		
		Sub-Total for output: 750,000									
TOTAL	**4,996,955,610**										

VIII. GOVERNANCE AND MANAGEMENT ARRANGEMENTS

The programme organisation structure may consist of a Project Steering Committee (PSC) as in figure 16 below.

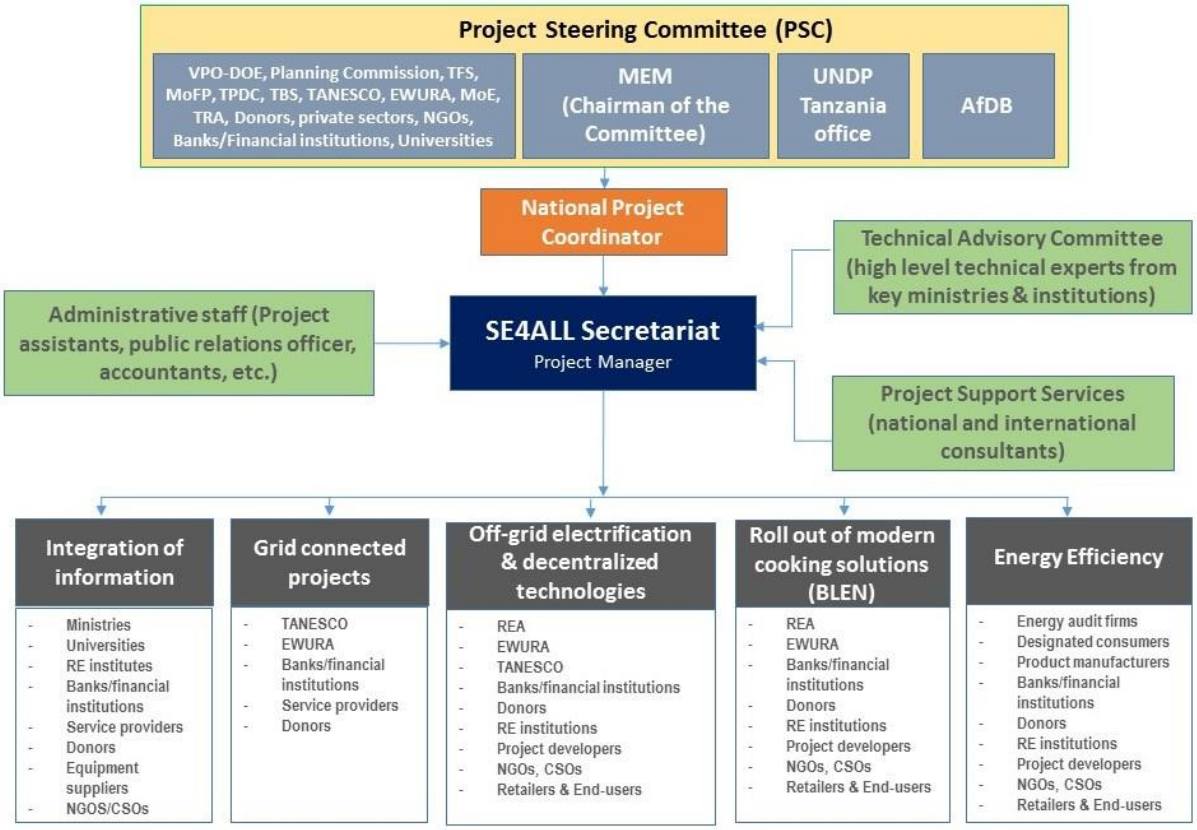


Figure 16: Project Management Structure with PSC

The SE4ALL Tanzania programme is implemented under MEM and the ministry will set up SE4ALL Secretariat, a dedicated unit to manage the programme and to oversee the programme implementation.

A PSC, chaired by the MEM will be established to provide strategic directions and management guidance to SE4ALL Secretariat in programme implementation. It will consist of representatives from relevant ministries and government departments/directorates, donors, banks and private sector representatives participating in the project, UNDP Country Office, AfDB, as well as representatives of the NGO community and women groups.

PSC will assign a senior officer as the National Project Co-ordinator (NPC). The main responsibilities of NPC will be to (i) coordinate the project activities with other government ministries (ii) certify the expenditures in line with approved budgets and work-plans; (iii) facilitate, monitor and report the procurement of inputs and delivery of outputs; (iv) approve the ToR for consultants and tender documents for sub-contracted assistance and inputs; and (v) report to PSC on the project delivery and impact.

The NPC will oversee the operation of SE4ALL Secretariat headed by a Project Manager (PM) recruited through a competitive process. The PM will be responsible for the overall programme coordination and implementation, consolidation of work plans and project papers, preparation of quarterly progress reports, reporting to the project supervisory bodies, and supervising the work of the project experts and project staff. The PM will also closely coordinate the project activities with the relevant government and other institutions and consult regularly with the project stakeholders.

In addition, Project Assistants (PAs) will be recruited for each focus sector to support the PM on administrative and financial issues. Other administrative staffs will also be provided to the SE4ALL Secretariat.

The Secretariat will be supported by a Technical Advisory Committee. This committee will include high level technical experts from key ministries, Universities and institutions which are members of the PSC. The PM may be supported by short-term international and national experts/consultants who will assist in implementation of specific technical assistance components of the programme.

Finally, the UNDP Country Office and AfDB will provide specific support services for proper project implementation, as required, through its administrative, programme and finance units. Specific support services will include support for annual project implementation review (PIR)), mid-term review and terminal evaluation.

Project implementation will be governed by the provisions of the present programme document. UNDP will maintain oversight and management of the overall project budget, utilizing a direct payment modality. Governance of the Project will be supported through annual work planning as well as reporting and monitoring the delivery of results and impact based on the results framework. The annual work plans as well as progress reporting will be the responsibility of the PM and will be approved by the NPC in close consultation with UNDP and AfDB.

The SE4ALL Secretariat will be the lead advocacy and managerial unit that:

- Monitors SE4ALL related projects and studies which are under the responsibility of the relevant government agencies and institutions.
- Takes the responsibility for realising SE4ALL goals, implementing the Investment Prospectus (IP) associated to AA, and applying the SE4ALL MER mechanism.
- Acts as the focal point for exchanging information with the SE4ALL global initiative, especially with the SE4ALLGlobal Facility Team, the SE4ALL Africa Hub and the SE4ALL Thematic Hubs.
- Presents draft modifications, to be reviewed by the SE4ALL PSC and GoT authorities to the AA and proposes any other “mid-course” adjustments that may be needed to keep Tanzania on its path to achieve its SE4ALL goals.
- Creates awareness on the focus sectors

Improved institutional cooperation for efficient RE service delivery will be the vision of SE4ALL Secretariat. To achieve this, SE4ALL Secretariat will conduct high level institutional and technical experts’ meetings on quarterly basis. It will work continuously for harmonization of policies and strategies related to RE energy. It will also ensure frequent sharing of latest information through TREMISRE portal.

IX. LEGAL CONTEXT

*[NOTE: Please choose **one** of the following options, as applicable. Delete all other options from the document]*

Option a. Where the country has signed the Standard Basic Assistance Agreement (SBAA)
This project document shall be the instrument referred to as such in Article 1 of the Standard Basic Assistance Agreement between the Government of (country) and UNDP, signed on (date). All references in the SBAA to "Executing Agency" shall be deemed to refer to "Implementing Partner."

This project will be implemented by [name of entity] ("Implementing Partner") in accordance with its financial regulations, rules, practices and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. Where the financial governance of an Implementing Partner does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition, the financial governance of UNDP shall apply.

Option b. Where the country has NOT signed the Standard Basic Assistance Agreement (SBAA)

The project document shall be the instrument envisaged and defined in the Supplemental Provisions to the Project Document, attached hereto and forming an integral part hereof, as "the Project Document".

This project will be implemented by [name of entity] ("Implementing Partner") in accordance with its financial regulations, rules, practices and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. Where the financial governance of an Implementing Partner does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition, the financial governance of UNDP shall apply.

Option c. For Global and Regional Projects
This project forms part of an overall programmatic framework under which several separate associated country level activities will be implemented. When assistance and support services are provided from this Project to the associated country level activities, this document shall be the "Project Document" instrument referred to in: (i) the respective signed SBAAs for the specific countries; or (ii) in the Supplemental Provisions to the Project Document attached to the Project Document in cases where the recipient country has not signed an SBAA with UNDP, attached hereto and forming an integral part hereof. All references in the SBAA to "Executing Agency" shall be deemed to refer to "Implementing Partner."

This project will be implemented by [name of entity] ("Implementing Partner") in accordance with its financial regulations, rules, practices and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. Where the financial governance of an Implementing Partner does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition, the financial governance of UNDP shall apply**.**

X. RISK MANAGEMENT

*[NOTE: Please choose **one** of the following options that corresponds to the implementation modality of the Project. Delete all other options.]*

Option a. Government Entity (NIM)

1. Consistent with the Article III of the SBAA *[or the Supplemental Provisions to the Project Document]*, the responsibility for the safety and security of the Implementing Partner and its personnel and property, and of UNDP's property in the Implementing Partner's custody, rests with the Implementing Partner. To this end, the Implementing Partner shall:
   a) put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;
   b) assume all risks and liabilities related to the Implementing Partner's security, and the full implementation of the security plan.

2. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the Implementing Partner's obligations under this Project Document.

3. The Implementing Partner agrees to undertake all reasonable efforts to ensure that no UNDP funds received pursuant to the Project Document are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the list maintained by the Security Council Committee established pursuant to resolution 1267 (1999). The list can be accessed via http://www.un.org/sc/committees/1267/aq_sanctions_list.shtml.

4. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

5. The Implementing Partner shall: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

6. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

7. The Implementing Partner will take appropriate steps to prevent misuse of funds, fraud or corruption, by its officials, consultants, responsible parties, subcontractors and sub-recipients in implementing the project or using UNDP funds. The Implementing Partner will ensure that its financial management, anti-corruption and anti-fraud policies are in place and enforced for all funding received from or through UNDP.

8. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to the Implementing Partner: (a) UNDP Policy on Fraud and other Corrupt Practices and (b) UNDP Office of Audit and Investigations Investigation Guidelines. The Implementing Partner agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

9. In the event that an investigation is required, UNDP has the obligation to conduct investigations relating to any aspect of UNDP projects and programmes. The Implementing Partner shall provide its full cooperation, including making available personnel, relevant documentation, and granting access to the Implementing Partner's (and its consultants', responsible parties', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with the Implementing Partner to find a solution.

10. The signatories to this Project Document will promptly inform one another in case of any incidence of inappropriate use of funds, or credible allegation of fraud or corruption with due confidentiality.

    Where the Implementing Partner becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, the Implementing Partner will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). The Implementing Partner shall provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

11. *Choose one of the three following options:*

    Option 1: UNDP shall be entitled to a refund from the Implementing Partner of any funds provided that have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document. Such amount may be deducted by UNDP from any payment due to the Implementing Partner under this or any other agreement. Recovery of such amount by UNDP shall not diminish or curtail the Implementing Partner's obligations under this Project Document.

    *Option 2:*The Implementing Partner agrees that, where applicable, donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities which are the subject of this Project Document, may seek recourse to the Implementing Partner for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

    Option 3: UNDP shall be entitled to a refund from the Implementing Partner of any funds provided that have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document. Such amount may be deducted by UNDP from any payment due to the Implementing Partner under this or any other agreement.

    Where such funds have not been refunded to UNDP, the Implementing Partner agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to the Implementing Partner for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

    Note: The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, subcontractors and sub-recipients.

12. Each contract issued by the Implementing Partner in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in contract execution, and that the recipient of funds from the Implementing Partner shall cooperate with any and all investigations and post-payment audits.

13. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP.

14. The Implementing Partner shall ensure that all of its obligations set forth under this section entitled "Risk Management" are passed on to each responsible party, subcontractor and sub-recipient and that all the clauses under this section entitled "Risk Management Standard Clauses" are included, *mutatis mutandis*, in all sub-contracts or sub-agreements entered into further to this Project Document.

Option b. UNDP (DIM)

1. UNDP as the Implementing Partner will comply with the policies, procedures and practices of the United Nations Security Management System (UNSMS.)

2. UNDP as the Implementing Partner will undertake all reasonable efforts to ensure that none of the [project funds][41] [UNDP funds received pursuant to the Project Document][42] are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the list maintained by the Security Council Committee established pursuant to resolution 1267 (1999). The list can be accessed via http://www.un.org/sc/committees/1267/aq_sanctions_list.shtml. This provision must be included in all sub-contracts or sub-agreements entered into under this Project Document.

3. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

4. UNDP as the Implementing Partner will: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

5. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

6. UNDP as the Implementing Partner will ensure that the following obligations are binding on each responsible party, subcontractor and sub-recipient:

a. Consistent with the Article III of the SBAA *[or the Supplemental Provisions to the Project Document]*, the responsibility for the safety and security of each responsible party, subcontractor and sub-recipient and its personnel and property, and of UNDP's property in such responsible party's, subcontractor's and sub-recipient's custody, rests with such responsible party, subcontractor and sub-recipient. To this end, each responsible party, subcontractor and sub-recipient shall:
   i. put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;
   ii. assume all risks and liabilities related to such responsible party's, subcontractor's and sub-recipient's security, and the full implementation of the security plan.

[41] To be used where UNDP is the Implementing Partner

[42] To be used where the UN, a UN fund/programme or a specialized agency is the Implementing Partner

b. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the responsible party's, subcontractor's and sub-recipient's obligations under this Project Document.

c. Each responsible party, subcontractor and sub-recipient will take appropriate steps to prevent misuse of funds, fraud or corruption, by its officials, consultants, subcontractors and sub-recipients in implementing the project or programme or using the UNDP funds. It will ensure that its financial management, anti-corruption and anti-fraud policies are in place and enforced for all funding received from or through UNDP.

d. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to each responsible party, subcontractor and sub-recipient: (a)UNDP Policy on Fraud and other Corrupt Practices and (b)UNDP Office of Audit and Investigations Investigation Guidelines. Each responsible party, subcontractor and sub-recipient agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

e. In the event that an investigation is required, UNDP will conduct investigations relating to any aspect of UNDP programmes and projects. Each responsible party, subcontractor and sub-recipient will provide its full cooperation, including making available personnel, relevant documentation, and granting access to its (and its consultants', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with it to find a solution.

f. Each responsible party, subcontractor and sub-recipient will promptly inform UNDP as the Implementing Partner in case of any incidence of inappropriate use of funds, or credible allegation of fraud or corruption with due confidentiality.

   Where it becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, each responsible party, subcontractor and sub-recipient will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). It will provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

g. Choose one of the three following options:

   Option 1: UNDP will be entitled to a refund from the responsible party, subcontractor or sub-recipient of any funds provided that have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of this Project Document. Such amount may be deducted by UNDP from any payment due to the responsible party, subcontractor or sub-recipient under this or any other agreement. Recovery of such amount by UNDP shall not diminish or curtail any responsible party's, subcontractor's or sub-recipient's obligations under this Project Document.

   *Option 2:*Eachresponsible party, subcontractor or sub-recipient agrees that, where applicable, donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities which are the subject of the Project Document, may seek recourse to such responsible party, subcontractor or sub-recipient for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

   Option 3: UNDP will be entitled to a refund from the responsible party, subcontractor or sub-recipient of any funds provided that have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document. Such amount may be deducted by UNDP from any payment due to the responsible party, subcontractor or sub-recipient under this or any other agreement.

Where such funds have not been refunded to UNDP, the responsible party, subcontractor or sub-recipient agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to such responsible party, subcontractor or sub-recipient for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

Note: The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, subcontractors and sub-recipients.

h. Each contract issued by the responsible party, subcontractor or sub-recipient in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in contract execution, and that the recipient of funds from it shall cooperate with any and all investigations and post-payment audits.

i. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project or programme, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP.

j. Each responsible party, subcontractor and sub-recipient shall ensure that all of its obligations set forth under this section entitled "Risk Management" are passed on to its subcontractors and sub-recipients and that all the clauses under this section entitled "Risk Management Standard Clauses" are adequately reflected, *mutatis mutandis*, in all its sub-contracts or sub-agreements entered into further to this Project Document.

Option c. CSO/NGO/Non-UN or other IGO with no signed SBEAA with UNDP

1. Consistent with the Article III of the SBAA *[or the Supplemental Provisions to the Project Document]*, the responsibility for the safety and security of the Implementing Partner and its personnel and property, and of UNDP's property in the Implementing Partner's custody, rests with the Implementing Partner. To this end, the Implementing Partner shall:
   a) put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;
   b) assume all risks and liabilities related to the Implementing Partner's security, and the full implementation of the security plan.

2. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the Implementing Partner's obligations under this Project Document and the Project Cooperation Agreement between UNDP and the Implementing Partner[43].

3. The Implementing Partner agrees to undertake all reasonable efforts to ensure that no UNDP funds received pursuant to the Project Document are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the list maintained by the Security Council Committee established pursuant to resolution 1267 (1999). The list can be accessed via http://www.un.org/sc/committees/1267/aq_sanctions_list.shtml.

[43] Use bracketed text only when IP is an NGO/IGO

4. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

5. The Implementing Partner shall: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

6. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

7. The Implementing Partner will take appropriate steps to prevent misuse of funds, fraud or corruption, by its officials, consultants, responsible parties, subcontractors and sub-recipients in implementing the project or using the UNDP funds. The Implementing Partner will ensure that its financial management, anti-corruption and anti-fraud policies are in place and enforced for all funding received from or through UNDP.

8. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to the Implementing Partner: (a) UNDP Policy on Fraud and other Corrupt Practices and (b) UNDP Office of Audit and Investigations Investigation Guidelines. The Implementing Partner agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

9. In the event that an investigation is required, UNDP has the obligation to conduct investigations relating to any aspect of UNDP programmes and projects. The Implementing Partner shall provide its full cooperation, including making available personnel, relevant documentation, and granting access to the Implementing Partner's (and its consultants', responsible parties', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with the Implementing Partner to find a solution.

10. The Implementing Partner will promptly inform UNDP in case of any incidence of inappropriate use of funds, or credible allegation of fraud or corruption with due confidentiality.

    Where the Implementing Partner becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, the Implementing Partner will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). The Implementing Partner shall provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

11. Choose one of the three following options:

    Option 1: UNDP shall be entitled to a refund from the Implementing Partner of any funds provided that have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of this Project Document. Such amount may be deducted by UNDP from any payment due to the Implementing Partner under this or any other agreement. Recovery of such amount by UNDP shall not diminish or curtail the Implementing Partner's obligations under this Project Document.

    *Option 2:*The Implementing Partner agrees that, where applicable, donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities

which are the subject of the Project Document, may seek recourse to the Implementing Partner for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

Option 3: UNDP shall be entitled to a refund from the Implementing Partner of any funds provided that have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document. Such amount may be deducted by UNDP from any payment due to the Implementing Partner under this or any other agreement.

Where such funds have not been refunded to UNDP, the Implementing Partner agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to the Implementing Partner for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

Note: The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with the Implementing Partner, responsible parties, subcontractors and sub-recipients.

12. Each contract issued by the Implementing Partner in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in contract execution, and that the recipient of funds from the Implementing Partner shall cooperate with any and all investigations and post-payment audits.

13. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP.

14. The Implementing Partner shall ensure that all of its obligations set forth under this section entitled "Risk Management Standard Clauses" are passed on to each responsible party, subcontractor and sub-recipient and that all the clauses under this section entitled "Risk Management" are included, *mutatis mutandis*, in all sub-contracts or sub-agreements entered into further to this Project Document.

Option d. UN Agency other than UNDP, and IGO with signed SBEAA with UNDP

1. [Name of UN Agency/IGO] as the Implementing Partner will comply with the policies, procedures and practices of the United Nations Security Management System (UNSMS.)

2. [Name of UN Agency/IGO] as the Implementing Partner will ensure that the following obligations are binding on each responsible party, subcontractor and sub-recipient that is not a UN entity:

    a. Consistent with the Article III of the SBAA *[or the Supplemental Provisions to the Project Document]*, the responsibility for the safety and security of each responsible party, subcontractor and sub-recipient and its personnel and property, and of [Name of UN Agency/IGO]'s property in such responsible party's, subcontractor's and sub-recipient's custody, rests with such responsible party, subcontractor and sub-recipient. To this end, each responsible party, subcontractor and sub-recipient shall:

        i. put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;

ii. assume all risks and liabilities related to such responsible party's, subcontractor's and sub-recipient's security, and the full implementation of the security plan.

b. [Name of UN Agency/IGO] reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the responsible party's, subcontractor's and sub-recipient's obligations under this Project Document.

3. [Name of UN Agency/IGO] agrees to undertake all reasonable efforts to ensure that none of the [project funds][44] [UNDP funds received pursuant to the Project Document][45] are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the list maintained by the Security Council Committee established pursuant to resolution 1267 (1999). The list can be accessed via: http://www.un.org/sc/committees/1267/aq_sanctions_list.shtml.

4. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

5. The Implementing Partner shall: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

6. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

7. The Implementing Partner will take appropriate steps to prevent misuse of funds, fraud or corruption, by its officials, consultants, responsible parties, subcontractors and sub-recipients in implementing the project or programme or using the UNDP funds. The Implementing Partner will ensure that its financial management, anti-corruption and anti-fraud policies are in place and enforced for all funding received from or through UNDP.

8. [*This text should be included when the Implementing Partner is a non-UN IGO*: The requirements of the following documents, then in force at the time of signature of the Project Document, apply to the Implementing Partner: (a)UNDP Policy on Fraud and other Corrupt Practices and (b)UNDP Office of Audit and Investigations Investigation Guidelines. The Implementing Partner agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.]

9. [*This text should be included when the Implementing Partner is a non-UN IGO*: In the event that an investigation is required, UNDP has the obligation to conduct investigations relating to any aspect of UNDP projects or programmes. The Implementing Partner shall provide its full cooperation, including making available personnel, relevant documentation, and granting access to the Implementing Partner's (and its consultants', responsible parties', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with the Implementing Partner to find a solution.]

10. The Implementing Partner and UNDP will promptly inform one another in case of any incidence of inappropriate use of funds, or credible allegation of fraud or corruption with due confidentiality.

[44] To be used where UNDP is the Implementing Partner

[45] To be used where the UN, a UN fund/programme or a specialized agency is the Implementing Partner

Where the Implementing Partner becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, the Implementing Partner will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). The Implementing Partner shall provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

11. *Choose one of the three following options:*

    Option 1: UNDP shall be entitled to a refund from the Implementing Partner of any funds provided that have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of this Project Document. Such amount may be deducted by UNDP from any payment due to the Implementing Partner under this or any other agreement. Recovery of such amount by UNDP shall not diminish or curtail the Implementing Partner's obligations under this Project Document.

    *Option 2:*The Implementing Partner agrees that, where applicable, donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities which are the subject of the Project Document, may seek recourse to the Implementing Partner for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

    Option 3: UNDP shall be entitled to a refund from the Implementing Partner of any funds provided that have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document. Such amount may be deducted by UNDP from any payment due to the Implementing Partner under this or any other agreement.

    Where such funds have not been refunded to UNDP, the Implementing Partner agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to the Implementing Partner for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

    Note: The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, subcontractors and sub-recipients.

12. Each contract issued by the Implementing Partner in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in contract execution, and that the recipient of funds from the Implementing Partner shall cooperate with any and all investigations and post-payment audits.

13. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP

14. The Implementing Partner shall ensure that all of its obligations set forth under this section entitled "Risk Management Standard Clauses" are passed on to each responsible party, subcontractor and sub-recipient and that all the clauses under this section entitled "Risk Management" are included, *mutatis mutandis*, in all sub-contracts or sub-agreements entered into further to this Project Document.

ANNEXES

1. **Project Quality Assurance Report**

2. **Social and Environmental Screening Template:** including additional Social and Environmental Assessments or Management Plans as relevant.*(NOTE: The SES Screening is not required for projects in which UNDP is Administrative Agent only and/or projects comprised solely of reports, coordination of events, trainings, workshops, meetings, conferences, preparation of communication materials, strengthening capacities of partners to participate in international negotiations and conferences, partnership coordination and management of networks, or global/regional projects with no country level activities).*

3. **Risk Analysis**. Use the standard Risk Log template. Please refer to the Deliverable Description of the Risk Log for instructions

4. **Capacity Assessment:** Results of capacity assessments of Implementing Partner (including HACT Micro Assessment)

5. **TORs for Project Board and key management positions**

6. **Information on installed power plant capacity in Tanzania**

7. **Information on cooking practices in Tanzania**

8. **Focus sectors, gaps and scope of interventions**

9. **Resource mobilisation**

10. **GHG emission reduction calculation sheet**

11. **Tanzania SE4ALL - Baseline, present projects and gaps estimation sheet**

PROJECT QA ASSESSMENT: DESIGN AND APPRAISAL

OVERALL PROJECT

EXEMPLARY (5) ◎◎◎◎◎	HIGHLY SATISFACTORY (4) ◎◎◎◎○	SATISFACTORY (3) ◎◎◎○○	NEEDS IMPROVEMENT (2) ◎◎○○○	INADEQUATE (1) ◎○○○○
At least four criteria are rated Exemplary, and all criteria are rated High or Exemplary.	All criteria are rated Satisfactory or higher, and at least four criteria are rated High or Exemplary.	At least six criteria are rated Satisfactory or higher, and only one may be rated Needs Improvement. The SES criterion must be rated Satisfactory or above.	At least three criteria are rated Satisfactory or higher, and only four criteria may be rated Needs Improvement.	One or more criteria are rated Inadequate, or five or more criteria are rated Needs Improvement.

DECISION

- **APPROVE** – the project is of sufficient quality to continue as planned**.** Any management actions must be addressed in a timely manner.
- **APPROVE WITH QUALIFICATIONS** – the project has issues that must be addressed before the project document can be approved. Any management actions must be addressed in a timely manner.
- **DISAPPROVE** – the project has significant issues that should prevent the project from being approved as drafted.

RATING CRITERIA

STRATEGIC

1. Does the project's Theory of Change specify how it will contribute to higher level change? (Select the option from 1-3 that best reflects the project):

- **3:** The project has a theory of change with explicit assumptions and clear change pathway describing how the project will contribute to outcome level change as specified in the programme/CPD, backed by credible evidence of what works effectively in this context. The project document clearly describes why the project's strategy is the best approach at this point in time.
- **2:** The project has a theory of change. It has an explicit change pathway that explains how the project intends to contribute to outcome-level change and why the project strategy is the best approach at this point in time, but is backed by limited evidence.
- **1:** The project does not have a theory of change, but the project document may describe in generic terms how the project will contribute to development results, without specifying the key assumptions. It does not make an explicit link to the programme/CPD's theory of change.

*Note: Management Action or strong management justification must be given for a score of 1

3	2
1	

Evidence

Theory of change is explained in Section II of the Prodoc

3	2
1	

2. Is the project aligned with the thematic focus of the UNDP Strategic Plan? (select the option from 1-3 that best reflects the project): • **3:** The project responds to one of the three areas of development work[1] as specified in the Strategic Plan; it addresses at least one of the proposed new and emerging areas[2]; an issues-. based analysis has been incorporated into the project design; and the project's RRF includes all the relevant SP output indicators *(all must be true to select this option)* • **2:** The project responds to one of the three areas of development work[1] as specified in the Strategic Plan. The project's RRF includes at least one SP output indicator, if relevant. *(both must be true to select this option)* • **1:** While the project may respond to one of the three areas of development work[1] as specified in the Strategic Plan, it is based on a sectoral approach without addressing the complexity of the development issue. None of the relevant SP indicators are included in the RRF. This answer is also selected if the project does not respond to any of the three areas of development work in the Strategic Plan.	**Evidence** The project responses to two of the three areas of development works, namely Sustainable development pathways and 2. Inclusive and effective democratic governance. It addresses issues faced in at least 5 of the proposed new and emerging areas, namely sustainable production technologies, access to modern energy services and energy efficiency, natural resources management, extractive industries and urbanization. An issues –based analysis is included into the project design. The Project RRF are in line with the relevant Strategy Plan output indictors. The results framework is provided in section V of Prodoc.
RELEVANT	
3. Does the project have strategies to effectively identify, engage and ensure the meaningful participation of targeted groups/geographic areas with a priority focus on the excluded and marginalized? (select the option from 1-3 that best reflects this project): • **3:** The target groups/geographic areas are appropriately specified, prioritising the excluded and/or marginalised. Beneficiaries will be identified through a rigorous process based on evidence (if applicable.)The project has an explicit strategy to identify, engage and ensure the meaningful participation of specified target groups/geographic areas throughout the project, including through monitoring and decision-making (such as representation on the project board) *(all must be true to select this option)* • **2:** The target groups/geographic areas are appropriately specified, prioritising the excluded and/or marginalised. The project document states how beneficiaries will be identified, engaged and how meaningful participation will be ensured throughout the project. *(both must be true to select this option)* • **1:** The target groups/geographic areas are not specified, or do not prioritize excluded and/or marginalised populations. The project does not have a written strategy to identify or engage or ensure the meaningful participation of the target groups/geographic areas throughout the project.	3 | **2** | 1 **Evidence** The details of potential target groups and the strategy for their effective engagement and participation is provided under section Stakeholder engagement in Prodoc.

[1] 1. Sustainable development pathways; 2. Inclusive and effective democratic governance; 3. Resilience building

[2] sustainable production technologies, access to modern energy services and energy efficiency, natural resources management, extractive industries, urbanization, citizen security, social protection, and risk management for resilience

*Note: Management Action must be taken for a score of 1, or select not applicable.	
4. Have knowledge, good practices, and past lessons learned of UNDP and others informed the project design? (select the option from 1-3 that best reflects this project): • **3:** Knowledge and lessons learned (gained e.g. through peer assist sessions) backed by credible evidence from evaluation, corporate policies/strategies, and monitoring have been explicitly used, with appropriate referencing, to develop the project's theory of change and justify the approach used by the project over alternatives. • **2:** The project design mentions knowledge and lessons learned backed by evidence/sources, which inform the project's theory of change but have not been used/are not sufficient to justify the approach selected over alternatives. • **1:** There is only scant or no mention of knowledge and lessons learned informing the project design. Any references that are made are not backed by evidence. *Note: Management Action or strong management justification must be given for a score of 1	3 **2** 1 **Evidence** The knowledge, good practices and lessons learnt from previous work by UNDP and others are explained in Section II of the Prodoc.
5. Does the project use gender analysis in the project design and does the project respond to this gender analysis with concrete measures to address gender inequities and empower women? (select the option from 1-3 that best reflects this project): • **3:** A participatory gender analysis on the project has been conducted. This analysis reflects on the different needs, roles and access to/control over resources of women and men, and it is fully integrated into the project document. The project establishes concrete priorities to address gender inequalities in its strategy. The results framework includes outputs and activities that specifically respond to this gender analysis, with indicators that measure and monitor results contributing to gender equality. *(all must be true to select this option)* • **2:** A gender analysis on the project has been conducted. This analysis reflects on the different needs, roles and access to/control over resources of women and men. Gender concerns are integrated in the development challenge and strategy sections of the project document. The results framework includes outputs and activities that specifically respond to this gender analysis, with indicators that measure and monitor results contributing to gender equality. *(all must be true to select this option)* • **1:** The project design may or may not mention information and/or data on the differential impact of the project's development situation on gender relations, women and men, but the constraints have not been clearly identified and interventions have not been considered. *Note: Management Action or strong management justification must be given for a score of 1	3 **2** 1 **Evidence** The necessary actions taken towards gender mainstreaming are explained under subtitle "Gender mainstreaming" in section III of the Prodoc.
	3 2 1

6. Does UNDP have a clear advantage to engage in the role envisioned by the project vis-à-vis national partners, other development partners, and other actors? (select from options 1-3 that best reflects this project): • **3:** An analysis has been conducted on the role of other partners in the area where the project intends to work, and credible evidence supports the proposed engagement of UNDP and partners through the project. It is clear how results achieved by relevant partners will contribute to outcome level change complementing the project's intended results. If relevant, options for south-south and triangular cooperation have been considered, as appropriate. *(all must be true to select this option)* • **2:** Some analysis has been conducted on the role of other partners where the project intends to work, and relatively limited evidence supports the proposed engagement of and division of labour between UNDP and partners through the project. Options for south-south and triangular cooperation may not have not been fully developed during project design, even if relevant opportunities have been identified. • **1:** No clear analysis has been conducted on the role of other partners in the area that the project intends to work, and relatively limited evidence supports the proposed engagement of UNDP and partners through the project. There is risk that the project overlaps and/or does not coordinate with partners' interventions in this area. Options for south-south and triangular cooperation have not been considered, despite its potential relevance. *Note: Management Action or strong management justification must be given for a score of 1	**Evidence** The list of partners and their roles under this programme is clearly listed under subtitle "Partnerships" in section III of the Prodoc. Also, UNDP has been in forefront in implementing sustainable energy projects in developing countries, Hence, it has clear advantage to engage in the role envisaged in the SE4ALL programme.
SOCIAL & ENVIRONMENTAL STANDARDS	
7. Does the project seek to further the realization of human rights using a human rights based approach? (select from options 1-3 that best reflects this project): • **3:** Credible evidence that the project aims to further the realization of human rights, upholding the relevant international and national laws and standards in the area of the project. Any potential adverse impacts on enjoyment of human rights were rigorously identified and assessed as relevant, with appropriate mitigation and management measures incorporated into project design and budget. *(all must be true to select this option)* • **2:** Some evidence that the project aims to further the realization of human rights. Potential adverse impacts on enjoyment of human rights were identified and assessed as relevant, and appropriate mitigation and management measures incorporated into the project design and budget. • **1:** No evidence that the project aims to further the realization of human rights. Limited or no evidence that potential adverse impacts on enjoyment of human rights were considered. *Note: Management action or strong management justification must be given for a score of 1	3 | 2 1 **Evidence** No direct plans are envisaged under the programme for further realization of human rights. However, the programme implementation approach will align with international human rights and labor guidelines in all its activities.
8. Did the project consider potential environmental opportunities and adverse impacts, applying a precautionary approach? (select from options 1-3 that best reflects this project): • **3:** Credible evidence that opportunities to enhance environmental sustainability and integrate poverty-environment linkages were fully considered as relevant, and integrated in project strategy and design. Credible evidence that potential adverse environmental impacts have been identified and rigorously assessed with appropriate management and mitigation measures incorporated into project design and budget. *(all must be true to select this option)*.	3 | 2 1 **Evidence** The programme envisages to reduce poverty levels in Tanzania through improved access to sustainable energy. Prevention of exploitation of natural resources and environment is an important aspect in providing sustainable energy. One direct

• **2:** No evidence that opportunities to strengthen environmental sustainability and poverty-environment linkages were considered. Credible evidence that potential adverse environmental impacts have been identified and assessed, if relevant, and appropriate management and mitigation measures incorporated into project design and budget. • **1:** No evidence that opportunities to strengthen environmental sustainability and poverty-environment linkages were considered. Limited or no evidence that potential adverse environmental impacts were adequately considered. *Note: Management action or strong management justification must be given for a score of 1	approach considered in Programme design is the sustainable energy plantation and controlled harvesting for production of charcoal. Indirectly, the access to sustainable energy reduces use of fossil fuels and thus leads to reduction of GHG emissions. Thus, the programme helps to enhance environmental sustainability while reducing the poverty levels.
9. Has the Social and Environmental Screening Procedure (SESP) been conducted to identify potential social and environmental impacts and risks? The SESP is not required for projects in which UNDP is Administrative Agent only and/or projects comprised solely of reports, coordination of events, trainings, workshops, meetings, conferences and/or communication materials and information dissemination. [if yes, upload the completed checklist. If SESP is not required, provide the reason for the exemption in the evidence section.]	Yes / No SESP Not Required
MANAGEMENT & MONITORING	
10. Does the project have a strong results framework? (select from options 1-3 that best reflects this project): • **3:** The project's selection of outputs and activities are at an appropriate level and relate in a clear way to the project's theory of change. Outputs are accompanied by SMART, results-oriented indicators that measure all of the key expected changes identified in the theory of change, each with credible data sources, and populated baselines and targets, including gender sensitive, sex-disaggregated indicators where appropriate. *(all must be true to select this option)* • **2:** The project's selection of outputs and activities are at an appropriate level, but may not cover all aspects of the project's theory of change. Outputs are accompanied by SMART, results-oriented indicators, but baselines, targets and data sources may not yet be fully specified. Some use of gender sensitive, sex-disaggregated indicators, as appropriate. *(all must be true to select this option)* • **1:** The results framework does not meet all of the conditions specified in selection "2" above. This includes: the project's selection of outputs and activities are not at an appropriate level and do not relate in a clear way to the project's theory of change; outputs are not accompanied by SMART, results-oriented indicators that measure the expected change, and have not been populated with baselines and targets; data sources are not specified, and/or no gender sensitive, sex-disaggregation of indicators. *Note: Management Action or strong management justification must be given for a score of 1	3 / 2 / 1 **Evidence** The project results framework with indicators in each output is provided in section 5 of Prodoc.
11. Is there a comprehensive and costed M&E plan in place with specified data collection sources and methods to support evidence-based management, monitoring and evaluation of the project?	Yes (3) / No (1)
	3 / 2 / 1

12. Is the project's governance mechanism clearly defined in the project document, including planned composition of the project board? (select from options 1-3 that best reflects this project): • **3:** The project's governance mechanism is fully defined in the project composition. Individuals have been specified for each position in the governance mechanism (especially all members of the project board.) Project Board members have agreed on their roles and responsibilities as specified in the terms of reference. The ToR of the project board has been attached to the project document. *(all must be true to select this option)*. • **2:** The project's governance mechanism is defined in the project document; specific institutions are noted as holding key governance roles, but individuals may not have been specified yet. The prodoc lists the most important responsibilities of the project board, project director/manager and quality assurance roles. *(all must be true to select this option)* • **1:** The project's governance mechanism is loosely defined in the project document, only mentioning key roles that will need to be filled at a later date. No information on the responsibilities of key positions in the governance mechanism is provided. *Note: Management Action or strong management justification must be given for a score of 1		**Evidence** The details of project management are explained under section VIII of the Prodoc. Project Board Terms of Reference and TORs of key management positions are provided in Annex 5.
13. Have the project risks been identified with clear plans stated to manage and mitigate each risks? (select from options 1-3 that best reflects this project): • **3:** Project risks related to the achievement of results are fully described in the project risk log, based on comprehensive analysis drawing on the theory of change, Social and Environmental Standards and screening, situation analysis, capacity assessments and other analysis. Clear and complete plan in place to manage and mitigate each risk. *(both must be true to select this option)* • **2:** Project risks related to the achievement of results identified in the initial project risk log with mitigation measures identified for each risk. • **1:** Some risks may be identified in the initial project risk log, but no evidence of analysis and no clear risk mitigation measures identified. This option is also selected if risks are not clearly identified and no initial risk log is included with the project document. *Note: Management Action must be taken for a score of 1	3 2 1	**Evidence** The risks identified and appropriate mitigation measures proposed are explained under sub-title "Risks and Assumptions" in section III of Prodoc.
EFFICIENT		
14. Have specific measures for ensuring cost-efficient use of resources been explicitly mentioned as part of the project design? This can include: i) using the theory of change analysis to explore different options of achieving the maximum results with the resources available; ii) using a portfolio management approach to improve cost effectiveness through synergies with other interventions; iii) through joint operations (e.g., monitoring or procurement) with other partners.	Yes (3)	No (1)

15. Are explicit plans in place to ensure the project links up with other relevant on-going projects and initiatives, whether led by UNDP, national or other partners, to achieve more efficient results (including, for example, through sharing resources or coordinating delivery?)	Yes (3)	No (1)
16. Is the budget justified and supported with valid estimates? • **3:** The project's budget is at the activity level with funding sources, and is specified for the duration of the project period in a multi-year budget. Costs are supported with valid estimates using benchmarks from similar projects or activities. Cost implications from inflation and foreign exchange exposure have been estimated and incorporated in the budget. • **2:** The project's budget is at the activity level with funding sources, when possible, and is specified for the duration of the project in a multi-year budget. Costs are supported with valid estimates based on prevailing rates. • **1:** The project's budget is not specified at the activity level, and/or may not be captured in a multi-year budget.	3	2 1 **Evidence** The multi-year work plan and budget is provided in section VII of the Prodoc.
17. Is the Country Office fully recovering the costs involved with project implementation? • **3:** The budget fully covers all project costs that are attributable to the project, including programme management and development effectiveness services related to strategic country programme planning, quality assurance, pipeline development, policy advocacy services, finance, procurement, human resources, administration, issuance of contracts, security, travel, assets, general services, information and communications based on full costing in accordance with prevailing UNDP policies (i.e., UPL, LPL.) • **2:** The budget covers significant project costs that are attributable to the project based on prevailing UNDP policies (i.e., UPL, LPL) as relevant. • **1:** The budget does not adequately cover project costs that are attributable to the project, and UNDP is cross-subsidizing the project. *Note: Management Action must be given for a score of 1. The budget must be revised to fully reflect the costs of implementation before the project commences.	3	2 1 **Evidence** The multi-year work plan and budget for each activity is provided in section VII of the Prodoc.
EFFECTIVE		
18. Is the chosen implementation modality most appropriate? (select from options 1-3 that best reflects this project): • **3:** The required implementing partner assessments (capacity assessment, HACT micro assessment) have been conducted, and there is evidence that options for implementation modalities have been thoroughly considered. There is a strong justification for choosing the selected modality, based on the development context. *(both must be true to select this option)* • **2:** The required implementing partner assessments (capacity assessment, HACT micro assessment) have been conducted and the implementation modality chosen is consistent with the results of the assessments.	3	2 1 **Evidence** The capacity assessment of implementing partners is provided in Annex 4 of the Prodoc.

• **1:** The required assessments have not been conducted, but there may be evidence that options for implementation modalities have been considered. *Note: Management Action or strong management justification must be given for a score of 1		
19. Have targeted groups, prioritizing marginalized and excluded populations that will be affected by the project, been engaged in the design of the project in a way that addresses any underlying causes of exclusion and discrimination? • 3: Credible evidence that all targeted groups, prioritising marginalized and excluded populations that will be involved in or affected by the project, have been actively engaged in the design of the project. Their views, rights and any constraints have been analysed and incorporated into the root cause analysis of the theory of change which seeks to address any underlying causes of exclusion and discrimination and the selection of project interventions. • 2: Some evidence that key targeted groups, prioritising marginalized and excluded populations that will be involved in the project, have been engaged in the design of the project. Some evidence that their views, rights and any constraints have been analysed and incorporated into the root cause analysis of the theory of change and the selection of project interventions. • 1: No evidence of engagement with marginalized and excluded populations that will be involved in the project during project design. No evidence that the views, rights and constraints of populations have been incorporated into the project.	3	2 1 **Evidence** The details of potential target groups and the strategy for their effective engagement and participation is provided under section Stakeholder engagement in Prodoc.
20. Does the project conduct regular monitoring activities, have explicit plans for evaluation, and include other lesson learning (e.g. through After Action Reviews or Lessons Learned Workshops), timed to inform course corrections if needed during project implementation?	Yes (3)	No (1)
21. The gender marker for all project outputs are scored at GEN2 or GEN3, indicating that gender has been fully mainstreamed into all project outputs at a minimum. *Note: Management Action or strong management justification must be given for a score of "no"	Yes (3)	No (1) **Evidence** The necessary actions taken towards gender mainstreaming are explained under subtitle "Gender mainstreaming" in section III of the Prodoc. In line with above, all project outputs will be monitored for the gender mainstreaming targets.
22. Is there a realistic multi-year work plan and budget to ensure outputs are delivered on time and within allotted resources? (select from options 1-3 that best reflects this project): • **3:** The project has a realistic work plan & budget covering the duration of the project *at the activity* level to ensure outputs are delivered on time and within the allotted resources. • **2:** The project has a work plan & budget covering the duration of the project at the output level. • **1:** The project does not yet have a work plan & budget covering the duration of the project.	3	2 1 **Evidence** The multi-year work plan and budget for each activity is provided in section VII of the Prodoc.
SUSTAINABILITY & NATIONAL OWNERSHIP		
	3	2

<table>
<tr><td rowspan="2">23. Have national partners led, or proactively engaged in, the design of the project? (select from options 1-3 that best reflects this project):
• <u>3:</u> National partners have full ownership of the project and led the process of the development of the project jointly with UNDP.
• <u>2:</u> The project has been developed by UNDP in close consultation with national partners.
• <u>1:</u> The project has been developed by UNDP with limited or no engagement with national partners.</td><td colspan="2">1</td></tr>
<tr><td colspan="2">Evidence

The programme will be completely managed under SE4ALL Secretariat, which will be created under the authority of MEM. UNDP provides only administrative assistance for programme implementation. Thus the national partners have full ownership of the programme.</td></tr>
<tr><td rowspan="4">24. Are key institutions and systems identified, and is there a strategy for strengthening specific/ comprehensive capacities based on capacity assessments conducted? (select from options 0-4 that best reflects this project):
• <u>3:</u> The project has a comprehensive strategy for strengthening specific capacities of national institutions based on a systematic and detailed capacity assessment that has been completed. This strategy includes an approach to regularly monitor national capacities using clear indicators and rigorous methods of data collection, and adjust the strategy to strengthen national capacities accordingly.
• <u>2.5:</u> A capacity assessment has been completed. The project document has identified activities that will be undertaken to strengthen capacity of national institutions, but these activities are not part of a comprehensive strategy to monitor and strengthen national capacities.
• <u>2:</u> A capacity assessment is planned after the start of the project. There are plans to develop a strategy to strengthen specific capacities of national institutions based on the results of the capacity assessment.
• <u>1.5:</u> There is mention in the project document of capacities of national institutions to be strengthened through the project, but no capacity assessments or specific strategy development are planned.
• <u>1:</u> Capacity assessments have not been carried out and are not foreseen. There is no strategy for strengthening specific capacities of national institutions.</td><td>3</td><td>2.5</td></tr>
<tr><td>2</td><td>1.5</td></tr>
<tr><td colspan="2">1</td></tr>
<tr><td colspan="2">Evidence

Capacity development and knowledge management is one the key outputs under the programme design. The details on activities are provided in section III of Prodoc.</td></tr>
<tr><td>25. Is there is a clear strategy embedded in the project specifying how the project will use national systems (i.e., procurement, monitoring, evaluations, etc.,) to the extent possible?</td><td>Yes (3)</td><td>No (1)</td></tr>
<tr><td>26. Is there a clear transition arrangement/ phase-out plan developed with key stakeholders in order to sustain or scale up results (including resource mobilisation strategy)?</td><td>Yes (3)</td><td>No (1)</td></tr>
</table>

OFFLINE RISK LOG

(see Deliverable Description for the Risk Log regarding its purpose and use)

Project Title: Implementation of the sustainable energy for all (SE4ALL) initiative in Tanzania	**Award ID:** 00094384	**Date: 17.03.2017**

#	Description	Date Identified	Type	Impact & Probability	Counter measures / Mngt response	Owner	Submitted, updated by	Last Update	Status
1	Change of Government plans and priorities	When was the risk first identified *(In Atlas, select date. Note: date cannot be modified after initial entry)*	Political	The government might change its views on priorities which might affect attaining the proposed programme outcomes or the programme itself P = 1 I = 4	GoT has already signed a letter for implementation of SE4ALL initiative. If there is change in government plans and priorities, the Project Steering Committee (PSC) will discuss with the Government representatives and do the needful for continued support and co-operation from the Government for this programme.	MEM	Who submitted the risk *(In Atlas, automatically recorded)*	When was the status of the risk last checked *(In Atlas, automatically recorded)*	e.g. dead, reducing, increasing, no change *(in Atlas, use the Management Response box)*
2	Inadequate financial resources to fund all the project components		Financial	The project might not achieve the proposed level of implementation and sustainable development outcomes P= 1 I =4	During project inception, Project partners will submit their financial commitments for the project activities	SE4ALL Secretariat			
3	Changes in policies of the government towards the project		Political	The changes might alter the legal procedures and approvals to be obtained for the project implementation resulting in delay of project completion	PSC will convene regularly to identify adverse policy changes and propose measures to overcome any potential	MEM			

#	Description	Date Identified	Type	Impact & Probability	Counter measures / Mngt response	Owner	Submitted, updated by	Last Update	Status
	implementation			P= 2 I =3	implications				
4	Delay in Programme implementation due to operational constraints		Operational	The programme completion timeline might be delayed, resulting in additional financial requirements to compensate for the extended project operations and personnel requirements. P= 1 I =2	Prudent Project planning will be done to ensure timely delivery of project activities, including procurement of services and goods. Regular monitoring and corrective actions will be taken accordingly in all the activities.	SE4ALL Secretariat			
5	Medium to large scale natural disasters distressing the project implementation		Environmental	The timeline for the completion of power plant construction might be prolonged if some natural disasters occur during the initial stages and/or might result in damage to the completed structures if occurs at later stages. Also, it will result in additional financial requirements to overcome the losses in terms of personnel requirement, infrastructure requirement and increased implementation timeline. P= 1 I = 5	Feasibility studies for power plants will foresee all possible natural disasters and construction will be done considering necessary precautions for any natural disaster like flood or drought. If any unforeseen disaster happens, immediate action will be taken accordingly based on the decision of PSC	Project developers			

Terms of Reference for Project Board and key management positions

1. Project Steering Committee

Roles of PSC

- Provide overall strategic vision of the programme
- Approve programme design and strategy, including annual work plan, project deliverables and timeline
- Appoint National Project Coordinator (NPC) and Project Manager (PM)
- Authorise any major deviations from the agreed programme plans
- Communicate about the programme to relevant organisations and key stakeholders regularly & timely
- Ensure resource availability (i.e., grants, loans, human resource, etc.)
- Ensure effective decision making and communications
- Resolve risks and deviations associated with the programme
- Ensure quality assurance of the programme
- Ensure accountability of the project budget and avoid budget overruns
- Ensure monitoring of the programme at the time of each deliverables

Eligibility of members of PSC:

Qualified senior personnel with 15 years' experience from government ministries, institutions, departments, UNDP, International donors, private sectors, NGOs, women's groups and banks/financial institutions

2. Chairperson to PSC

Scope of work:

- Convene meeting of the PSC
- Present the agenda and lead the discussions
- Present annual project work plan to PSC
- Consolidate the views and concerns of members
- Convey decisions taken during the meetings to NPC
- Ensure consistent communications, discussion and meeting of PSC

Eligibility of Chairperson:

Qualified senior person from MEM with 20 years' experience in public administration, national level project management, stakeholder engagement, policy planning, etc.

3. National Project Coordinator (NPC)

Scope of work:

- Coordinate the project activities with other government ministries
- Provide directions to PM as per decisions taken by PSC
- Facilitate, monitor and report the procurement of inputs and delivery of outputs
- Approve the ToR for consultants and tender documents for sub-contracted assistance and inputs
- Certify the expenditures in line with approved budgets and work-plans
- Report to PSC on the project delivery and impact

Eligibility:

Qualified senior person from MEM with 20 years' experience in public administration, national level project management, stakeholder engagement, policy planning, etc.

4. Project Manager (PM)

Scope of work:

- Plan schedule and deliver the overall programme and its activities in accordance with the mission and goals
- Ensure operation of the program activities within the national policies and procedures
- Develop project monitoring and evaluation strategy to support the strategic direction of the programme
- Develop annual budget and operating plan to support the programme
- Develop program evaluation framework to assess the strengths of the programme and to identify areas for improvement
- Ensure accountability of the programme for continuous delivery of services
- Ensure effective communication and reporting with NPC and PSC
- Ensure effective time and conflict management
- Recruit, interview and select well-qualified programme staff
- Ensure proper maintenance of the programme documents
- Establish and implement a performance management process for all programme staff

Eligibility:

- Master degree with 15 years of experience in Renewable Energy/Sustainable Energy/Environment Engineering
- Minimum 10 years of experience in project management with proven record
- Excellent interpersonal and communication skills
- Strong knowledge on MS office applications

5. Project Assistant (PA)

Scope of work:

- Support PM in day to day activities of the programme
- Ensure timely delivery of the assigned tasks
- Prepare reports and presentations on the project progress and submit to PM
- Keep records of all information related to the project for documentation, clarification and presentation
- Maintain effective communication with project partners
- Make site visit to project implementation sites for monitoring
- Undertake any other duties as assigned by the PM

Eligibility:

- Master/Bachelor degree in Energy/Environment Engineering with 5 years of work experience
- Excellent skills with MS Office applications
- Excellent interpersonal, documentation and communication skills

6. Public Relation Officer (PRO)

Scope of work:

- Support PM in maintaining good relationship with the Public and stakeholders
- Assist determining the most effective ways of communication with stakeholders and beneficiaries of the programme
- Coordination support and facilitation of press releases, publications, etc.
- Prepare information media kits like journals, leaflets, miniatures, etc.
- Support the programme by raising awareness through media, publications, leaflets, social networking, etc.
- Monitor media, including newspapers, magazines, journals, broadcasts, social media and blogs for project outreach
- Assist formulating policies and procedures related to public information programs
- Support facilitating production of in house publicity and promotional materials, films and other video products and regulate their distribution
- Supervise & guide the programme staff effectively
- Undertake any other duties assigned by the PM

Eligibility:

- Master degree in Mass communication or Journalism from a recognized university
- 10 years of work experience in Public relations or Journalism
- Minimum 07 years of experience in project management with proven record
- Strong knowledge on media industry and social networking
- Excellent interpersonal and communication skills

7. Finance Officer

Scope of work:

- Be responsible for all financial decisions and reporting done in the programme
- Keep permanent records of grants, loans received, project disbursements, etc. on annual basis
- Keep accurate record of all receipts and payments authorized
- Monitor cash reserves and investments of the programme
- Maintain work order system for the programme
- Be responsible for financial transactions (e.g. cheques, electronic transfer, cash, etc.) of the programme
- Prepare monthly/quarterly/annual financial statement of the programme and submit to PM
- Undertake any other duties as assigned by the PM

Eligibility:

- Master degree in Finance or Commerce from a recognized university
- 10 years of work experience in finance with proven record
- Strong knowledge on accounting tools and Microsoft office applications
- Excellent interpersonal and communication skills

8. Information and Communication Technology (ICT) Officer

Scope of work:

- Monitor and maintain the computer systems and networks
- Support project staff in completion of day-to-day tasks
- Troubleshoot and diagnose hardware, software and network problems
- Support procurement of hardware, software and roll out of new applications/software
- Train programme staff on computer software and applications
- Create and maintain inventory, which may include hardware, software and various items such as laser printer cartridges and peripheral equipment.
- Maintain documentation of processes, procedures, and troubleshooting guides.
- To undertake any other duties assigned by the PM

Eligibility:

- Master/Bachelor degree in Computer science/Information Technology from a recognized university
- 5 years of work experience in ICT systems
- Excellent interpersonal and communication skills

Annex 6. Installed power plant capacity in Tanzania

The latest installation data of power plants in grid and mini-grid was requested from TANESCO during the stakeholder validation meetings. Accordingly, TANESCO had provided the below information in Jan 2017.

Table 1 provides the installed grid connected power plant capacity (Jan 2017).

Table 1. Installed grid power capacity

No.	Plant name	Capacity (MW)	Fuel Type	Ownership
1.	UBUNGO GAS PLANT - I	102.00	Gas	TANESCO
2.	TEGETA-GAS WATSILA	45.00	Gas	TANESCO
3.	UBUNGO GAS PLANT - II	129.00	Gas	TANESCO
4.	KINYEREZI I	150.00	Gas	TANESCO
5.	SONGAS	189.00	Gas	IPP
	Subtotal	**615.00**		
6.	KIDATU	204.00	Large Hydro	TANESCO
7.	KIHANSI	180.00	Large Hydro	TANESCO
8.	MTERA	80.00	Large Hydro	TANESCO
9.	HALE	21.00	Large Hydro	TANESCO
10.	NYUMBA YA MUNGU	8.00	Large Hydro	TANESCO
11.	NEW PANGANI FALLS	68.00	Large Hydro	TANESCO
	Subtotal	**561.00**		
12.	ZUZU DODOMA	7.40	IDO	TANESCO
13.	NYAKATO PLANT	63.00	HFO	TANESCO
14.	IPTL	103.00	HFO	IPP
	Subtotal	**173.40**		
15.	UWEMBA MIN HYDRO	0.84	Small Hydro	TANESCO

No.	Plant name	Capacity (MW)	Fuel Type	Ownership
16.	MWENGA MIN HYDRO	4.00	Small Hydro	SPP
17.	YOVI	0.95	Small Hydro	SPP
18.	DARAKUTA	0.32	Small Hydro	SPP
19.	TANWAT	1.50	Biomass	SPP
20.	TPC	9.00	Biomass	SPP
	Subtotal	**16.61**		
	Total	**1,366.01**		

Table 2 provides the installed mini-grid connected power plant capacity under TANESCO (Jan 2017).

Table 2. Installed mini-grid power capacity under TANESCO

No.	Plant name	Capacity (MW)	Fuel Type	Ownership
1.	BIHARAMULO	1.59	IDO	TANESCO
2.	BUKOBA	2.56	IDO	TANESCO
3.	KASULU	2.50	IDO	TANESCO
4.	KIBONDO	2.50	IDO	TANESCO
5.	KIGOMA	11.86	IDO	TANESCO
6.	LIWALE	0.85	IDO	TANESCO
7.	LOLIONDO	5.00	IDO	TANESCO
8.	LUDEWA	1.27	IDO	TANESCO
9.	MAFIA	2.13	IDO	TANESCO
10.	MBINGA	2.26	IDO	TANESCO
11.	MPANDA	2.98	IDO	TANESCO
12.	NGARA	0.95	IDO	TANESCO

No.	Plant name	Capacity (MW)	Fuel Type	Ownership
13.	SONGEA	8.33	IDO	TANESCO
14.	SUMBAWANGA	5.00	IDO	TANESCO
15.	TUNDURU	2.07	IDO	TANESCO
16.	MTWARA G/PLANT	18.00	Gas	TANESCO
17.	NAMTUMBO	0.34	GO	TANESCO
18.	SOMANGA G/PLANT	7.50	Gas	TANESCO
	Total	**77.69**		

Table 3 provides the installed mini-grid connected power plant capacity under TANESCO (Jan 2017).

Table 3. Installed mini-grid power capacity under IPP

No	Plant name	Capacity (MW)	Fuel type
1.	MATEMBWE (IKONDO)	0.50	Small Hydro
2.	TULILA	5.00	Small Hydro
3.	ANDOYA	0.50	Small Hydro
4.	NGOMBENI	2.50	Biomass
	Total	**8.50**	

The overall installed capacity considering grid and mini-grid installations is provided in table 4.

Table 4. Overall installed power plant capacity

No.	Power source	Grid power (MW)	Mini-grid power (MW)	Total (MW)	Percentage
1.	Natural gas	615.00	25.84	640.84	44
2.	Large hydro	561.00	0	561.00	38
3.	Other fossil fuel (HFO, IDO)	173.40	51.85	225.25	16
4.	Biomass power	10.50	2.50	13.00	1
5.	Small hydro	6.11	6.00	12.11	1
		1,366.01	**86.19**	**1,452.20**	**100**

Annex 7. Information on cooking practises in Tanzania

Table 1 provides a brief overview of energy consumption in a typical Tanzanian urban household.

Table 1: Share of different fuels in urban household cooking energy consumption (2012)[1]

No.	Primary cooking & heating energy	Percentage of population (%)	Number of households (millions)	Population (millions)
1	Charcoal	71	1.97	8.29
2	Firewood	20	0.56	2.34
3	Paraffin, Kerosene	7	0.19	0.82
4	Electricity	1	0.03	0.12
5	Biogas	0.4	0.01	0.05
6	Other	0.4	0.01	0.05
7	Crop Residues	0.1	0	0.01
8	LPG	0.1	0	0.01
	Total	**100**	**2.77**	**11.69**

The following table 2 summarizes the average fuel consumption for different cooking fuels in Tanzanian household.

Table 2: Tanzania cooking cost scenario (traditional cooking method)[2]

Fuel	Purchase unit	Usage	Cost of purchase	Fuel use per day	USD per week
Firewood	3 pieces	1 day	500 TZS (USD 0.31)	3 pieces	2.19
Charcoal	40 kg sack	25 days	30,000 TZS (USD 19)	1.6 kg/day	5.25
LPG	13 kg cylinder	30 days	56,000 TZS (USD 35)	0.44 kg/day	8.12
Kerosene	1 litre	2 days	2,500 TZS (USD 1.6)	0.5 litre/day	5.46

Note: Fuel usage value will vary depending on family size, location, stove and fuel mix.

[1] National Bureau of Statistics 2012 Census, 2007 Household Budget Survey and NBS-Tanzania Commission for AIDS-Tanzania HIV/AIDS and Malaria Indicator Survey 2011-12 (NBS, 2009, 2012a, 2013 a&b, NBS-TACAIDs, 2013).

[2] http://www.cleancookstoves.org/resources_files/tanzania-market-assessment-mapping.pdf

Annex 8. Focus sectors, gaps and scope of interventions

Activities	Gaps Identified	Scope for interventions	Stakeholders[1]
1.1. Integration of available information resources on renewable energy[2]	• Though various initiatives and works are being done related to RE and EE within the country, there is no integration of information resources/services, so that the information is accessible to relevant stakeholders as per their requirement from one location. • The web portal that is being developed through TREMIS requires to be updated and maintained regularly. There is no plan for that right now. • TIC is unable to get the latest information from various relevant institutions to offer updated service accordingly. This is essential to attract investments in the country • Mapping of potential biomass and wind mini-grid sites need to be updated in the www.minigrids.go.tz portal. • Conduct of baseline study on ICS/clean cook stove market demand assessment at the district and regional levels with special focus on women preferences. ICS market intelligent study has been conducted so far in Arusha, Manyara and Mtwara regions only.	• Web portals of TREMIS to be integrated with all other resources such as TIC (one-stop-shop for investment), mini-grids information hub (www.minigrids.go.tz), ICS Task force, NEMC procedures, TRA information, resource assessments or anything done by donors, etc. • Ensure continuous update of TREMIS website and its integrated information • The gaps in the information available will also help to decide the scope of intervention • The short term goal can be that the website can be made as A-Z renewable energy portal, i.e., which can give information on all aspects related to renewable energy. • The long time goal can be one-stop-shop for energy	MEM, MOIT, VPO-DoE, TFS, TBS, PORALG, NEMC, MoF UNDP LGAs Universities & research institutions Banks, financial institutions International donor agencies Equipment suppliers Private sectors NGOs/CSOs
1.2. Addressing knowledge gaps in	• No consolidated data/information on institutions, courses, curriculum, training facilities available for	• Support NEMC with technical capacity for making the checklist for hydro and biomass mini-grid projects. Similarly, such gap can be identified for other businesses as well and	Ministries (MEM, MIT, VPO-DOE, TFS, TBS, PORALG, REA,)

[1] Tentative list only. The actual could be more or less depending upon the actual implementation activities.
[2] Beneficiaries: Entrepreneurs, business developers, private sectors, international and local investors, funding agencies

Activities	Gaps Identified	Scope for interventions	Stakeholders[1]
RE, EE, modern cooking[3]	ICS/LEDs/CFLs/BLEN[4]/SHS/solar PV/solar thermal/small hydro power turbines/other EE sectors • No centralised approved minimum quality standard set for the various training courses, facilities, trainers' qualification, trainees' qualification, etc. The present status is that, different institutions are providing different levels of training courses with different training facilities in the renewable energy and cooking sector • Limited training courses/facilities for EE sector • Need for capacity building for public, private sector, financial institutions, NGOs, etc., as per their requirements/involvement in RE/EE sector	support can be provided subsequently for one by one depending upon the priorities then. • Assessment of facilities/resources available at Universities and institutions in line with SE4ALL focus sector • Assessment of capacity building in various focus sector needed for women and men in ministries, LGAs, private sectors, financial institutions, entrepreneurs, artisans, NGOs, technical institutes, etc. in their own levels • Identification of nodal training institute for each sector • Design of curriculum for training/certification courses based on assessment in these sectors and stakeholders • Development of minimum standards for setting up training centres and trainings • Selection of partner channels[5] to provide these training/certification courses at large scale • Setting up of training/skill development centres at the selected Universities/institutes and partner channels • Preparation of guidebooks, manuals, training materials, etc. • Trainings/certification courses conducted and certificates provided to trainees • Establishment of central accreditation unit to ensure quality training partnership for all training centres under the program • Development of quality assurance procedure and continuous monitoring	UNDP Universities & research institutions International donor agencies LGAs

[3] Beneficiaries: Ministries, local government authorities, banks, financial institutions, private sectors, entrepreneurs, NGOs/CSOs, women development organisations, skilled and semi-skilled labours

[4] Biogas, LPG, ethanol, natural gas

[5] Partner channels are authorized private sector partners who will set up training centers and provide trainings within the curriculum and minimum standard set by the nodal institute

Activities	Gaps Identified	Scope for interventions	Stakeholders[1]
1.3. Institutional framework for EE promotion[6]	• Institutional framework needed for integrated approach on promotion/enforcement of nation-wide EE activities across the sectors • Need for capacity building for government ministries, private sectors, financial institutions, entrepreneurs, technical institutes, skilled labours, NGOs, etc. on EE and energy conservation • Inadequate availability of EE or energy audit professionals • Need for benchmarking baseline energy consumption in industries (mining, small and medium agro-based), transport, households and commercial building	• Design of roles, responsibilities, budget and organisation structure for formation and operation of dedicated EE unit/team within the existing policies and regulations • Assessment of energy use in different sectors and establishment of benchmark for energy consumption • Identification of designated consumers who must achieve energy consumption reductions (voluntarily or mandatorily) • Assessment of capacity building needed for ministries, private sectors, financial institutions, entrepreneurs, artisans, NGOs, institutes, etc. • Identification of institution to support EE courses and conduct of energy auditor exams • Design of curriculum for training/certification courses • Preparation of guidebooks, manuals, training materials, etc. • Conduct of energy auditor exams and certificates provided to the successful candidates • Development of a dedicated website and update of online resources • Linking it with TREMIS energy portal	MEM TBS PORALG Universities & research institutions International donor agencies LGAs
2.1 Prioritization of RE/EE sector in investment promotion, commercial lending and donor funds	• No special consideration is given for the RE projects in the banks for credit facilities • Lack of long-term funding for RE projects is a major constraint • Inability of the project developers to match the high equity requirement	• Workshops and discussion forums between financial institutions, donors and RE/EE sector companies • Establish green funds for finance and credits to RE/EE sector projects • Developing bank guarantee schemes for RE/EE projects	

[6] Beneficiaries: Ministries, Local government authorities, banks, financial institutions, energy intensive industries, energy engineers, skilled and semi-skilled labors

Activities	Gaps Identified	Scope for interventions	Stakeholders[1]
	(30% of total investment) of the local banks. • More documentation and longer process time		
2.2. Quality standards and testing regulations[7]	• Need for quality standards on: – Biogas cook stoves – LPG stoves – Natural gas stoves – Ethanol stoves – Pressure cookers – Fuel grade ethanol – LED/CFL Lightings – Small hydro turbines – Wind turbines – Solar PV panel and major components of SHS • Need to study Regional/international standards available and their possible adaptation for above products • Inadequate testing facilities for product testing • Poor enforcement of product standards through market surveillance • Inadequate consumer awareness in the purchase of quality products • Lack of enforcement on the use of LA products until additional standards products are described. This can also avoid the low quality products coming into the market.	• Study the existing Regional standards availability and their quick adaptation for Tanzania • Preparation of quality standards for renewable energy products, if regional standards are not available • Strengthening of existing testing centres at TBS and/or other Universities/institutions • Establishment of additional testing facilities (for solar panel, LED lighting, etc.) at TBS and/or other Universities/institutions • Encouraging private sector to publish quality approvals of product through website/newspapers/advertisements/ banners for imported products • Advertisements in newspapers, radio, TV channels to purchase only quality approved products • Only approved quality products to be promoted through SE4ALL	MEM TBS Universities & research institutions International donor agencies Private sectors NGOs/CSOs LGAs
2.3. Promotion of RE industries/businesses	• Limited industries in the local manufacturing/assembling, spare parts	• Promotion of assembling/production of products/spare parts, after sales and services, O&M related companies dealing with RE/EE components within SEZ (For example: solar panels, batteries, SHS peripherals, energy	Ministries (EPZA, MEM, MIT, VPO-DOE, TFS, TBS)

[7] Beneficiaries: Product manufacturers, entrepreneurs, project developers, testing labs, end users

Activities	Gaps Identified	Scope for interventions	Stakeholders[1]
[8]	and O&M related industries in RE/EE sector • High cost of products due to existing overheads and • Limited funds and resources for development of infrastructure services for SEZ • Limited awareness in establishing industries within SEZ	efficient TVs, lighting products, other components used with SHS, wind turbine related components, biomass power plant based components, biogas based components, spares of any such systems, relevant after sales services and O&M firms, etc.) • Collaboration with EPZA on infrastructure development of SEZ sites	Banks, financial institutions International donor agencies Private sectors LGAs
3.1. Increasing electricity access through grid expansion[9] (activities under this will also contribute to increasing RE share in overall power mix in country)	• Inadequate finance/grants for the programs • Additional capacity building for TANESCO and REA for the improved project implementation and monitoring • Inadequate land use allocation for transmission lines • Inappropriate synchronization of the RE grid connected plants • No compensation for the loss occurred to the RE based IPPs, when they are ready to export and could not be taken by TANESCO • Delayed payment from TANESCO to RE based IPPs	Around 85%[10] of target electricity access (to 75% of total population as per AA) to be provided through the main grid connecting those settlements/villages/ households that are within 10 km of the existing grid infrastructure (Reference: Action Agenda, page 21) • Capacity building for TANESCO and REA on project monitoring • Support for effective land use allocation for transmission lines • Finance for grid expansion programs	TANESCO EWURA REA LGAs International donor agencies
3.2 Increasing electricity access through decentralised RE (activities under this will also contribute to increasing RE share	• No detailed assessment of potential sites with RE resources (Like municipal abattoir waste water treatment, agro processing industries like coffee pulp washing, coffee curing, cashew processing industries, kitchen wastes in residential institutions, etc.), load	Around 15% of target electricity access (75% of the total population as per AA) to be provided through rural electrification by mini-grid, off-grid and decentralised RE power for settlements/villages/households located beyond 10 km from the main grid and not expected to be	MEM TANESCO EWURA REA PORALG

[8] Beneficiaries: Private sectors, entrepreneurs, project developers

[9] Beneficiaries: Households, small and medium scale industries, entrepreneurs, schools, hospitals, etc.

[10] 85 % of 75% SE4ALL target

Activities	Gaps Identified	Scope for interventions	Stakeholders[1]
in overall power mix in country)	demand, time for grid reach to the region, etc. • No special consideration for RE projects for credit facilities in banks/financial institutions (not to be compared with other general business models) • No successful business models for mini-grids • Inadequate standalone electricity access to schools, health centres, irrigation water pumps, domestic water supply pumps, business units, etc. • Inadequate utilization of biomass for community power generation • Inadequate availability of skilled technicians for after sale servicing in each locality • Inadequate funds/grants for the projects • Inadequate availability of technical expertise in relevant RE technology • Inadequate land use allocations for the project sites • Inadequate utilization of Municipal solid wastes, waste water disposal from municipal abattoirs, bio-wastes from community kitchens, etc.	connected to the grid until 2030 (Reference: Action Agenda, page 21) • Solar/wind/small hydro based community mini-grids supplying to communication towers, small businesses and households • Solar/wind based power for schools, health care units, hospitals, etc. • Solar based street lighting, irrigation water pumping and domestic water supply pumping • SHSs • SWHs in commercial buildings, hotels, schools, hospitals, etc. • Pay-as-you-go model/pay-as-you-use model • Waste to power in community kitchens, municipal abattoirs, agro processing units like coffee pulp washing, coffee curing, etc. • Research and survey to be conducted to collect statistics on power access in street lighting, irrigation water pumping, domestic water supply pumping, schools, primary health care units, etc. • After sale service centres (for example: Solar Kiosks) or product servicing of all/most common brands under one roof in each region/town with special focus for women entrepreneurs. Local technicians will be encouraged. It would be win-win for customers, local technician and private sector • Awareness campaigns • Solar panel and battery recycling	Banks, financial institutions International donor agencies LGAs Equipment suppliers Private sectors NGOs/CSOs

Activities	Gaps Identified	Scope for interventions	Stakeholders[1]
3.3. Increasing RE share in thermal requirements[11]	• No nationwide data available on the exact requirement of firewood, charcoal, use of ICS, etc. • Inadequate finance/grants for the programs • Inadequate stakeholder engagement at all levels (ministries, local government, communities, NGOs, etc.) • Inadequate awareness to end users on the benefits of using good quality and improved charcoal stoves • Lack of sustainable charcoal supply chain networks • Inadequate effective land use planning for sustainable harvesting of wood for fuel (fire wood and/or charcoal) • Unexplored Waste to energy potential in municipal abattoirs, agro industries, etc. • Inadequate village land use planning to be done in all villages • Limited awareness raising about the sustainable forest use, sustainable harvesting, importance of preserving local bio diversity, separate grazing, sustainable harvesting for fuel wood, etc.	• Increase sustainable charcoal production capacity (energy plantations/sustainable harvesting/biomass briquettes) • Effective land use planning to identify and allocate lands for sustainable biomass cultivation • Assistance in village land use planning to cover more villages within a shorter period • Develop improved cook stove production and market network • Co-operate and co-ordinate with activities of Global Alliance for Clean Cook stoves and ICS Task Force to improve ICS market • Effective engagement of Local government authorities for implementation support • Awareness campaigns • Assessment of WTE (for heat) potential in all Municipality abattoirs, coffee pulp and cashew mango process waste, community kitchens, etc.	Ministries (VPO-DOE, TFS, TBS, PORALG, Land use planning commission) LGAs Banks, financial institutions International donor agencies Equipment suppliers Private sectors NGOs/CSOs

[11] Beneficiaries: Households, energy intensive industries, small and medium scale industries, commercial buildings, schools, hospitals, etc.

Activities	Gaps Identified	Scope for interventions	Stakeholders[1]
3.4. Reducing energy intensity through EE measures[12]	• Inadequate finance/grants for the programs • Inadequate stakeholder engagement at all levels (ministries, local government, communities, NGOs, etc.) • Need of Minimum Energy Performance Standards (MEPS) • Awareness to end users on the benefits of using energy efficient products • Technical assistance on energy efficient technologies	• Technical assistance (for resource assessments, feasibility study and development of business plans through international agencies), grants/loans to TANESCO for improving grid efficiency • Alignment of efficient lighting initiatives with other activities such as SHSs, solar lanterns, grid connection to households under RE mini-grids and EE improvement in industries/commercial buildings • Engagement of experts/consultants for preparation of Minimum Energy performance (MEPS) of electrical • Product testing and certification for MEPS and performance rating • Conduct of nationwide awareness campaigns • Training and accreditation of energy audit firms • Conduct of energy audits at industries of designated consumers • EE promotion in large buildings (public, private and commercial) • Subsidy scheme for conduct of energy audit and implementation of EE/energy saving measures by designated consumers	Ministries TANESCO REA LGAs Private sectors Banks, financial institutions International donor agencies
3.5. Roll out of modern cooking solutions[13]	• Inadequate grant/fund for promoting BLEN & compensate for the high cost involved in using these technologies. • Lack of assessment of exact number of modern cook stoves in use and market	• Digester quality assessment/assurance system (Ex. mason incentive programme) • Encourage sale of standard/approved stove, cylinder, regulators models, • Pressure cooker + separators for fuel savings • Pay-as-you-use model • Nation-wide awareness campaigns • Education to women on decision making for choosing cooking options	MEM REA EWURA TPDC PORALG LGAs

[12] Beneficiaries: Households, energy intensive industries, small and medium scale industries, commercial buildings, schools, hospitals, etc.
[13] Beneficiaries: Households, community kitchens, hotels, etc.

Activities	Gaps Identified	Scope for interventions	Stakeholders[1]
	potential for different clean cooking fuels • Inadequate micro-finance for end user to overcome high cost of modern cooking solutions • No successful business model for development of supply chain network for modern cook stoves and related fuel supply • No quality assessment for biogas digesters • No quality standards for cook stoves and cooking fuels (like LPG stoves, ethanol stoves, biogas stoves, natural gas stoves, etc. and fuel ethanol, natural gas, for cooking.) • Inadequate awareness creation on health and environmental benefits of modern cooking solutions • Inadequate education to women on decision making for choosing cooking options • Lack of data on LPG penetration rate, which could help in information to the reduction on wood fuel. • Inadequate promoting biogas cooking solutions in residential schools, hotels, communities, etc.	• Finance/grants/incentives for project developers • Micro-finance options for end users for stove and fuel. • Women self -help group, NGOs, SACCOs, etc. can be involved.	Private sectors Banks, financial institutions Micro-finance institutions International donor agencies

Annex 9. Resource mobilization initiatives

Tanzania SE4ALL – Tentative Sources for Resource Mobilization

(Based on personal discussions and email communications)

No.	Agency	Focus Area	Fund available (million USD) *(tentative)*
1	DFID	• Solar based mini-grid and off-grid solutions • Adapting the quality standards for solar products; use of already available standards like Lighting Africa project quality standards, or EAC standards, etc. • Capacity building for the solar entrepreneurs; Women involvement in solar business • Assurance of quality on curriculum, training activities, physical infrastructure, trained personnel qualification, etc. of training centres/institutions. • Awareness raising and campaigning activities	6.20
2	EIB	Natural gas based cooking solutions	Expressed Interest
3	EU	• Biomass development • Sustainable charcoal harvesting, ICS	5.25
4	GIZ	• Dedicated unit for EE • EE and performance labelling • Energy audits and energy conservation	8.60
5	SIDA	• Energy efficient lightings • green RE mini-grids projects in Tanzania • technical assistance to REA on energy in gender, productive uses and solar home systems • RE/EE mapping • EE lightings • Support TBS for testing of lights and mobiles • Support NEMC for technical capacity building	5.65
6	UNDP	• High impact activities like solar water pump installations for irrigation, solar power for health centres and schools in rural areas, etc. • Solar kiosks, • Training, awareness raising, etc. • MER • Project Management *UNDP will mobilize around additional USD 14 million through various sources for this program.*	1
7	USAID	Bank guarantees for project developers	Expressed Interest